

DETAILED PROJECT REPORT

FOR THE PROPOSED
ARUNAI DEEMED TO BE UNIVERSITY
THIRUVANNAMALAI

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ARUNAI DEEMED UNIVERSITY



THE ORGANIZATION OF THE PROPOSAL (REPORTS)

The Saraswathi Ammal Educational Trust is applying for the Deemed University Status under the Distinct Category. The application, as per the Deemed University Regulation 2023, is submitted online, and the following reports are developed as per the requirements.

- I. The Self-Study-Report (DPR)
- ii. The Strategic Plan
- iii. The Institute Development Plan (IDP)

THE ANNEXURES ACCOMPLISH THE SELF-STUDY REPORT

THE SELF-STUDY REPORT

HIGHLIGHTS ON THE FOLLOWING INFORMATION

- ▶ The profile of the project institutes.
- ▶ The Educational programs offered by the project institutions
- ▶ The existing resources include, Human Resources, Land, Buildings, Machinery, Books, Furniture, and others. The purpose is to demonstrate the financial investment in the project institutes.
- ▶ The research profile.
- ▶ The industry institute interactions and MOU.
- ▶ The Community Development activities.
- ▶ The outcomes of the educational programs. The placements and the progression to higher education.
- ▶ The sports and cultural activities conducted in the past.
- ▶ The inclusiveness in students' admissions.
- ▶ The diversity and scholarships for the students.
- ▶ Alumni Contributions.
- ▶ The income and Expenditure statements of all project institutes together.
- ▶ The achievements and awards.
- ▶ The statutory compliances and certifications (including NAAC/NBA accreditation)

THE STRATEGIC PLAN (SP)

INCLUDES THE FOLLOWING INFORMATION.

- ▶ Introduction
- ▶ Preamble
- ▶ Vision
- ▶ Mission
- ▶ Short-term goals
- ▶ Medium-term Goals
- ▶ Long-Term-Goals
- ▶ The program and projects to meet the goals
- ▶ The strategies
- ▶ Key stages in the project development
- ▶ The financial planning
- ▶ The operational planning.

THE INSTITUTE DEVELOPMENT PLAN (IDP)

INCLUDES THE FOLLOWING INFORMATION.

- ▶ Introduction
- ▶ Preamble
- ▶ The Goals
- ▶ Mapping of university goals with the UGC-recommended strategic goals
- ▶ The Enablers
- ▶ The mapping of Goals and projects with the Enablers
- ▶ Developmental Objectives of the Arunai University
- ▶ The key performance indicators

These three documents are connected and describes the institutional planning at the different stages along with the strategies. All three documents are supported to each other to establish the objectives of the project, operational strategies, planning and addressing the enablers of higher education. The annexures are attached as the evidences for the relevant information.

CHAPTER-I

1.1. PREAMBLE

The University Grants Commission (Institutions Deemed to be Universities) Regulations, 2023 outline the criteria and benefits for institutions seeking "Deemed-to-be University" status under the distinct category. This status is granted to institutions that demonstrate high academic standards and a commitment to excellence in education.

For the Saraswathi Ammal Educational Trust, achieving "Deemed-to-be University" status would signify national recognition of its educational excellence and autonomy in academic and administrative matters. This autonomy would allow the Trust to design innovative curricula, establish interdisciplinary programs, and foster research initiatives tailored to societal needs. Additionally, the Trust could expand its reach by establishing off-campus centers after five years, further contributing to regional and national educational development. Attaining "Deemed-to-be University" status would enhance the Trust's ability to provide high-quality education, promote research, and contribute significantly to societal advancement.

Saraswathi Ammal Educational Trust is poised to achieve Deemed-to-be University status under the Distinct Category as per the UGC (Institutions Deemed to be Universities) Regulations, 2023. This recognition would enable the trust to create a pioneering academic ecosystem that integrates education, research, innovation, and societal impact at an exceptional level.

Alignment with the Objectives of a Distinct University

The UGC Deemed University Regulations (2023) define a Distinct Category institution as one that:

- ▶ Offers unique and interdisciplinary programs addressing futuristic academic and research needs.
 - ▶ Has a strong focus on innovation, entrepreneurship, and skill development aligned with national priorities.
 - ▶ Demonstrates regional, national, and international impact in education and research.
 - ▶ Contributes significantly to society, sustainable development, and social transformation.
- Saraswathi Ammal Educational Trust meets these criteria through its long-standing commitment to academic excellence, industry collaboration, and community upliftment.

Capabilities of Saraswathi Ammal Educational Trust to Become a Distinct University

A. Academic Excellence and Innovative Programs

- ▶ Establishment of multidisciplinary Schools in Engineering, Technology, Healthcare, Law, Management and Arts and Science with cutting-edge curricula.
- ▶ Introduction of AI-driven learning models for adaptive and personalized education.
- ▶ Development of skill-oriented certification programs in Industry 4.0 technologies, digital transformation, and sustainable development.
- ▶ Integration of Ayurveda and modern medicine research, leveraging the trust's expertise in health sciences institutions.

B. Research, Innovation, and Entrepreneurship

- ▶ Creation of specialized research centers in semiconductor technology, AI, renewable energy, and biotechnology.
- ▶ Development of an Incubation and Entrepreneurship Hub, supporting startups, patent development, and technology transfer.
- ▶ Encouragement of faculty and student participation in international research collaborations and government-funded projects.
- ▶ Establishment of a Technology Business Incubator (TBI) under Atal Innovation Mission and DST grants.

C. Industry Collaboration and Global Outreach

- ▶ Establishment of strong linkages with industries to create an industry-academia ecosystem.
- ▶ Offering dual-degree and twinning programs in collaboration with top global universities.
- ▶ Expansion of MOUs with multinational corporations (MNCs) and research organizations for joint R&D projects.
- ▶ Participation in national and international policy development in higher education and research.

D. Social and Regional Impact

- ▶ Rural Development Initiatives – Providing skill training programs, vocational education, and rural entrepreneurship support.
- ▶ Public Health Contributions – Extending healthcare services through telemedicine, mobile health units, and community health initiatives.
- ▶ Women Empowerment Programs – Offering scholarships, leadership development workshops, and microfinance support for women entrepreneurs.
- ▶ Sustainable Development Goals (SDGs) Implementation – Working on clean energy projects, smart villages, and environmental conservation initiatives.

E. Digital and Smart Campus Infrastructure

- ▶ Establishing a fully digitalized campus with AI-driven academic and administrative management.
- ▶ Implementation of Learning Management Systems (LMS), Virtual Labs, and Smart Classrooms.
- ▶ Developing a Green Campus with sustainable energy and eco-friendly infrastructure.

Achieving Deemed-to-be University status would empower Saraswathi Ammal Educational Trust to:

1. Exercise Academic Autonomy – Develop and deliver innovative, interdisciplinary, and customized programs aligned with global education trends.
2. Expand Research and Innovation – Attract top-tier researchers, faculty, and students to work on national and global challenges.
3. Enhance Industry and International Collaborations – Strengthen ties with corporates, research bodies, and foreign universities to create a world-class learning environment.

4. Transform Societal Impact – Scale up initiatives in education, healthcare, entrepreneurship, and sustainability, benefitting thousands in the local and regional communities.
5. Become a Knowledge Hub for Future Technologies – Establish a global research destination in AI, IoT, robotics, and emerging sciences.

With its visionary leadership, academic innovation, research excellence, and societal commitment, Saraswathi Ammal Educational Trust is well-positioned to emerge as a Distinct University under the UGC Deemed University Regulations (2023). This transition would not only elevate the institution's national and global standing but also create transformative educational and research opportunities for future generations.

1.2. ABOUT THE FOUNDER AND MANAGEMENT OF THE TRUST

The Saraswathi Ammal Educational Trust was founded with a mission to provide accessible, high-quality education to students, particularly in rural and semi-urban areas of Tamil Nadu. The trust has set up multiple institutions, including Arunai Engineering College (AEC) and other higher education institutions, to cater to the growing demand for technical and professional education.

The key objectives behind establishing these institutions include:

1. Empowering Youth through Education – Providing students with knowledge and technical skills to enhance employability.
2. Promoting Innovation and Research – Encouraging students to engage in innovation, research, and entrepreneurial ventures.
3. Enhancing Societal Development – Contributing to the socio-economic development of the region by producing competent professionals.
4. Integrating Ethics and Values – Fostering moral and ethical education to develop responsible citizens.
5. Expanding into Health Sciences – Through a sister organization, the trust has also ventured into health sciences, offering medical and allied health programs to meet the growing healthcare needs.

The recognition by Frontline magazine further validates the trust's impact in shaping Tamil Nadu's education sector. Today, under the leadership of Thiru E. V. Velu, the institutions continue to evolve, aligning with global education standards and advancing interdisciplinary learning. Thiru E. V. Velu, through Saraswathi Ammal Educational Trust, has made a significant impact on education, societal development, and the upliftment of local communities in Tamil Nadu. His visionary leadership has transformed the region by establishing quality institutions and empowering students with knowledge, skills, and values. Under the leadership of Thiru E. V. Velu, Saraswathi Ammal Educational Trust has revolutionized education, uplifted communities, and contributed to the economic growth of Tamil Nadu. Through its institutions, research initiatives, and social welfare programs, the trust has empowered thousands of students and professionals, ensuring that they contribute to the progress of society. The recognition by Frontline magazine as a key institution shaping Tamil Nadu's education system further highlights the immense contribution of the trust in transforming lives through education and community service. Today, AEC and its allied institutions continue to innovate and expand, with a vision to become a globally recognized center of excellence while staying committed to regional and societal development. Major Contributions and Achievements of Saraswathi Ammal Educational Trust The Saraswathi Ammal Educational Trust, under the leadership of Thiru E. V. Velu, has made remarkable contributions to education, research, social development, and healthcare.

The Saraswathi Ammal Educational Trust, under the leadership of Thiru E. V. Velu, has made remarkable contributions to education, research, social development, and healthcare. The trust has transformed the educational landscape of Tamil Nadu, particularly in rural and semi-urban areas, and has earned national recognition for its impact. Later the trust is led by Mrs. Shankari, the wife of Tamil Nadu Minister E.V. Velu. Together, they have expanded the trust's reach, establishing institutions that cater to various educational needs. Their son, Er. E.V. Kumaran, holds the position of Vice Chairman of SAET, contributing to the trust's ongoing development. As Vice Chairman, He plays a pivotal role in overseeing the trust's various educational institutions, ensuring they maintain high standards of academic excellence. Under Mrs. Sankari's leadership, the trust manages several institutions, including, Arunai Engineering College, Karan Arts and Science College, Kamban Arts College for Women.

1. Key Contributions in Education

a. Establishment of Leading Educational Institutions

- ▶ Arunai Engineering College (AEC) (Founded in 1993) – One of the top engineering institutions in Tamil Nadu, offering diverse technical and professional courses.
- ▶ Other Higher Education Institutions – The trust manages three additional colleges and a school, covering disciplines such as engineering, management, technology, and healthcare.
- ▶ Expansion into Health Sciences – Through a sister organization, the trust has established health science institutions to train future medical professionals.

b. Bridging the Urban-Rural Education Gap

- ▶ Focuses on providing quality education to students from rural and semi-urban backgrounds.
- ▶ Offers affordable education and scholarships to deserving students.
- ▶ Provides holistic education that integrates academics, research, ethics, and values.

c. Industry Collaboration and Research Excellence

- ▶ Strong ties with industries and research organizations for curriculum development, internships, and job placements.
- ▶ Encourages entrepreneurship and innovation, leading to the creation of startups and research projects.
- ▶ Supports faculty and student research in engineering, management, and health sciences.

2. Achievements in Community Development and Social Impact

a. Upliftment of Local Communities

- ▶ Provides employment and skill development opportunities for youth in the region.
- ▶ Supports women empowerment programs and self-employment training.
- ▶ Conducts community service programs, such as rural education and vocational training.

b. Contribution to Healthcare and Public Welfare

- ▶ Established health science institutions to train professionals in medical and allied health fields.
- ▶ Conducts free medical camps, blood donation drives, and public health awareness programs.
- ▶ Supports maternal and child healthcare programs in rural areas.

c. Infrastructure Development in the Region

- ▶ Developed modern educational infrastructure, including state-of-the-art classrooms, laboratories, and research centers.
- ▶ Provides student hostels, libraries, sports complexes, and digital learning resources.
- ▶ Promotes eco-friendly and sustainable practices within its campuses.

3. Recognition and Achievements

a. National and Media Recognition

- ▶ Featured in Frontline Magazine as a key institution shaping Tamil Nadu's education system.
- ▶ Recognized for academic excellence, research contributions, and innovative education models.

b. Academic Excellence and Accreditations

- ▶ Institutions under the trust have secured accreditations and affiliations from reputed academic bodies.
- ▶ High success rates in placements and competitive exams, with graduates excelling in top companies, research organizations, and government services.

c. Strategic Expansion and Vision for the Future

- ▶ Aiming to transform Arunai Engineering College into a Distinct Deemed University as per UGC guidelines.
- ▶ Plans to establish new interdisciplinary and industry-oriented programs for future technological advancements.
- ▶ Strengthening international collaborations and research partnerships to elevate the trust's institutions to global standards.

The Saraswathi Ammal Educational Trust, under Thiru E. V. Velu's visionary leadership, has revolutionized education, empowered communities, and contributed to regional development. Its institutions continue to shape future leaders, innovators, and professionals while staying committed to quality education, research, and social welfare. With a strong foundation and a clear vision for the future, the trust is poised to expand its impact nationally and internationally, ensuring continued growth in education, healthcare, and community development. The Saraswathi Ammal Educational Trust, under the leadership of Thiru E. V. Velu, has played a transformative role in Tamil Nadu's education sector, healthcare, and community upliftment. The Saraswathi Ammal Educational Trust is not just an educational management body; it is a catalyst for societal transformation. Through its innovative programs, industry-driven research, sustainable development efforts, and community welfare initiatives, the trust continues to redefine education and social progress in Tamil Nadu and beyond. With an unwavering commitment to excellence, the organization is poised to set new benchmarks in global education, research, and social innovation while remaining deeply rooted in its mission of empowering communities and transforming lives.

The management of Saraswathi Ammal Educational Trust, led by chairperson Thiru E. V. Velu, comprises visionary leaders, philanthropists, academicians, and industry experts who have significantly contributed to the development of society beyond education. Their individual and collective efforts have impacted economic development, social welfare, public policy, and infrastructure growth across Tamil Nadu. The management members of Saraswathi Ammal Educational Trust have not only shaped higher education in Tamil Nadu but have also played a pivotal role in policy reforms, economic development, public health, social justice, and environmental sustainability. Their continued commitment to societal progress ensures that their impact extends beyond classrooms, fostering a better future for generations to come.

CHAPTER-II

2.1 THE LIST OF CONSTITUENT UNITS (DEPARTMENTS, SCHOOLS, CENTERS).

The present proposal for the establishment of Deemed University under Distinct category includes following three institutions of the same management.

1. Arunai Engineering College.
2. Kamban College of Arts and Science for Women.
3. Karan Arts and Science College.

Arunai Engineering College (AEC) is approved by AICTE, affiliated with Anna University, and recognized by the UGC Act of 1956 u/s 2(f) and under section 12(B) in the year 2013. The College was accredited by NAAC with an 'A' Grade in the year 2015 and a B+ in the year 2022, 'A' grade in 2024 and certified for Quality Management System by DNV, Netherlands under ISO 9001:2015 standards. The College offers Eight undergraduate engineering programmes, 5 Postgraduate programmes, MBA and 2 Doctoral Programmes.

Kamban College of Arts and Science for Women was established in the year 1996 and affiliated with Thiruvalluvar University. The founder of the Institution is Thiru. E.V. Velu has a long-time desire to provide good education to rural women students. Empowering the rural women sector is the major aim of college. The college was incepted in the year 1996 with the strength of 18 students. Now the Institution is running successfully with 3250 students and 115 faculties. The college currently offers 10 Undergraduate (UG) programs, 7 Postgraduate (PG) programs, 7 M. Phil programs, and a few certificate courses.

Karan Arts and Science College established in the year 2016 and affiliated with Thiruvalluvar University to foster education in the arts and science domain for the rural youth. The college is offering 12 UG courses and 4 PG courses.

2.2 INSTITUTIONAL PROFILE

The proposal encompasses three institutions. Arunai Engineering College (AEC), Kamban College of Arts and Science for Women and Karan Arts and Science College. All three project institutes are sponsored by one organization/management, Saraswathi Ammal Educational Trust. The Arunai Engineering College established in 1993, has been a beacon of quality engineering education and a transformative force in the region. With a mission to develop technically proficient, morally upright, and socially responsible professionals, AEC has consistently delivered a high standard of education, supported by a holistic approach to student development. Located on the Chittoor-Cuddalore national highway, the campus provides a serene, inspiring learning environment that fosters intellectual curiosity and personal growth.

2.3 ABOUT ARUNAI ENGINEERING COLLEGE

Situated in a rural area of the state, the Institute is committed to providing affordable and high-quality technical education to the local community. Over its 30-year history, the institution has produced numerous engineers and technocrats who have gone on to work dedicated to this noble cause for the past 30 years. The institution is committed

to aligning in prestigious firms worldwide. With a dedication to enhancing educational standards, the institute actively seeks to add value through certifications from esteemed organizations.

Established in the rural part of the state with a philanthropic vision to provide technical education to the underserved rural population, Arunai Engineering College has been itself with the nation's development, consistently excelling in education, research, and social activities. Guided by the mission, "Our products are our ambassadors," the institute actively practices this ethos, seeking recognition in NIRF, AISHE, NAAC, NBA, and IIC (AICTE).

Emphasizing holistic development, students are trained in curricular, co-curricular, extracurricular, cultural, and sporting activities while actively engaging in social initiatives for the local community's upliftment. Over the course of three decades, the institute has produced numerous engineers and technologists, placing them in prestigious positions. The pride lies in contributing to the upliftment of the rural community and playing a role in nation-building. The global placement of our students in renowned companies attests to the quality of education we provide.

LIST OF COURSES OFFERED	
SR. NO.	COURSES
1	B.E. - Agricultural Engineering
2	B.E - Artificial Intelligence and Machine Learning
3	B.E - Civil Engineering
4	B.E - Computer Science and Engineering
5	B.E - C.S.E-(Cyber Security)
6	B.E - Electrical and Electronics Engineering
7	B.E - Electronics and Communication Engineering
8	B.E - Mechanical Engineering
9	B.TECH - Artificial Intelligence and Data Sciences
10	B.TECH - Biotechnology
11	B.TECH - Chemical Engineering
12	B.TECH - Information Technology
13	M.E. - Computer Science and Engineering
14	M.E - Power Electronics and Drives
15	M.E - Thermal Engineering
16	M.TECH - Biotechnology
17	MBA - Master of Business Administration

Arunai Engineering College (AEC), a Co-Educational institution was established in the year 1993 under Saraswathi Ammal Educational Trust in Tiruvannamalai, a rural area of Tamilnadu to provide access to higher education for the people of this part. The college is spread out on serene sylvan settings on the Chittor - Cuddalore national highway, providing a holistic environment, ideal for dedicated study with discipline, away from the hustle and bustle

of the modern cities. The serene spiritual atmosphere at AEC helps an individual to discover him/her and the contribution he/she can make to the world. We, at Arunai Engineering College, strive to bring out and enlighten the hidden technical skills and abilities of youth with the highest quality technical education and proper discipline.

AEC is approved by AICTE, affiliated with Anna University, and recognized by the UGC act of 1956 u/s 2(f) and under section 12(B) in the year 2013. The College was accredited by NAAC with an 'A' Grade in the year 2015 and a B+ in the year 2022, 'A' grade in the year 2024 and certified for Quality Management System by DNV, Netherlands under ISO 9001:2015 standards. The College offers 12 undergraduate engineering programmes, 5 Postgraduate programmes, including MBA and 2 Doctoral Programmes. The institution has a student strength of 2683 as of today and a staff strength of 195.

The teaching-learning process in the college is essentially student-centric with an emphasis on the overall, holistic development of the student community. The teaching-learning process is fortified by ICT facilities, updating the library resources with the successful mentoring system in place. Our students bag university ranks every year. At present, the institution has shown immense and magnificent infrastructural growth on a spanning area of 24.66 acres with a built-up area of 27819 sq.m. for instructional and administrative buildings, 22786 sq.m. for amenities, and 31,893 sq.m. for playgrounds. It has spacious lecture halls, ICT-enabled Smart rooms, and Tutorial rooms. Well-equipped Laboratories, English language labs, Computer labs, Open Air Auditorium, Air-conditioned Seminar halls, Common Room, Sports and Gymnasium. There is also a cafeteria, a general store, a Bank, a Post office, an ATM, a Temple, and a Dispensary.

The students have flexibility in choosing their elective subjects. Students have the option to select and study 2 open elective courses across other departments. The Students may optionally undergo Value Added Courses and the credits earned through the Value-Added Courses shall be over and above the total credit requirement prescribed in the curriculum for the award of the degree. 2 departments were recognized as research centers by the affiliating University to offer PhD degrees. At the start of every semester, the affiliating University will provide an academic schedule that has curricular activities, and the institute will prepare the academic calendar based on this with co-curricular and extracurricular activities. The academic calendar will be followed by the departments to ensure the effective participation of all the stakeholders. For an efficient curricular prospect, each department will prepare separate semester plans which include lesson plans, question banks, and laboratory manuals of individual courses. The semester plan will be approved by the principal for effective implementation. Feedback are collected from the stakeholders to identify the curricular gaps and actions to rectify were taken accordingly. To strengthen the curricular outcome, the institute is organizing various co-curricular events such as Guest lectures, Seminars, conferences, symposiums, workshops, and additional courses.

Arunai Engineering College admits students based on the eligibility criteria fixed by the regulatory bodies. Undergraduate admissions were fulfilled by both Government Quota seats and Management Quota seats, 65% of the seats were filled by the Single Window Admission system framed by the Directorate of Technical Education, Government of Tamilnadu, and the remaining 35% of seats by Management through the Consortium of Self-Financing Professional, Arts and Science Colleges in Tamilnadu. Postgraduate admissions were fulfilled by the Tamilnadu Common Entrance Test and the Entrance Exam by the consortium.

To motivate the students towards their academic duty, bridge courses will be conducted for the first-year students. The institute follows an assessment methodology framed by the affiliating University. 20% of the marks were awarded based on the Continuous Internal Assessment Tests and 80% by the end semester examinations conducted by the University. Subjects were allocated to the faculty members based on their specialization. Staff handling courses will be maintaining their course file in which subject-level documents like lesson plans, records of class work, course materials, question bank, internal assessment marks, and assignments were maintained. This course file will be a key to monitoring specific outcomes of the course.

Staff are advised to use ICT resources to enhance the learning ability of the students. Students will be given assignments, presentations, and mini-projects from time to time which will increase the participative learning of the students. To inculcate the practical knowledge Industrial visits were arranged for the students when and where possible. Guest lectures, Seminars, conferences, symposiums, workshops, and additional courses were organized by the institute to enhance the students' knowledge. Students were allowed to participate in co-curricular events outside the campus, those students will be given full On-duty permission and financial aid is given when and where needed. Slow learners were identified by their performance in internal assessments and Remedial classes were conducted to improve the results. Feedback will be collected from the students and staff. Based on the feedback and end-semester exam results an analytical report on corrective and preventive actions will be prepared which will help to eradicate the shortfalls.

The institution has a well-established Research and Development cell. The Institute promotes research by motivating the faculties with various facilities. 2 Departments of the Institute were recognized as authorized research centers by the Anna University and Annamalai University. 45 staff had completed their PhD degree and 12 staff were currently pursuing. An active research forum of the Institute plays a significant role in motivating staff and students towards research. They are motivated to attend events outside the campus which will help to enhance their research skills. The Institute received Rs.16.4 Lakhs as research funds in the past 5 years. The Institute has 175 publications in the last 5 years with most of the papers published in journals in the UGC CARE list and indexed in SCOPUS, SCI, and included in Web of Science. 50 papers published in National/International conference proceedings. The Institute has an active Institute Innovation Council which is articulated and monitored by the Ministry of Education, Government of India. The Institute organizes project expos in which innovative students' projects are exhibited to the general public to make them aware of technological developments. Society-oriented projects by departments were taken and 2 of the projects Online Car Parking Management System and Missing Children Tracking System were implemented to benefit the District Police Department. Students and staff of the Institute consistently bag National awards for their innovative research work. More than 22 awards were bagged by them during the last 5 years. Staff were encouraged to apply for various internship programs and they were allowed to attend those with on duty and required financial support. More than 25 functional MoUs with various industries help the student and staff to carry out their research work, industrial visits, internships, training programs, etc.,

The institute actively involved in extension activities through its NSS, YRC wing, which regularly conducts various programs. Some of the remarkable activities are NSS camp in nearby villages, Blood Donation, Tree Plantation, Road safety awareness, Swatch Bharat, etc. The Institute coordinated with ISRO and organized World Space Week program in which more than 12000 school students and general public attended the event.

It has spacious lecture halls, ICT-enabled Smart rooms and tutorial rooms. Well-equipped Laboratories, English language labs, Computer labs, an Open -Air Auditorium, Air-Conditioned Seminar halls, Common Room, Sports and Gymnasium. There is also a cafeteria that serves good food, a general store, a bank, a Post office, an ATM, a Temple, and a Dispensary. Uninterrupted power supply through generator sets with a capacity of 1000KvA is installed on the campus. Hygienic RO-treated drinking water facilities were provided to the stakeholders. Restrooms were available in prime locations. There are a total of 97 classrooms, 48 laboratories, 2 indoor auditoriums, and 1 open auditorium on the campus. Basketball, Volleyball, Football, and Cricket grounds were dedicatedly used for the sporting activities of the students, apart from these students had the option to play indoor games in the open auditorium. There are 3 men's hostels and 1 women's hostel with mess facilities available inside the campus. To inculcate ICT-enabled education LCDs and other audio-visual facilities are provided in necessary places. The Institute's library can house 200 students at a time, it also has computers with internet connectivity which helps the students and staff to utilize the e-Resources. The library has a massive number of 45842 books in it. It also houses 220 journals (print and e-journals) and 3199 CDs. The computer lab has having 100MBPS leased line with 871 computers. Transport facilities for the students and staff with 30 buses connecting around 60 Km from the Institute.

As the institute is located in the rural part it actively involves in providing support to the students by offering Scholarships. It offered Rs.781250 in 2022-23. Apart from this the Institute initiates and helps students to avail scholarships offered by various government schemes, around 635 students availed Rs.2.90 Cr in 2022-23.

The training and placement cell of the Institute is committed to providing placement training for the students. Apart from this various other skill development programs were organized by the cell. Guidance to the students for competitive exams and higher studies was given both at the department level and the Institute level. Feedbacks were collected from time to time from the students and the concerns raised were rectified to satisfy the students' needs. Students' progression was closely monitored and best-performing alumni were called to the campus to interact with the students.

The institute has a Committee for SC/ST, a Minority Cell, a Grievance Redressal Committee, an Internal Compliant Committee, an Anti-Ragging Committee, an OBC Cell, and a Women empowerment cell which will actively involve in progression of the students. To provide a platform to exchange their knowledge, subject experts from both academia and industry were called to the campus for guest lectures, seminars, workshops, conferences, and value-added courses. The Institute is a member of the Institute of Electrical and Electronics Engineers (IEEE), Indian Society for Technical Education (ISTE), Computer Society of India (CSI), Indian Institute of Chemical Engineers (IICHE), Biotech Research Society of India (BRSI), Engineers Without Borders (EWB).

An alumni association has been registered to provide an opportunity to the students to interact with their predecessors which may help them to better understand the social and industrial needs. The visionary Management of the institute is participative in all academic and official activities.

The governing council of the Institute meets once a year to measure the performance and to devise the growth strategy. The management of the Institute is headed by the dynamic Vice Chairman Er. E.V. Kumaran who gives regular, comprehensive, Strategic Plans to the Registrar. The Registrar monitors the implementation of the plans through a structured mechanism. The principal is head of all the academic activities, he supervises and directs the subordinates to effectively implement the policies framed by the management. The HoDs and functional heads directly report to the Principal which were communicated to the Registrar. The Internal Quality Assurance Cell (IQAC) of the Institution is actively involved in the design and implementation of the quality policy of the Institution through its meetings, recommendations, data collection, and storage, observation of evolving trends in Higher Education, the IQAC plans the institutional activities accordingly. The quality policy was framed as per the ISO-9001:2015 standards. This provides various templates for various activities to set the operations as chronologically recorded and audited every six months. Several committees are constituted under the leadership of the Principal of the Institution to continuously guide, monitor, and assess the execution of various activities as planned and scheduled. The Management strictly follows the regulations provided by regulatory bodies such as AICTE, the Directorate of Technical Education, and Anna University (Affiliating University). The Governing Body ensures that the policy statements and action plans are fulfilled by the stated mission of the institution.

The institute is known for its social responsibility. The campus is maintained clean and green which made the slogan "Clean Arunai Green Arunai". The scenario of the college exhibits the importance given to an eco-friendly campus. Rainwater harvesting systems were installed in all major places. The RRR concept of Reduce, Reuse, and Recycle was taught to the students to minimize plastic and other hazardous wastes. Social responsibility was taught to the students. The institute is known for discipline, and the positive attitude of the students was improved by maintaining discipline both inside the campus and the outside world. The institute is an equal-opportunity employer which follows gender equality. The Institute is constantly involved in the growth of rural communities, as it is located in a Rural place it provides quality education and infrastructural facilities to the students around and indirectly plays a role in the economic upliftment of surrounding areas.

2.4 The Kamban College of Arts and Science for Women

The second institute that is part of Deemed University is Kamban College of Arts and Science for Women which is affiliated to Thiruvalluvar University. The curriculum and syllabus prescribed by Thiruvalluvar University are strictly adhered to. A multi-level system has been evolved in the college for planning and implementation of the curriculum transparently and effectively. The courses, pedagogy, and infrastructure have been upgraded regularly to remain responsive to changing needs.

Kamban College of Arts and Science for Women was established in the year 1996 and affiliated with Thiruvalluvar University. The founder of the Institution is Thiru. E.V. Velu has a long-time desire to provide good education to rural women students. Empowering the rural women sector is the major aim of college. At present, the institution has shown immense and magnificent infrastructural growth on a spanning area of 6.25 acres with a built-up area of 133561 sq.m. Now the Institution is running successfully with 2710 students and 145 faculties. The college currently offers 16 Undergraduate (UG) programs, 8 Postgraduate (PG) programs, and few certificate courses.

It has spacious 35 lecture halls, 11 Well-equipped Laboratories, 2 Auditoriums, Common Rooms and Gymnasium. There is also a cafeteria that serves good food, a general store, a bank, a Post office, an ATM, a Temple, and a Dispensary. Hygienic RO-treated drinking water facilities were provided to the stakeholders. Restrooms were available in prime locations. The academic system of subject allotment, lesson plan, work diary and feedback from students are developed and monitored regularly. Curriculum includes courses on environmental science and human rights. Effective curriculum delivery is ensured through advance planning of academic activities, formulation of objective driven teaching plan, and preparation of adequate learning materials.

Every staff maintains a course file & remedial classes are given for slow learners. The principal ensures the quality of teaching and learning through periodic reviews. Trust in the development of knowledge and skills are provided through various programs in the college. Though the college is affiliated with Thiruvalluvar University, we ensure student-centric learning. Bridge courses and enrichment courses are strategies to bridge the gap of learning. Bright students are encouraged to go beyond the syllabus and explore their potential by participating in seminars, paper presentations, and project work. The remedial classes are arranged for slow learners.

2.5 Karan Arts and Science College

Karan Arts and Science College is a Co-Ed college was established in the year 2017 and affiliated with Thiruvalluvar University. The institution spreads on a area of 7.26 acres of land with a built-up area of 9777.21sq.m. Now the Institution is running successfully with 1532 students and 50 faculties. The college currently offers 13 Undergraduate (UG) programs, 4 Postgraduate (PG) programs. The college provides ample opportunities for the faculty members to excel in their teaching by facilitating them to conduct and attend workshops/seminars and pieces of training. Experimental learning at every level is achieved through industrial projects, guest lectures, and interaction with eminent personalities from various fields. The continuous assessment of teacher quality is done through student feedback and evaluation by the principal. Assessment of students is done through periodic tests, preparatory examinations, assignments, and final examinations as per the university norms. The college adopts the pattern of the examination and evaluation prescribed by the affiliated university which involves continuous assessment.

The very need for higher education promotes research. The concept of research should be in practice and a norm in higher education. The college atmosphere has encouraged the faculty members to organize seminars and conferences to strengthen the research culture. The students are also encouraged to take up real-time projects. The institution is committed to inculcating a strong sense of social responsibility among its students and faculties. The college has a few major forums like NSS, YRC, and ECO-CLUB, these forums organize programs that educate the community through gatherings and participating jointly with NGOs. The institution offers moral support for the faculties and students for research and community extension work. The college imbibes the research culture to develop analytical skills and critical thinking for practical learning.

The college provides a well-equipped infrastructure, spacious building, huge ground, computer laboratories, ICT-enabled classes, and library facilities to the learners. The college has three computer labs, a Microbiology lab, an Electronic lab, a Physics lab, and a Digital library which are being continuously upgraded and modernized. Time table is designed to ensure the computer-student ratio is 1:1 during practicals. The growth of the infrastructure keeps pace with the academic growth of the college.

The entire computer is networked through LAN and the campus is equipped with Wi-Fi, generators, and a reprographic machine facility.

The institution has a spacious air-conditioned seminar hall and an open auditorium. Our college has a canteen on the campus at subsidized rates. Keeping in mind the overall development of the students our institution provides access to various sports facilities such as basketball, cricket, football, volleyball kho-kho, etc. The institution runs 35 buses connecting different points in and around all villages. Well-facilitated hostel facilities for girls are provided with an adequate water supply and power supply. The college makes adequate provision in the budget for upkeeping of infrastructure. The college has formed a committee supported by technical staff, an engineer, a sergeant electrician, and a supervisor to look into maintenance on a day-to-day basis. The college plans and executes all its plans and programs keeping students' betterment in focus. The admission process follows a roster system as per the rules, this is managed by the admission committee which councils every student during admission. Students from the same institution are supported with fee concessions. Remedial coaching is conducted for slow learners with an active backlog to improve their academic performance. Students are given on-duty permission to participate in tournaments and inter-collegiate competitions, to develop leadership quality, teamwork & competitive spirit. The Fine Arts Association organizes intercollegiate cultural festivals. On and off-campus guidelines are given to students by placement cell. Pre-placement training is arranged to help the students to face the interview with confidence. There is a successful campus recruitment process in place with a substantial number of students being placed every year in leading companies. The college has an active alumni association; they are invited as chief guests to give a motivational address to students. MoUs have been signed with ICTACT, Eden Bridge Skill Solution Pvt Ltd, Chennai, and T.I.M.E, Vellore for coaching students to face various competitive exams. An effective tutorial system is followed for individual student monitoring. Students feedback are given weightage and appropriate remedial measures are ensured. The statutory and non-statutory committees function effectively for the welfare of the students.

2.6 The Activities at the Project Institutes

The management is supportive of all the endeavors of the institution. The Institution has a governing council comprising management members, heads of the institution, and senior faculties. All major decisions regarding institutional functioning are taken by the governing council. Proactive roles are played by the principal in designing and implementing academic policies. The post of the Head of the Department gives room to groom leaders in academic and administration work of the departments in particular and of the college in general. The faculty members who produce 100% results in their subjects are appreciated and awarded. The IQAC conducts meetings regularly and suggestions are given to various committees to ensure smooth and effective operations. Students inculcate their leadership quality through co-curricular activities like sports, NSS, YRC, Consumer Club, Eco-club Rotary Club, etc. Leadership quality is promoted among faculties & students through committees. Suggestions of both students and staff are taken into consideration in the implementation of the college activities.

The college echo club inculcates an environmentally friendly atmosphere in students and disseminates the message of conservation of nature. The students are encouraged to adopt clean and green initiatives for energy conservation. Awareness is created on reduced usage of plastics on campus. The students are motivated to plant saplings to

promote afforestation. We conduct a food festival for the students to teach the traditional and nutritional values of the food. Student assembly is a special feature of the college. Assembly promotes, motivates, and shares the information with the students imparting ethical values. Recognizing & appreciating students in the assembly encourages the students to participate and win laurels in Intra and Intercollegiate competitions. The placement committee makes necessary arrangements to conduct campus interviews and recruitment drives. On all special occasions, students wear uniform coats. NSS, YRC, Consumer Club, Rotatory Club, Red Cross, and other clubs encourage the students to learn leadership qualities. All National and religious festivals are celebrated. Special lectures are provided on subject and general topics through guest lectures and State-level / National / International seminars. Faculties and research students have published articles in reputed journals. Students also have carved their names in Guinness records with their special talents. Sports students have not only been selected by the university but also represented themselves in national tournaments. Students are academically coached to achieve more centum and a minimum of 20 ranks per year. Slow learners are concentrated through remedial classes and 100% result is achieved.

2.7 Details of governance, leadership, and administrative structure.

The visionary Management of the institute is participative in all academic and official activities. The governing council of the Institute meets once a year to measure the performance and to devise the growth strategy. The management of the Institute is headed by the dynamic Vice Chairman who gives regular, comprehensive, Strategic Plans to the Registrar. The Registrar monitors the implementation of the plans through a structured mechanism. The principal is head of all the academic activities, he supervises and directs the subordinates to effectively implement the policies framed by the management. The HoDs and functional heads directly report to the Principal which were communicated to the Registrar. The Internal Quality Assurance Cell (IQAC) of the Institution is actively involved in the design and implementation of the quality policy of the Institution through its meetings, recommendations, data collection, and storage, observation of evolving trends in Higher Education, and planning institutional activities accordingly. The quality policy was framed as per the ISO9001:2015 standards. This provides various templates for various activities to set the operations as chronologically recorded and audited every six months. Several committees are constituted under the leadership of the Principal of the Institution to continuously guide, monitor, and assess the execution of various activities as planned and scheduled. The Management strictly follows the regulations provided by regulatory bodies such as AICTE, the Directorate of Technical Education, and Anna University (Affiliating University). The Governing Body ensures that the policy statements and action plans are fulfilled by the stated mission of the institution.

Policies for faculty recruitment, retention, and development. The service rules, the recruitment procedure, and the promotion policies are framed by the norms of AICTE, and/or affiliating universities. The institute has well-developed HR policies for faculty recruitment, retention, and development, the service rules, and promotion policies are framed by the norms of AICTE and UGC.

2.8 THE EDUCATIONAL & RESEARCH PROGRAMMES OFFERED BY THE PROJECT INSTITUTE

ARUNAI ENGINEERING COLLEGE

FEE STRUCTURE FOR EXISTING COURSES

SR.NO	COURSE	DURATION	INTAKE	FEE PER ANNUM	
				MANAGEMENT	GOVT. QUOTA
1.	B.E CSE	4 Years	240	85000	50000
2.	B.Tech AIDS	4 Years	120	85000	50000
3.	B.Tech IT	4 Years	120	75000	50000
4.	B.E ECE	4 Years	120	50000	50000
5.	B.Tech Cyber Security	4 Years	60	50000	50000
6.	B.E AIML	4 Years	60	50000	50000
7.	B.E EEE	4 Years	60	50000	50000
8.	B.E Mech	4 Years	60	50000	50000
9.	B.Tech Biotech	4 Years	60	50000	50000
10.	B.E Civil	4 Years	30	50000	50000
11.	BTech Chemical	4 Years	30	50000	50000
12.	B.Tech Agri.	4 Years	60	50000	50000
13.	MBA, M.E and M.Tech	4 Years	60	30000	30000
14.	M.Tech Biotech	2 Years	18	30000	30000
15.	M.E Power Electronics and Drives	2 Years	12	30000	30000
16.	M.E Thermal Engineering	2 Years	12	30000	30000
17.	PhD			34000 (PT)	34000 (FT)

KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN

FEE STRUCTURE FOR EXISTING COURSES

SR.NO	COURSE	DURATION	INTAKE	FEE PER ANNUM	
				MANAGEMENT	GOVT. QUOTA
1.	B.A., Tamil	3 Years	70	18,000	18,000
2.	B.A., English	3 Years	70	18,000	18,000
3.	BBA	3 Years	70	18,000	18,000
4.	BCA	3 Years	100	29,000	29,000
5.	B.Com., (General)	3 Years	140	18,000	18,000
6.	B.Com., (Computer Application)	3 Years	70	22,000	22,000
7.	B.Com (FA)	3 Years	70	18,000	18,000
8.	B.Sc., Computer Science	3 Years	100	29,000	29,000
9.	B.Sc., Mathematics	3 Years	70	18,000	18,000
10.	B.Sc., Microbiology	3 Years	50	29,000	29,000
11.	B.Sc., Chemistry	3 Years	100	22,000	22,000
12.	B.Sc., Physics	3 Years	50	20,000	20,000

SR.NO	COURSE	DURATION	INTAKE	FEE PER ANNUM	
				MANAGEMENT	GOVT. QUOTA
13.	B.Sc., Nutrition, Food Service Management and Dietetics	3 Years	50	29,000	29,000
14.	B.Sc., Visual Communication	3 Years	50	20,000	20,000
15.	B.Sc., Artificial Intelligence	3 Years	50	22,000	22,000
16.	B.Sc., Data Science	3 Years	50	22,000	22,000
17.	M.A., Tamil	2 Years	40	19,000	19,000
18.	M.A., English	2 Years	40	19,000	19,000
19.	M.Com.,	2 Years	40	20,000	20,000
20.	M.Sc., Computer Science	2 Years	26	30,000	30,000
21.	M.Sc., Mathematics	2 Years	40	22,000	22,000
22.	M.Sc., Microbiology	2 Years	26	24,000	24,000
23.	M.Sc., Chemistry	2 Years	26	24,000	24,000
24.	M.Sc., Physics	2 Years	26	24,000	24,000

KARAN ARTS AND SCIENCE COLLEGE

FEE STRUCTURE FOR EXISTING COURSES

SR.NO	COURSE	DURATION	INTAKE	FEE PER ANNUM	
				MANAGEMENT	GOVT. QUOTA
1.	B.A., Tamil	3 Years	70	14,000	14,000
2.	B.A., English	3 Years	70	19,000	19,000
3.	BBA	3 Years	70	14,000	14,000
4.	BCA	3 Years	100	26,000	26,000
5.	B.Com., (General)	3 Years	70	17,000	17,000
6.	B.Com., (Computer Application)	3 Years	70	18,000	18,000
7.	B.Com (FA)	3 Years	70	17,000	17,000
8.	B.Sc., Computer Science	3 Years	100	26,000	26,000
9.	B.Sc., Mathematics	3 Years	70	19,000	19,000
11.	B.Sc., Chemistry	3 Years	70	19,000	19,000
12.	B.Sc., Physics	3 Years	50	19,000	19,000
15.	B.Sc., Artificial Intelligence	3 Years	50	26,000	26,000
16.	B.Sc., Data Science	3 Years	50	26,000	26,000
18.	M.A., English	2 Years	40	17,000	17,000
19.	M.Com.,	2 Years	40	17,000	17,000
20.	M.Sc., Computer Science	2 Years	26	30,000	30,000
21.	M.Sc., Mathematics	2 Years	40	17,000	17,000

2.11. THE INFRASTRUCTURE & RESOURCES WITH THE COLLEGE

Arunai Engineering College building sits on a campus measuring 24.66 acres with a built area of 52,142 sq.m.

The institution has shown immense and magnificent infrastructural growth in a built-up area of 27,819 sq.m. for instructional and administrative buildings, 22,786 sq.m. for amenities, and 12,900 sq.m for circulation area also 31,893 sq.m. for playgrounds. It has spacious lecture halls, ICT-enabled Smart rooms, Tutorial rooms, Well-equipped Laboratories, English language labs, Computer labs, an Open Air Auditorium, Air Conditioned Auditorium & Seminar halls, Common Room, Sports and Gymnasium. There are a total of 97 classrooms, 48 laboratories, 2 indoor auditoriums, and 1 open auditorium on the campus. To inculcate ICT-enabled education LCDs and other audio-visual facilities are provided in necessary places. The Institute library can house 200 students at a time, it also has computers with internet connectivity which helps the students and faculty to utilize the e-Resources. The library holds 45,842 books in it. It also houses 118 National Journals, 103 International Journals in print, 860 e-journals, and 3223 CDs. The library is automated using MODERNLIB library management software. The computer lab has a 100 MBPS leased line with 627 computers.

The institute has 2 indoor air-conditioned auditoriums with seating capacity of more than 500 and a massive open auditorium with a seating capacity of around 2500. The auditorium is used for organizing various events of the Institute.

The institute campus has Basketball, Volleyball, Football, and Cricket grounds, these playgrounds are dedicatedly used for the sporting activities of the students. Apart from this, the students are permitted to play indoor games in the open auditorium. Various intramural, zonal, state, and national level competitions are held on these. The institute has a dedicated gymnasium in the women's hostel and men's hostel that caters to the needs of the hostel students. Students are allowed daily in the morning and evening time to work out.

The Kaban College Campus is 6.25 acres with an area of about 2,72,250 sq. ft. at Tiruvannamalai. The campus comprises five academic blocks with spacious classrooms, well-equipped laboratories, a spacious library, a large auditorium, air-conditioned seminar halls, a well-maintained hostel, a hygienic mess, a canteen, and an administrative block. The institution has a well-built infrastructure. The institution always decides to provide and enhance the infrastructure facilities for effective teaching and learning. The management of the college keeps in mind the infrastructure and development policies and plans to create the required and adequate infrastructural facilities when there is a need and demand. The institution has Totally 35 classrooms.

- ▶ Well-furnished and fully equipped staff room that includes a computer. One fully air-conditioned seminar hall.
- ▶ High-configured computer laboratories consisting of 171 systems with the latest software. 32 college buses with first aid boxes and emergency exit windows. The buses ply to various areas around the college.
- ▶ 32 college buses with first aid boxes and emergency exit windows. The buses ply to various areas around the college.
- ▶ The alumni association constitutes a network of KCAS graduates for career opportunities. Various information is informed to the students by the principal during a common prayer conducted every Monday.
- ▶ The management periodically discusses with principal and HODs for the needed infrastructural facilities and enhancement to provide for effective teaching and learning.
- ▶ The placement cell organizes pre-placement training on Soft & Interview Skills, Projects, and Campus Recruitments for students.
- ▶ One Generator (30kW) facility.
- ▶ RO Drinking water facilities to the students.
- ▶ Canteen with stationary facility, spacious library, adequate rest rooms, Suggestion Box, First Aid Box are available for the benefit of students.
- ▶ The institution has all the necessary facilities for sports and games.
- ▶ A spacious play ground with indoor and outdoor games facilities provided. A playground with open stage for conducting various sports events in the premises. The playing field is well maintained for practice and play.
- ▶ The institute established fine arts club. They are conducting cultural activities periodically and our students are participating in intercollegiate cultural competitions. Store room, rest room, Xerox copying room, vehicle shed, stores for office stationary are available.
- ▶ Auditorium with 3500 seating capacity and a seminar hall with audio visual resources are established for the conduction of cultural programmes, workshops, seminars, conferences, debates, college celebrations, various meetings and inter collegiate meet.

Arunai Engineering college has a dedicated massive library building with an area of 1120sq.m that can house 200 students at a time. Huge volumes of textbooks, reference books, magazines, journals, and periodicals were available in the library. Apart from this number of e-resources are available for the students and staff with good internet connectivity. The library has a massive number of 45,842 books in it. It also houses 118 National Journals, 103 International Journals in print, 860 e-journals, and 3223 CDs. The library is automated using MODERNLIB library management software.

The Kamban College Library is a Learning Resource that provides several books, periodicals, News Clippings, Reference books, and Net facilities. It enables them to gather information and update themselves to the current world. The library uses INFLIBNET Software, as part of ILMS. It is very helpful to manage the circulation of books, member's details entry and Catalogues and Accessioning, etc. This is one of the latest web-centric library management software incorporating latest technologies that enables library to serve its users more efficiently using.

IT facilities and services are available in all areas of the engineering institute. The institute has having 100 MBPS leased line for internet connectivity, apart from this the institute has Jio internet on the campus. Wi-Fi connectivity is provided in all the blocks of the institute for seamless connectivity. Every department and functional unit is provided with computers with internet connectivity for efficient communication. The IT infrastructure is developed periodically based on the requirement. The internet connectivity has been increased from 100 MBPS in 2018 to 300 MBPS various connections in 2023, this is based on the needs of the stakeholders. Students and faculty can utilize the Wi-Fi internet connectivity at no cost. Apart from this, the computer labs function after the regular class hours to help the students to browse the internet.

At Kaban College, the Internet in education has been incredibly useful as it facilitates both information and communication. The internet has increased the accessibility of education at all levels and has turned out to be a giant repository of knowledge. Considering this, our institution provides up-to-date computer facilities to the students as well as the teachers. 3 separate computer labs are available on our college campus. Systems are provided in the faculty rooms, the administration Officer's room, and the Principal's room. Different types of system configurations are provided in our college systems which are of the latest configuration. A WIFI facility is provided in the institution for the faculty to download the necessary subject content to handle the class with the PowerPoint presentation. PowerPoint is prepared by our faculty members and is shown in the Audio Visual theatre to make the students understand their lesson easily. The students get motivated by the staff and prepare PowerPoint Presentations during seminars or workshops organized by the concerned department or other colleges. Apart from the PowerPoint Presentation, the faculty uses LCD projectors as a supplement for the lecture method and the college has Eleven halls equipped with LCD projectors. Each hall can accommodate a minimum of 60 and a maximum of 100 students. The rapid development of IT has shrunk the world into a global village. It is the e-world everywhere and all the way and the concept of e-learning has become a prominent feature in the field of education. Realizing this, our institution has introduced e-learning in our day-to-day classroom teaching, by which the latest information on various subjects is downloaded from the internet and presented in AV theatre for better understanding and enrichment of the knowledge of the students. As we are a self-financing institution whenever we face a problem it will be rectified immediately by the systems team.

LAND DETAILS

SR. NO.	DESCRIPTION / DOCUMENT DETAILS	AREA (ACRES)
31/9	Arunai Engineering College	0.07
32/3B	Arunai Engineering College	0.19
35/5	Arunai Engineering College	2.17
32/6	Arunai Engineering College	2.37
32/7	Arunai Engineering College	1.56
32/8	Arunai Engineering College	1.83
49/1	Arunai Engineering College	0.40
49/2	Arunai Engineering College	1.36
50/1	Arunai Engineering College	1.14
50/2	Arunai Engineering College	1.43
50/3	Arunai Engineering College	0.33
50/4	Arunai Engineering College	1.43
50/5	Arunai Engineering College	1.64
51/1	Arunai Engineering College	1.85
51/2	Arunai Engineering College	2.59
54/4B	Arunai Engineering College	0.46
54/5B	Arunai Engineering College	1.41
55/1D2	Arunai Engineering College	0.43
55/2	Arunai Engineering College	0.89
55/3	Arunai Engineering College	1.11
14/1	Kamban College of Arts and Science for Women	4.90
14/3	Kamban College of Arts and Science for Women	1.35
41/2B	Karan Arts and Science College	1.98
4/2	Karan Arts and Science College	1.36
4/2	Karan Arts and Science College	0.68
4/3A	Karan Arts and Science College	0.74
4/3B	Karan Arts and Science College	0.73
4/3C	Karan Arts and Science College	0.73
1/2A2	Karan Arts and Science College	0.45
1/2B2	Karan Arts and Science College	0.14
2/1B	Karan Arts and Science College	0.45

THE BUILDINGS AND AREA OF CONSTRUCTION UNDER PROJECT INSTITUTES

INSTITUTE	BUILDING NO. / NAME	AREA IN SQ. M	TOTAL INSTITUTE -WISE BUILT-UP AREA IN SQ. M
ARUNAI ENGINEERING COLLEGE	Arunai Centre & Admin Block	2350.52	52142.3
	HAS Block	3150.78	
	Workshop	531.22	
	Security Block	40.23	
	Power House	69.58	
	Mother Theresa Women's Hostel	5243.50	
	Gateway, Library & Computer Centre	3362.60	
	EEE, CSE & EIE Block	3527.13	
	ECE Block	3130.39	
	Auditorium (Sheet)	2043.85	
	Mechanical Block	2476.03	
	MBA Block	1989.87	
	BT Block	1551.37	
	Civil Block	2984.39	
	Mess Block	3628.30	
	APJ Abdul kalam Mens Hostel	4018.12	
	Thanthai Periyar Mens Hostel	3206.89	
	Physical Education Block	136.43	
	HAS Block - II	1536.48	
	Artificial Intelligence Block	2764.86	
	Kalaingner Mens Hostel	2503.44	
	Canteen	419.45	
	Bank and Post Office Block	598.10	
	Guest House	751.67	
	Mens Toilet	63.55	
	Women's Toilet	63.55	
	ADMIN BLOCK	22381	
KAMBAN COLLEGE OF ATRS AND SCIENCE FOR WOMEN	MAIN BLOCK	24692	133561
	ARTS BLOCK	15773	
	SCIENCE BLOCK - I	10872	
	SCIENCE BLOCK - II & CANTEEN BLOCK	43574	
	PG BLOCK	16269	
KARAN ARTS AND SCIENCE COLLEGE	Admin Block	2739.21	9777.21
	Main Block	7038	

LIST OF PLAYGROUNDS (Common to All Institutions)

SR.NO	NAME OF GROUNDS	NUMBERS	AREA (SQ.M)
Outdoor Games			
1	Basketball Court	1	608
2	Cricket Ground	1	14950
3	Football Ground	1	7700
4	Volleyball court	4	2400
5	Handball Court	1	800
6	Kho-Kho Ground	1	510
7	Ball Badminton	2	576
Indoor Games			
1	Badminton Court	2	144
2	Table Tennis	4	200
3	Chess 15	5	
Athletic Games			
1	400m Track	1	4000
	TOTAL AREA		31893

DETAILS OF THE ASSETS

PARTICULARS	AEC	KCASW	KASC	TOTAL
A) Immovable Properties Building	605751025	46229792	58062740	710043557
Total - A	605751025	46229792	58062740	710043557
B) Movable Properties				
Computers & Accessories	24527425	13107619	6249295	43884339
Electrical Installation	7980037	7185626	4391216	19556879
Furniture Account	722486	520190	346793	1589469
Kitchen Equipment	9643229	1343218	1028749	12015196
Lab. Equipment	14925461	12364702	6298719	33588882
Library Books Account	4375905	3150653	2100435	9626993
Office Equipment	8941280	6437220	4291814	19670314
Sports Goods Account	420163	302517	201678	924358
Vehicles	155725130	26090710	14060473	195876313
Total - B	227261116	70502455	38969172	336732743
Total - A+B	833012141	116732247	97031912	1046776300

THE PHYSICAL INFRASTRUCTURE: THE EQUIPMENT, FURNITURE, BOOKS, ICT, ETC.

INSTITUTE	MACHINERY & EQUIPMENT	FURNITURE	BOOKS	COMPUTER	ICT
Arunai Engineering College	₹ 43,631,261.00	₹ 13,800,000.00	₹ 38,787,350.00	₹ 13,653,000.00	₹ 1,500,000.00
Kamban College of Arts and Science for Women	₹ 2,621,514.00	₹ 9,240,000.00	₹ 5,448,343.00	₹ 7,790,000.00	₹ 1,000,000.00
Karan Arts and Science College	₹ 2,022,414.00	₹ 8,236,000.00	₹ 3,115,500.00	₹ 7,200,000.00	-
GYMNASIUM	750,000.00				
GRAND TOTAL	49,025,189.00	31,276,000.00	47,351,193.00	28643000.00	2500000.00

GRAND TOTAL IN Crore.: 15.90 LAKHS (Except Land and Building Construction)

2.12 COMPLIANCE WITH ENVIRONMENTAL AND SUSTAINABILITY STANDARDS

The Arunai Engineering College is compiling to the requirements of the environment and sustainability standards. The following are the few examples from the certificate/report of the GREENAUDIT.

Segregation of waste as Bio-Degradable and Non-Bio-Degradable in the Campus The campus is neat and clean in all areas of the campus. The waste segregation shall be improved in the campus, by providing waste bins at appropriate places in the campus, like the canteen area. The waste shall be collected as Bio-Degradable and Non-Bio-Degradable and disposed of. This will improve waste management further and upgrade the campus to the best level. In this regard awareness campaign shall be conducted for the students and employees. Installation of Sanitary Napkins disposal equipment in the campus the Girl's students' strength is about 950 nos. and the ladies' staff

Arunai Engineering College has put in place a new STP to meet the requirements. The treatment scheme consists of screening, primary sedimentation, biological treatment, aeration, secondary treatment and disinfection. Sewage treatment plants help protect water bodies and ecosystems by treating wastewater before discharge. By removing pollutants and harmful substances, treatment plants reduce the risk of water pollution and ecosystem degradation, promoting environmental sustainability.

Overall, having a sewage treatment plant provides multiple benefits, including environmental protection, public health improvement, regulatory compliance, resource recovery, water reuse, odor control, community well-being, and long-term cost savings. These advantages make sewage treatment plants essential components of sustainable urban development and wastewater management strategies.

At Kamban College of Arts and Science for Women, the following initiatives are taken to maintain environmental and sustainability standards.

WASTE MANAGEMENT STEPS INCLUDING SOLID WASTE MANAGEMENT:

The Green-colored dustbins are meant for wet and biodegradable waters. Other color dustbins are meant for disposal of plastic wrappers and non-biodegradable wastes. Cleaning or emptying of the dustbins is being done on a regular basis from 7.00 am to 9.00 am and at 4.30 pm every day.

SOLID WASTE MANAGEMENT:

1. The major solid waste which is generated on the Campus includes Paper, Leaf litter, and Kitchen waste.
2. Leaves being shed are a major solid waste generated.
3. Metal and wooden waste is given to authorized scrap agents for recycling.
4. The chemical waste generated in Life Science and Bioscience Laboratories is segregated regularly.
5. The vegetable scraps and fruit peels are used in vermin composting.
6. Single papers are reused for writing and printing in all departments.

LIQUID WASTE MANAGEMENT:

1. A proper system has been implemented to discharge the liquid waste generated on campus for watering the saplings and trees planted.
2. Liquid waste from the points of generation like the canteen, laundry, and toilet is segregated and disinfected and let out as effluent into a common drainage facility.
3. This liquid waste effluent could trigger various infections and can cause disease outbreaks among people
4. Periodic check-up of plumbing lines is undertaken to ensure minimal loss of water due to leakage. Effective reuse of waste water from RO plant is being used for irrigating the green areas.
5. All electronic equipment used on the campus is regularly maintained and repaired to ensure minimum e-waste.

E-WASTE MANAGEMENT:

The E-waste generated in the college premises is very less in quantity. The cartridges of Laser Printers are refilled outside the college campus.

In case of any repairs, computers are collected from different locations on the campus by designated technicians and sent to the institute computer center. At the center, the computers are sometimes repaired and some are likely not repaired due to the level of damage. The unrepaired are inspected by the authorities and its open for disposal through contractors or vendors, who pay a certain amount of money to dispose of them.

2.13 TECHNOLOGICAL INTEGRATION IN THE CAMPUS ECOSYSTEM (ERP, LMS, ETC.).

Arunai Engineering College effectively uses ICT Tools to integrate technology, communication, and information to create, manage, and disseminate information. ICT encompasses a broad range of technologies, including computers, software, networks, and telecommunication systems. integration of technology, communication, and information to create, manage, and disseminate information. ICT encompasses a broad range of technologies, including computers, software, networks, and telecommunication systems.

The institute is using Net Campus ERP Software is built on the latest web technology and is a web-based software. Since it does not need to be installed, it can be accessed from anywhere and at any time with authorized credentials and a cost value of 3 Lakhs per year

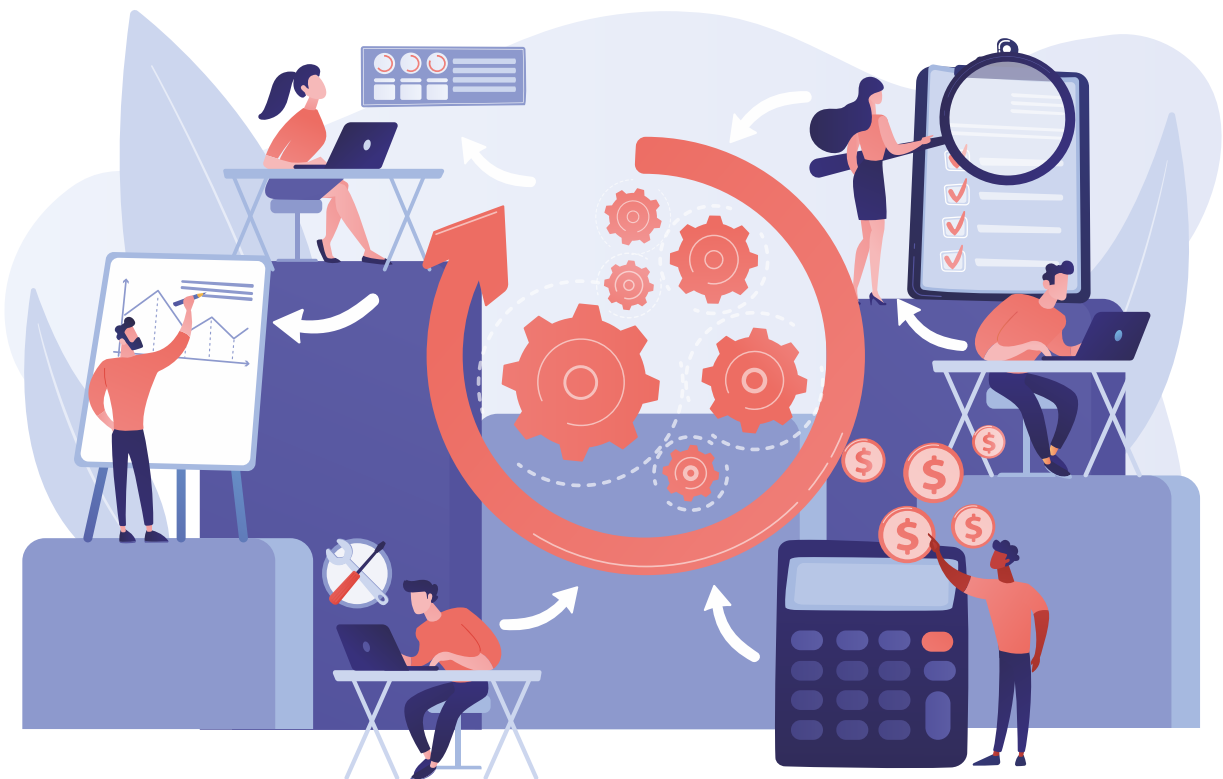
Major modules include:

Campus Management System:

Admission and inquiry, HRM, Leave Management, Payroll, expense accounts, Fees, Library, Hostel, Transport, Stock and Inventory, and many more

Learning Management System:

Academics, Placements, clubs and cells, Event, certificate, Content management, Student grievances, CBCS and feedback.



2.14. THE INCOME AND EXPENDITURE SCENARIO FOR PROJECT INSTITUTES

INCOME AND EXPENDITURE (LAST THREE YEARS DATA)
ARUNAI ENGINEERING COLLEGE

<u>YEAR</u>	2021-22	2022-23	2023-24
<u>INCOME</u>	Rs.	Rs.	Rs.
Tuition Fees	129271000	70706707	129762475
Hostel, Lunch & Transport Fees	6257106	67197602	21905650
Total	135528106	137904309	151668125
<u>Expenditure</u>	452653	337300	900000
Advertisement Expenses			
Affiliation & Approval fees	5652842	5926583	6008325
Book & Periodicals	2568456	271928	561900
Text Books for Students	68541	84523	365000
College Maintenance	405658	488954	512138
Electricity Charges	4856523	6982379	80,80,169
Function & Seminar Expenses	284514	382891	11,14,664
Laboratory Consumables & Services A/c	2,56,238	2,75,015	3,02,725
Mess Expenses	8452523	9654295	16658643
Miscellaneous Expenses	35846	152360	585638
Printing & Stationery Expenses	1845252	360497	421098
Professional Charges	1534000	1980000	1862989
Rent, Rate & Taxes	415268	398100	348787
Repairs & Maintenance	685236	40900	894292
Salaries and Allowances	42587563	58554121	58589222
EPF Paid	0	1350750	994266
Subscription	315276	203358	66134
Telephone, Post & Internet exp	1455863	1228487	1441015
Travelling & Conveyance	548567	69000	405646
Vehicle Maintenance Expenses	18525741	24892492	24225852
Bank Interest & Charges	265823	232478	258867

KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN

PARTICULARS	2021-2022	2022-2023	2023-2024
<u>INCOME</u>	Rs.	Rs.	Rs.
Tution Fees, Hostel & Transport Fee	102161100	105689261	115313317
Total	102161100	105689261	115313317
<u>Expenditure</u>	284523	298526	461172
Advertisement Expenses			
Affiliation & Approval fees	285450	315270	495700
Book & Periodicals	89563	94523	117660
College Maintenance	105623	49527	53029
Electricity Charges	985623	1003250	11,33,358
Function & Seminar Expenses	78563	284574	4,12,000
Laboratory Consumables & Services A/c	1,85,636	1,95,287	65,440
Printing & Stationery Expenses	365823	185522	128000
Repairs & Maintenance	189523	106538	148900
Salaries and Allowances	24248563	25575852	28537935
EPF Paid	0	0	102972
Subscription	165723	105336	187640
Telephone, Post & Internet exp	685673	542536	737985
Travelling & Conveyance	150537	75850	45263
Vehicle Maintenance Expenses	8765236	19854254	27026229
Bank Interest & Charges	986596	2398587	2377010
Total	37572655	51085432	62030293

KARAN ARTS AND SCIENCE COLLEGE

PARTICULARS	2021-2022	2022-2023	2023-2024
INCOME	Rs.	Rs.	Rs.
Tuition Fees, Hostel & Transport Fee	20010852	23304300	39979856
Total	20010852	23304300	39979856
<u>Expenditure</u>	184520	385620	397520
Advertisement Expenses			
Affiliation & Approval fees	352674	1743640	2052404
Book & Periodicals	98563	105238	108410
College Maintenance	67436	45653	48247
Electricity Charges	735234	832573	11,23,423
Function & Seminar Expenses	285630	385640	5,01,256
Laboratory Consumables & Services A/c	48,756	56,234	59,308
Printing & Stationery Expenses	114520	119248	123470
Repairs & Maintenance	138563	141536	146320
Salaries and Allowances	10785623	11856423	19013280
Subscription	136458	165247	167842
Telephone, Post & Internet exp	548546	652410	682360
Travelling & Conveyance	45868	52364	57423
Vehicle Maintenance Expenses	4051360	4353260	9652380
Bank Interest & Charges	545476	741371	742530
Total		21636457	34876173
18139227			
	90.64694996	92.84319632	87.23436373

THE CUMULATIVE INCOME AND EXPENDITURE OF PROJECT INSTITUTES (Last Three-Year Data)

Head Income/ Expenditure	Arunai Engineering College	Kamban College of Arts and Science for Women	Karan Arts and Science College
Income	257700058	266897870	306961298
Expenditure	146924265	186588300	221503836
Surplus	110775793	80309570	85457462

Total Surplus: Rs. 276,542,825.00 in last three years

i.e. Rs. 27.65 Cr in three years

Average Surplus per year: 9.07 Cr

2.15. DETAILS OF RESEARCH CENTERS AND INITIATIVES.

Arunai Engineering College promotes entrepreneurship, innovations, IPR, and NSS/YRC-based activities both on-campus as well as off-campus. The institution has a well-established Intellectual Property Rights (IPR) cell and Entrepreneurship Development Cell (EDC) motivating faculties and students to organize various events in the trust areas like patents, start-ups, incubation facilities, etc. EDC started in 2019 provides an awareness among students of the institution regarding entrepreneurship as a career option. IPR cells foster innovation and cultivate a culture of research and development among the students' community. Our institution has a well-established Research and Development cell. The institute promotes research by motivating the faculties with various facilities. Two departments of the Institute were recognized as authorized research centers by Anna University. 41 faculties have completed their Ph.D. degree in different universities and 15 faculties are currently pursuing.

Fully established laboratories cater to the physical needs for research of both the faculties and the students. An active research forum of the Institute plays a significant role in motivating the faculties and students towards research. They are motivated to attend events outside the campus which helps to enhance their research skills. The Institute has received Rs.16.4 Lakhs as research funds in the past 5 years (2018-19 to 2022-23). The Institute has done 175 publications in the last 5 years (2018-19 to 2022-23), with most of the papers published in journals in the UGC CARE list and indexed in SCOPUS, SCI, and included in Web of Science. 29 research works were published as book chapters and conference proceedings in various National/International conference proceedings. Three books were drafted and published.

The Institute has an active Institute Innovation Council (IIC) which is articulated and monitored by the Ministry of Education, Government of India. The Institute organizes project

expos in which innovative student projects are exhibited to the general public to make them aware of technological developments. More than 25 functional MoUs with various industries help the students and faculties to carry out their research work, industrial visits, internships, industrial training programs, etc.,

Kamban College of Arts and Science for Women encourages students to take up entrepreneurship as career. Hosts National Service Scheme (NSS) which brings awareness to build environment cleanliness. The undergraduate and postgraduate students are encouraged to provide innovative strategies for development. The **Department of Mathematics** conducted a special lecture on ideas of mathematics by Dr. Lau Gre Choon, which initiated student's knowledge. Inaugurated Ramanujam's Association and arranged guest lecture on "Application of Mathematics in our day to day life" by Dr. A. Annamalai M.Sc., M.Phil., Ph.D., The **Department of Bachelor of Computer Applications** initiated Programmes like Network Security's advantages and communication programs to enhance the student's knowledge. Codemania'18 was a technological meeting, the objective of conducting Codemania'18 is a technical meet to expose their talents, utilize their knowledge, and test the skills of the students. Through this event, the students can able to learn about new ideas and current trends in the IT field. The **Department of Computer Science** organized a workshop on Personal Career Profiling to develop student's talents. Conducted technofun'18, which allowed the students to learn about new ideas and expose their talents in different ways. The **Department of Tamil emphasizes** the story of ethics which brings prosperity to life. The **Department of Physics** conducted State state-level seminar on Recent trends in Nano Science and Technology by Dr.R. Jayavel Professor and Director, at Anna University.

The following two tables shows the details of paper publications and book publications by the faculty members of the institute.

PAPER PUBLICATIONS

2022-23	2021-22	2020-21	2019-20	2018-19
33	32	31	29	50

BOOK PUBLICATIONS

2022-23	2021-22	2020-21	2019-20	2018-19
6	5	4	6	8

2.16 EVIDENCE OF FOSTERING INNOVATION, ENTREPRENEURSHIP, AND INDUSTRY PARTNERSHIPS.

ENTREPRENEURSHIP DEVELOPMENT CELL

The Entrepreneurship Development Cell (EDC) was started in Arunai Engineering College since 2019 with the object of promoting specialized knowledge in the field of entrepreneurship development. The Entrepreneurship Development Cell has been started to create job providers rather than job seekers among our students. The cell strives to identify talented youth to entrepreneurial works. The cell plans to organize various programs regarding entrepreneurship development.



VISION:

The Entrepreneurship Development Cell is committed to nurturing and developing entrepreneurial skills experientially among the student's community.



MISSION:

The mission of EDC is to mould entrepreneurial thinking in our students and to create a business opportunity in and out of the campus with a great stability for achieving social and economic progress through entrepreneurship.

OBJECTIVES:

- ▶ To create awareness among the students of the Institution regarding entrepreneurship as a career option.
- ▶ To provide guidance and facilities to first-time entrepreneurs during gestation.
- ▶ To conduct programs in Entrepreneurship enabling skills.
- ▶ Motivate students to develop their startups.
- ▶ Develop business incubators.
- ▶ To provide a platform for interaction with entrepreneurs.
- ▶ Impart entrepreneurial education/skills amongst students through various trainings and exercises.



LIST OF EVENTS

DATE	TITLE OF THE EVENT	CHIEF GUEST
12-07-2019	Challenges And Opportunities In Engineering	Dr.P.Karupannan, PhD, Associate Professor, Motilal Nehru National Institute of Technology, Allahabad.
27-05-2020	A Techno Talk On How To Become A Successful Entrepreneur	Dr.Suresh.A shan, Head-Digital Innovation & Future Technology, BITS, Mahindra & Mahindra finance services Ltd., Mumbai.
20-01-2021	Webinar On “National Education Policy With A Focus On Entrepreneurship And Innovation”	Dr.Suresh.A shan, Head-Digital Innovation & Future Technology, BITS, Mahindra & Mahindra finance services Ltd., Mumbai.
06-02-2021	Webinar on “innovation and technological entrepreneurship”	Mr.Sampath Mohan Raj, MD-Coffee N Recharge, Coffee Day shop.
12-03-2021	Webinar On “Session For Student Entrepreneurs Through Innovation Agent”	Mr. Gyanprakash, Cofounder, Emsoft Technology, Chennai.
27-11-2021	Webinar on “utilization of opportunities for startups and entrepreneurs”	Mr. R. Rajesh, Entrepreneur Businessman.
07-02-2022	Workshop on “transformation from smart dream into a startup”	Mr.N.Gowtham Kumar, Founder & CEO, Universal Power Supplies, Bangalore.
30-11-2022	Awareness Programme On Innovation Voucher Program	Er.GokulRaj.S, chief technical officer and co-founder of Tavisha Robotics, Salem

17-02-2023	Awareness Programme On Tamilnadu Student Innovators	Dr.G.Seenuvasan Ph.D, founder of Ellarukkum Ellam Educational Trust.
22-09-2023	Guest Lecture On Job Opportunities In The Government Sector	Mr.K.Tamilvendhan,Director,Chennai study center(TNPSC & UPSC),Tiruvanamalai
6-10-2023	Guest Lecture On Time And Money Management	Mr.J.Karthick, entrepreneur, edible oil manufacturer, and cattle feed wholesaler, Tiruvanamalai.
25-10-2023	Workshop On Digital Marketing	Mr.S.Chennakrishnan, Deputy General Manager, JIO platform Ltd, Bangalore.
15-11-2023	2 nd National Level Meet And Arunai Conclave	Mr. K.B. Kannan, Regional Coordinator and Master Trainer, EDII university college of engineering, Villupuram.
5-12-2023	Workshop On The Future Of Business In The Digital Age	Mr.Renganathan.V, Head of sales and business development Birdscale technology and services(P) Ltd.
4-4-2024	Guest Lecture On Business Plan And Linkage With Innovation Ambassadors	Dr. R. Baskaran, Dean of PG studies at ARM Engineering College, M.M.Nagar.
21-05-2024	Workshop On Advanced Research Methodology And Data Analysis Using SPSS	Prof.D.Yuvaraj, C.Abdul Hakeem college of engineering and technology, Vellore.
24-05-2024	Workshop on the inauguration of finance club and workshop of financial awareness and consumer training.	Dr.R.Venkatesan,Professor in the department of business administration ,Government Arts and Science College,Gingee,Villupuram.

IIC – INSTITUTION'S INNOVATION COUNCIL [IIC ID:IC202015453]

ABOUT IIC

MIC will focus on creating a complete ecosystem which will foster the culture of Innovation across all educational institutions from ideas generation to pre-incubation, incubation and graduating from the incubator as successful start-ups. MIC will also work on designing ranking system to identify institutions in the forefront of innovation.

Ministry of Education, Govt. of India has established an 'Innovation cell' with the purpose of systematically fostering the culture of Innovation in all Higher Education Institutions (HEIs) across the country.

In the year 2018, the Ministry of Education (MoE) through MoE's Innovation Cell (MIC) launched the Institution's Innovation Council (IIC) program in collaboration with AICTE for Higher Educational Institutions (HEIs) to systematically foster the culture of innovation and start-up ecosystem in education institutions. Primarily, IICs' role is to engage a large number of faculty, students, and staff in various innovation and entrepreneurship-related activities such as ideation, Problem-solving, Proof of Concept development, Design Thinking, IPR, project handling and management at Pre-incubation/Incubation stage, etc., so that innovation and entrepreneurship ecosystem gets established and stabilized in HEIs. The IIC model is designed to address the existing challenges/issues in HEIs such as fewer numbers, occasional and unplanned Innovation & Entrepreneurship (I&E) activities organized in HEIs with low involvement of top leadership, lack of coherence and absence of synergy in resource mobilization, deployment and underutilization of creative potential of youths as a major barrier for vibrant I&E ecosystem to emerge from HEIs.

POLICY APPROACH & ACTION PLAN FOR I&E DEVELOPMENT IN HEIS

GOAL

- ▶ Ecosystem Preparedness
- ▶ Ecosystem Responsiveness
- ▶ Ecosystem Performance

OBJECTIVE

- ▶ Creating, Streamlining and Strengthening I&E Ecosystem in HEIs & Schools
- ▶ Pathways for Quality Innovation and Entrepreneurship
- ▶ Sustained Generation of Innovations and Entrepreneurs from HEIs & Schools

PROGRAMS AND SCHEMES FOR IIC INSTITUTIONS

- ▶ Innovation Ambassador Program
- ▶ Building In-house Human Resource Capacity to Engage, manage, and Generate I&E: Training & mentoring support to Students & Faculties
- ▶ Funding Support to HEIs located far away from the I&E Ecosystem to Organize Impact Lectures: Institute Performance Improvement

- ▶ 360 Degree Experiential Learning Program for Potential Student Innovators and Entrepreneurs Students
- ▶ Mentor-Mentee Program: Peer Group Mentoring Support with Funding Support – Institute Improvement & Inter-Institutional Collaborations
- ▶ Yukti National Innovation Repository: A System of Building Repository of Ideas, Innovations, and Starts ups – campus to commercialization.

EVENT IN 2023 - 2024
(STATUS - Q1 TO Q4)

SR.NO.	EVENT / ACTIVITY TYPE	TOTAL	
1	IIC Driven Activities	21	68
2	MIC Driven Activities	03	
3	Self Driven Activities	27	
4	Celebration Activites	04	
5	I & E Exposure Visit	01	
6	I & E Value Added Courses	04	
7	I & E Collaborations (MoU)	04	
8	IPR Cell	01	
9	EDC Cell	01	
10	Innovation Cell	01	
11	NISP Cell (Registered)	01	
12	New HEI Referred (Kumaran Polytechnic, Karan College)	02	02

**KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN:
ENTREPRENEURSHIP DEVELOPMENT CELL AND IIC**

ABOUTED:

The Entrepreneurship Development Cell is committed to nurturing and developing entrepreneurial skills experientially among the student's community. The mission of EDC is to mold entrepreneurial thinking in our students and to create a business opportunity in and out of the campus with great stability to achieve social and economic progress through entrepreneurship. This cell educates various activities such as Entrepreneurship education and training courses like Learn Wise, Entrepreneurship Games and exercises, workshops, Guest lectures, seminars/conferences, Skill Development programs, Panel discussions, Intra and inter-institutional competitions, and get-togethers with entrepreneurs.

OBJECTIVES OF THE ED CELL:

1. Create awareness on Entrepreneurship among the students through training programs and camps.
2. To identify and motivate budding entrepreneurs.
3. To facilitate budding entrepreneurs by providing information on entrepreneurial opportunities.
4. To create a database and networking to help entrepreneurs.
5. To assist entrepreneurs in product development.



ED CELL IMPLEMENTATION & PROGRAM LIST:

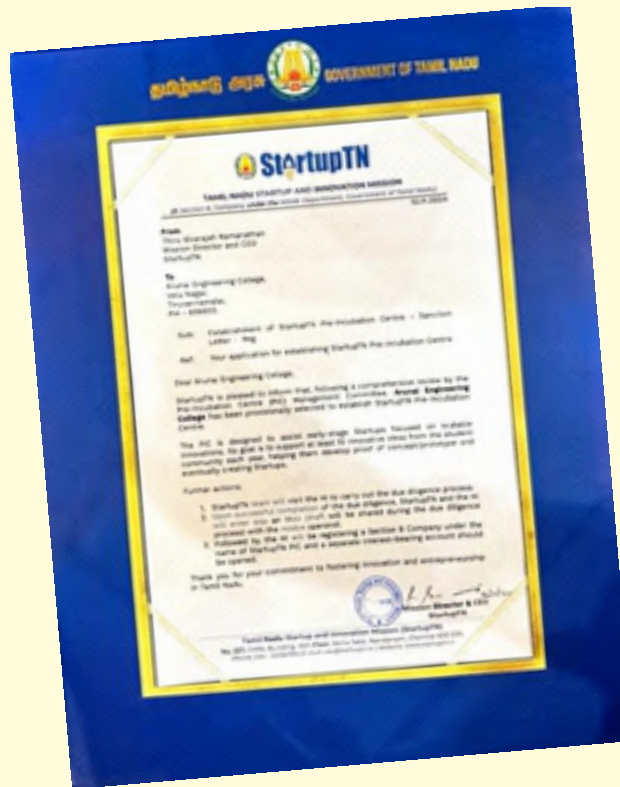
The Impact Series lecture was organized by Kamban College of Arts & Science for Women (ED & IIC Cell) and funded Under the Impact Series Lecture Scheme. The program started with a Prayer song followed by the lighting of a Lamp by Chief Guest Mrs. Gandhimathi (Advocate), Mrs. Gowri Sathyaprakash, and Dr. S. Seethalakshmi (Principal). Welcome Address was given by Ms. S Padmapriya and the stage was enlightened by the guest speech about the IPR Act & Women's entrepreneurship. Finally, it came to an end with a Vote of Thanks given by Ms. V Dhivya proceeded by the National Anthem.

PROGRAM

S.NO	DATE	YEAR	NAME OF THE EVENT
1	20.09.24	2024	ORIENTATION ON DIGIIT
2	22.8.2024 & 23.08.24		COLLEGE BAZAAR JOINTLY WITH SELF GROUP
3	25.03.2024		IMPACT SERIES LECTURE - Received Rs.20000 fund from AICTE-IIC
4	15-02-24 & 16-02-2024		CROCHET KNITTING -TRAINING PROGRAM
5	13.04.2024 to 16.04.2024		MEGA EXHIBITION
6	21.09.2023	2023	ENTREPRENEURSHIP IN INDIA IN 2023 -CHALLENGES, OPPORTUNITIES & GROWTH
7	10.08.2023 & 11.08.2023		WORKSHOP ON BASIC MAKE-UP TECHNIQUE
8	17.05.2023		AWARENESS PROGRAM ON SOCIAL ENTREPRENEURSHIP
9	20.03.2023		AWARENESS PROGRAM ON ENTREPRENEURSHIP & INNOVATION
10	17.02.2023		ORGANIC GARDEN FARMING SEMINAR
11	16.02.2023		TEXTILE PRINTING TRAINING PROGRAM
12	15.09.2022 to 17.09.2022	2022	MEGA EXHIBITION SALE
13	22.07.2022		ONLINE WEBINAR ON THE IPR ACT
14	09.06.2022		ONLINE WEBINAR ON IPR ACT
15	29.04.2022		ONLINE WEBINAR ON CRPTO CURRENCY
16	07.04.2022	2022	SEMINAR ON" OPPORTUNITY THROUGH TECHNOLOGY BUSINESS INCUBATOR FOR BUDDING ENTREPRENEURS"
17	23.03.2022		ONLINE WEBINAR ON THE IPR ACT
18	25.03.2022		ONLINE WEBINAR ON SUSTAINABLE WASTE MANAGEMENT INDEX
19	19.12.2021 & 17.12.2021	2021	MEGA EXHIBITION

IMPLEMENTATION AND FUTURE GOALS:

In connection with all ongoing efforts ED cell at Kamban College of Arts & Science for Women, develops a various activity for students. Each Academic year quarterly Meetings are conducted with staff and students to encourage the student's activities and also provide various opportunities regarding entrepreneurship programmes for promoting their product to market.



Establishment of Startup TN Preincubation Centre



2.17 INTELLECTUAL PROPERTY RIGHTS (IPR) POLICY AND ACHIEVEMENTS.

INTELLECTUAL PROPERTY RIGHTS (IPR)

The IPR Cell serves as a pivotal hub for fostering innovation, protecting intellectual property, and cultivating a culture of research and development. Established with the core objectives of raising awareness about intellectual property rights and facilitating the protection of innovative ideas, the IPR Cell engages in a range of activities. Structured with a dedicated team of professionals, the cell organizes workshops, seminars, and training sessions to educate faculty, researchers, and students on patent filing processes and related aspects. Through its comprehensive support, the IPR Cell assists in patent searches, guides through the filing process, and celebrates successes in obtaining patent grants for noteworthy innovations emerging from the college. In addition to its educational endeavours, the IPR Cell actively promotes collaboration and partnerships. It engages with industries, research organizations, and other academic institutions to stimulate research and development initiatives. The cell plays a crucial role in identifying funding opportunities and fostering technology transfer, contributing significantly to the growth and commercialization of intellectual property. Despite its achievements, the IPR Cell faces challenges, and the report suggests strategic recommendations for overcoming these hurdles. Looking forward, the report envisions the IPR Cell as a key driver for future growth and enhancement of intellectual property-related activities at Arunai Engineering College.

OBJECTIVES OF THE IPR CELL:

The objectives of the Intellectual Property Rights (IPR) Cell include:

- ▶ Raise Awareness ▶ Facilitate Innovation
- ▶ Support Patent Filing ▶ Promote Technology Transfer
- ▶ Enhance Research and Development (R&D) Activities
- ▶ Protect and Manage Intellectual Property
- ▶ Create a Supportive Ecosystem
- ▶ Collaborate with Legal Experts
- ▶ Educate and Train
- ▶ Appreciation of Achievements

IPR CELL INAUGURATION & WORKSHOP:

The IPR Cell was started in January 2021. IPR Cell was funded by the Tamil Nadu State Council for Science and Technology (TNSCST) in July 2023. The inauguration of an Intellectual Property Rights (IPR) initiative at Arunai Engineering College, Tiruvannamalai, on July 13, 2023, signifies the institution's commitment to fostering innovation, protecting intellectual property, and promoting a culture of research and development. Such initiatives play a crucial role in creating an environment that encourages students and faculty to explore new ideas, engage in research, and contribute to the advancement of knowledge. In continuation to the inauguration IPR awareness workshop was also conducted. The Intellectual Property Rights (IPR) awareness workshop was held on July 13 & 14, 2023, at Arunai Engineering College, Tiruvannamalai, witnessed an impressive turnout with active participation from over 50 staff members and 250 students. The workshop organized as a continuation of the IPR initiative's inauguration, aimed to deepen understanding and instill awareness of intellectual property concepts within the academic community it was a highly informative and successful event.

IMPLEMENTATION:

Legal Framework: The program implements and updates laws related to patents, and copyrights, trademarks, and trade secrets to provide comprehensive protection to various forms of intellectual property.

Awareness and Education: Public awareness campaigns and educational initiatives are conducted to inform individuals, businesses, and communities about the importance of intellectual property and the consequences of infringement.

Enforcement and Litigation: The program emphasizes strengthening enforcement mechanisms, including cooperation with law enforcement agencies, to combat piracy, counterfeiting, and intellectual property infringement cases.

International Cooperation: Collaboration with international organizations and participation in treaties and agreements facilitate the harmonization of intellectual property laws and global protection.

IMPROVEMENTS AND FUTURE GOALS:

In conjunction with the ongoing efforts to bolster the Intellectual Property Rights (IPR) Cell at Arunai Engineering College, a crucial enhancement involves the dissemination of insights gained from the annual review meetings with TNSCST (Tamil Nadu State Council for Science and Technology) directly to college staff and students. Simultaneously, the IPR Cell Coordinator, acting as a proactive conduit, shares vital information, including awareness initiatives and step-by-step guidelines for filing IPR, with the college community through the principal. This twofold communication strategy not only establishes a transparent and inclusive channel but also empowers the academic community with the knowledge and tools necessary to actively participate in intellectual property processes. By aligning with the college's commitment to fostering innovation and collaboration, these initiatives ensure that every staff member and student remains well-informed, engaged, and motivated to contribute meaningfully to the evolving landscape of intellectual property at Arunai Engineering College.

CALENDAR OF EVENT SCHEDULE

SR NO	ACADEMIC YEAR	DATE	TOPIC
1	2020-2021	2 nd Sep 2021	FDP on Intellectual Property Rights
2	2021-2022	4 th Aug 2022	Workshop on Intellectual Property Rights
3	2022-2023	13 th & 14 th July 2023	Inaugural function of IPR CELL & A two Day State Level Technical workshop
4	2023-2024	14 th Feb 2024 26 th March 2024	1. One-day Workshop on "Various Aspects of Intellectual Property Rights (IPR)" 2. Celebration of World Intellectual Property Day 2024 • E-QUIZ Competition • Essay Writing Competition
5	2024-2025	6 th Dec 2024	Workshop on Patent Filing

LIST OF PATENTS

SR. NO	INVENTOR NAME	DEPARTMENT	PATENT APPLICATION NO.	TOPIC	RESULTS
Dr.S.Sivakumar	EEE Dept	202141018763	23.04.2021	An Apparatus and a Method for the Automatic Segregation and Recycling of Waste Material	Granted
Mr. A.Eakambaram	Automobile Engg.	202141046883	14-10-2021	An Intelligent Road Speed Breaker System	PUBLISHED UNDER EXAMINATION FOR GRANT
Mr. A.Eakambaram	Automobile Engg.	202141047286	18-10-2021	An Apparatus and a method for Achieving a Zero-Fade in Brake pads	PUBLISHED UNDER EXAMINATION FOR GRANT
Mr. A.Eakambaram	Automobile Engg.	202141048870	26-10-2021	A Responsive Braking System	PUBLISHED UNDER EXAMINATION FOR GRANT
Dr.S.Selvarasu, Dr.S.Thirumalvalavan	Mechanical Engg.	202131054037	23.11.2021	Multi Axis Dedicated Machine for Machining Silicon	PUBLISHED UNDER EXAMINATION FOR GRANT
Mr.A.Eakambaram	Automobile Engg.	202211006188	04-02-2022	A Method for Developing a Low-Metallic Brake Pad Mixture.	PUBLISHED UNDER EXAMINATION FOR GRANT
Dr.S.Elango	ECE Dept	2022410153	19.03.2022	AI, IOT Based Real-Time Condition Monitoring of Electric Machines Using Python languages	PUBLISHED UNDER EXAMINATION FOR GRANT
Dr.S.Thirumalvalavan	Mechanical Engg.	202241012929	25.03.2022	Optimization of Thermal Conductivity and Viscosity of Refrigerants	PUBLISHED UNDER EXAMINATION FOR GRANT
Dr.S.Mohanarangan	Cyber Security Dept.	02241037755	22.07.2022	Designing a WSN Platform for Long Term Environmental Monitoring for IOT Applications	PUBLISHED UNDER EXAMINATION FOR GRANT
Dr.C.Kannan, Dr.S.Priyadharsini	EEE Dept.	202341014546A	17.03.2023	Design Modified Rotor for Wind Turbine to Enhance the Efficiency	PUBLISHED
Mr.A.Eakambaram	Chemical Engg.	202341040255	16.06.2023	An Intelligent turn indicator system for Road Transport Vehicle	PUBLISHED
Dr.S.Selvarasu, Dr.S.Thirumalvalavan	Mechanical Engg.	399732001	12.01.2024	Organic Waste Sorting Machine	PUBLISHED
Mr.A.Eakambaram	Chemical Engg.	202441014309	27.02.2024	Trailer/Tow Hitch Remote Access & Control	PUBLISHED

SR. NO	INVENTOR NAME	DEPARTMENT	PATENT APPLICATION NO.	TOPIC	RESULTS
Dr.S.Selvarasu	Mechanical Engg.	404231001	09.04.2024	IOT Based Agriculture Robot for Pesticides Spraying	PUBLISHED
Mr.S.Venkatesan, Mr.R.Senthil	Mechanical Engg.	202441033737	10.05.2024	Next-Generation Robotics Arm with Adaptive Gripping Technology	PUBLISHED
Mr.S.Venkatesan, Mr.R.Senthil	Mech. Engg.	202441041557	31.05.2024	Development of Low Cost Floor Cleaning Machine	PUBLISHED
Dr.S.Sivakumar	EEE Dept.	414286001	05.06.2024	Autonomous Navigation for Electric Bike	GRANTED
Dr.C.Kannan	EEE Dept.	202441058608A	09.08.2024	Optimizing Electric Vehicle production with Fourier Analysis and Laplace transform	PUBLISHED
Dr.C.Kannan, Dr.S.Priyadharsini	EEE Dept.	202441062454A	30.08.2024	3D Modelling and Printing in Smart Transportation by Deep Learning for Agriculture	PUBLISHED

2.18. TECHNOLOGY INTEGRATION AND INNOVATION

Arunai Engineering College effectively using ICT Tools to the integration of technology, communication, and information to create, manage, and disseminate information. ICT encompasses a broad range of technologies, including computers, software, networks, and telecommunication systems. integration of technology, communication, and information to create, manage, and disseminate information. ICT encompasses a broad range of technologies, including computers, software, networks, and telecommunication systems.

We are using Net Kampuss ERP Software is built on the latest web technology and is a web-based software. Since it does not need to be installed, it can be accessed from anywhere and at any time with authorized credentials
Cost value 3 Lakhs per year

Major modules include:

CAMPUS MANAGEMENT SYSTEM:

Admission and inquiry, HRM, Leave Management, Payroll, expense accounts, Fees, Library, Hostel, Transport, Stock and Inventory and many more

LEARNING MANAGEMENT SYSTEM:

Academics, Placements, clubs and cells, Events, certificates, Content management, Student grievances, CBCS and feedback

2.19. STUDENT SUPPORT AND PROGRESSION

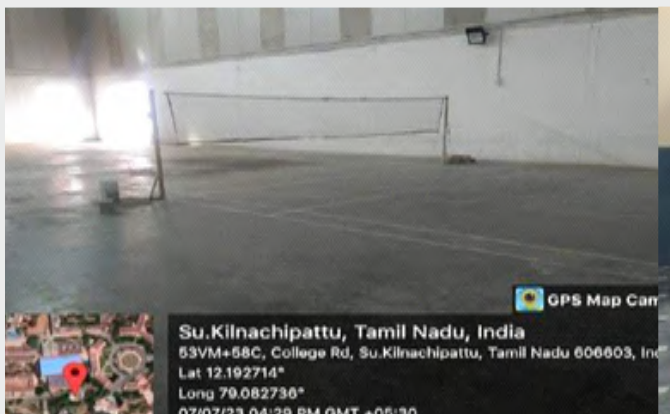
Evidence of holistic development initiatives (extracurricular and co-curricular). Arunai Engineering College building sits on a campus measuring 24.66 acres with a built area of 52,142 sq.mts. The institution has shown immense and magnificent infrastructural growth in a built-up area of 27,819 sq.m. for instructional and administrative also 31,893 sq.m for playgrounds. The institute has Basketball, Volley ball, Football, Cricket grounds, these playgrounds are dedicatedly used for the sporting activities of the students. Apart from this, the students are permitted to play indoor games in the open auditorium. Various intramural, zonal, state and national level competitions are held on these. The institute has a dedicated gymnasium in women's hostel and men's hostel that caters to the needs of the hostel students. Students are allowed daily in the morning and evening time to work out

LIST OF PLAY GROUNDS

Sr.No	Name of Grounds	Numbers
	OUTDOOR GAMES	
1	Basketball Court	1
2	Cricket Ground	1
3	Football Ground	1
4	Volleyball court	4
5	Handball Court	1
6	Kho-Kho Ground	1
7	Ball Badminton	2
	INDOOR GAMES	
1	Badminton Court	2
2	Table Tennis	4
3	Chess	15
	ATHLETIC GAMES	
1	400m Track	1

FACILITIES FOR SPORTS





FACILITIES FOR CULTURAL ACTIVITIES-OPEN AIR AUDITORIUM



FACILITIES FOR CULTURAL ACTIVITIES

Cultural Significance

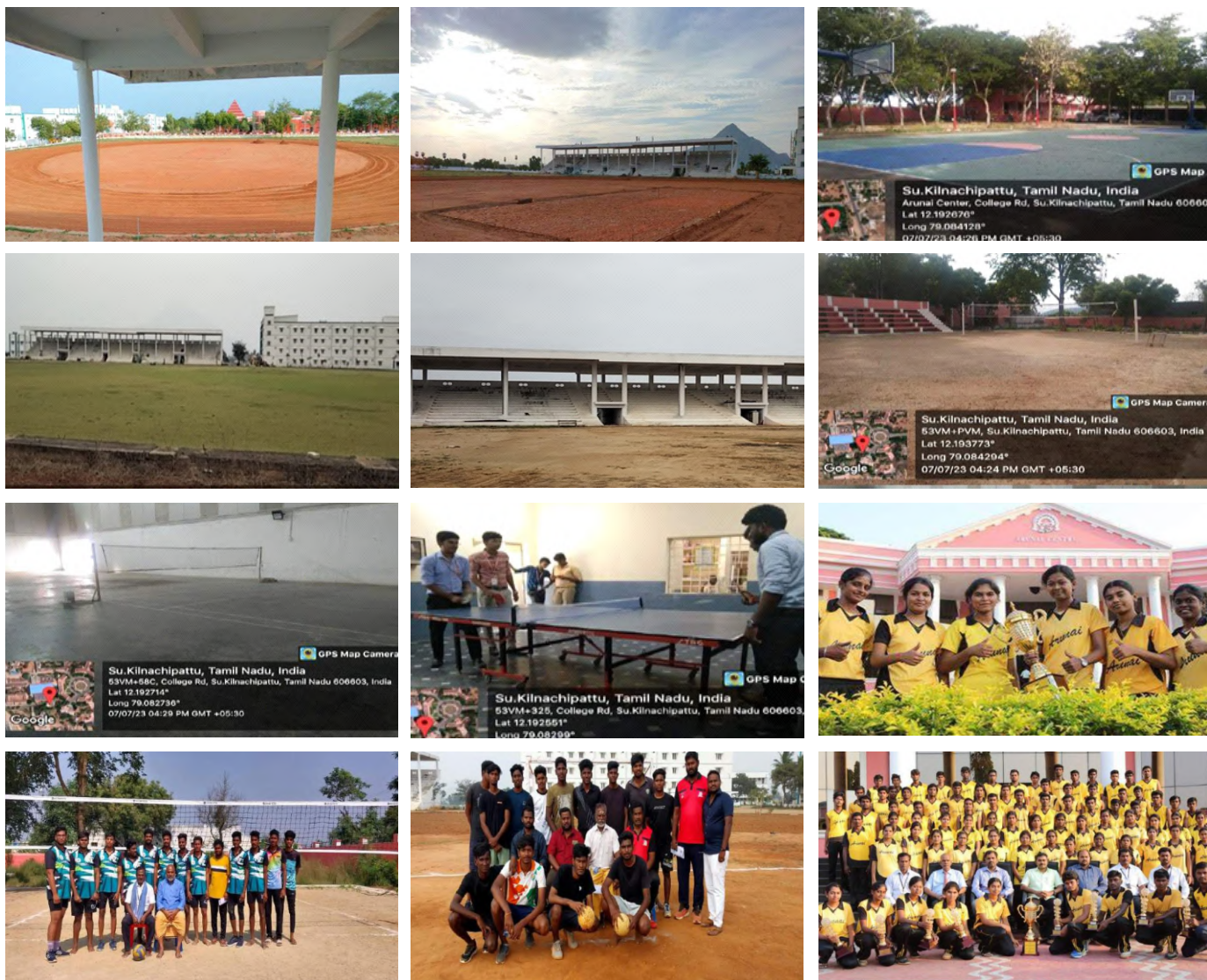
The event celebrated the diversity of Indian culture, showcasing various art forms, music, and dance. It provided a platform for students to express themselves creatively and showcase their talents.



FACILITIES FOR GYMNASIUM



FACILITIES FOR SPORTS



Facilities for Cultural Activities

AECOFEST is the annual day celebration of Arunai Engineering College in Tiruvannamalai, Tamil Nadu. It is a cultural festival and competition that celebrates excellence and leadership. The college cultural event, "AECOFEST," was a grand celebration of music, dance, and art. The event was organized by the Arunai Engineering College Cultural Committee, Students, faculty, and teams participate in the festival, there are competitions and cultural events

Event Highlights

- 1. Inaugural Ceremony:** The event began with a traditional lamp-lighting ceremony, followed by a welcome address by the college principal.
- 2. Music Performances:** The event featured a range of music performances, including solo and group singing, instrumental performances, and a DJ night.
- 3. Dance Performances:** Students showcased their dancing skills through various performances, including classical, contemporary, and folk dances.
- 4. Art Exhibition:** An art exhibition was also organized, featuring paintings, sketches, and sculptures created by students.

KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN

The institution has all necessary facilities for sports and games. The college ground is spread across an area of 11,683 sq. feet. A spacious play ground with indoor and outdoor games facilities provided. A playground with open stage for conducting various sports events in the premises. The play field is well maintained for practice and play. We have established fine arts club. They are conducting cultural activities periodically and our students are participating in intercollegiate cultural competitions.

A spacious play ground with indoor and outdoor games facilities provided. A playground with open stage for conducting various sports events in the premises. The play field is well maintained for practice and play. Auditorium with 3500 seating capacity and a seminar hall with audio visual resources are established for the conduction of cultural programmes, workshops, seminars, conferences, debates, college celebrations, various meetings and inter collegiate meet.

FACILITIES FOR SPORTS





KARAN ARTS AND SCIENCE COLLEGE (Co-Ed)

Facilities for Sports

The institute has Basketball, Volley ball, Football, Cricket grounds, these playgrounds are dedicatedly used for the sporting activities of the students. Apart from this, the students are permitted to play indoor games in the open auditorium. Various intramural, zonal, state and national level competitions are held on these. The institute has a dedicated gymnasium in women's hostel and men's hostel that caters to the needs of the hostel students. Students are allowed daily in the morning and evening time to work out

FACILITIES FOR SPORTS



FACILITIES FOR CULTURAL ACTIVITIES

KARANFEST is the annual day celebration of Karan Arts and Science College in Tiruvannamalai, Tamil Nadu. It is a cultural festival and competition that celebrates excellence and leadership.

The college cultural event, "KARANFEST," was a grand celebration of music, dance, and art. The event was organized by the Karan Arts and Science College Cultural Committee. Students, faculty, and teams participate in the festival, there are competitions and cultural events.

EVENT HIGHLIGHTS

1. **Inaugural Ceremony:** The event began with a traditional lamp-lighting ceremony, followed by a welcome address by the college principal.
2. **Music Performances:** The event featured a range of music performances, including solo and group singing, instrumental performances, and a DJ night.
3. **Dance Performances:** Students showcased their dancing skills through various performances, including classical, contemporary, and folk dances.
4. **Art Exhibition:** An art exhibition was also organized, featuring paintings, sketches, and sculptures created by students.

CULTURAL SIGNIFICANCE

The event celebrated the diversity of Indian culture, showcasing various art forms, music, and dance. It provided a platform for students to express themselves creatively and showcase their talents.



ALUMNI ACHIEVEMENTS AND CONTRIBUTIONS.

Since its inception in 1993, Arunai Engineering College has been a beacon of education and innovation, dedicated to shaping bright minds and fostering a culture of excellence in engineering education. Over three decades, the institution has proudly graduated 27 remarkable batches, each leaving an enduring legacy that enriches the college's fabric. Central to this legacy is a vibrant alumni community of over 15,000 accomplished individuals, whose contributions span industries and continents.

Our alumni's journeys—from the classrooms of Arunai Engineering College to the global stage—exemplify the transformative power of education. They have excelled not only as professionals but also as pioneers, innovators, and leaders in their respective fields. Whether conducting groundbreaking research at esteemed institutions like ISRO and DRDO or leading entrepreneurial ventures that redefine industries, their achievements inspire future generations.

Beyond professional success, what truly sets our alumni apart is their unwavering commitment to their alma mater. Through financial support, mentorship programs, and industry collaborations, they actively enhance the educational experiences of current students. This enduring engagement ensures that the legacy of excellence at Arunai Engineering College continues to flourish.

Moreover, our alumni exemplify social responsibility and compassionate leadership. They spearhead initiatives that drive positive change, from sponsoring scholarships for deserving students to supporting environmental conservation efforts. These endeavors reflect their dedication to uplifting communities and addressing societal challenges.

As ambassadors of Arunai Engineering College, our alumni play a pivotal role in bridging academia and industry. By facilitating internships, job placements, and networking opportunities, they empower aspiring engineers to navigate an ever-evolving global landscape. Their mentorship and guidance significantly shape the trajectories of our students, enabling them to reach their full potential.

In essence, Arunai Engineering College's legacy is intrinsically tied to the collective achievements of its alumni. As we continue to pursue excellence in education and innovation, we do so with deep gratitude for their contributions and a steadfast commitment to nurturing minds and transforming lives. Together, we build a future where the spirit of Arunai Engineering College thrives for generations to come.

2.20. COMMUNITY ENGAGEMENT AND SOCIETAL IMPACT

- ▶ Evidence of initiatives aimed at societal development, such as rural outreach programs, environmental sustainability projects, or skill development initiatives.
- ▶ Collaboration with government agencies, NGOs, and industries for community welfare.

Arunai Engineering College's National Service Scheme (NSS) Special Camp stands as a beacon of hope and transformation, poised to deliver a week-long cascade of benefits to the nearby villages and beyond. With diverse range of activities planned, including blood-donation camps, tree-plantations, voter- awareness drives, sanitary and fire-safety awareness programs, women's-safety initiatives, and medical- camps, this endeavor embodies the spirit of service and community empowerment.

The tree plantation drive is set to green barren landscapes and combat the looming specter of deforestation and climate change. By engaging volunteers in tree plantation, it endeavors to foster environmental stewardship and cultivate a sustainable future for future generations.

Voter awareness initiatives are poised to ignite a spark of civic engagement and democratic participation among public, empowering them with knowledge and resources to exercise their right to vote. Through interactive workshops and informative sessions, it seeks to strengthen the foundation of democracy and promote inclusive governance.

Sanitary and fire safety awareness programs aim to educate public on best practices for hygiene and disaster preparedness, equipping them with vital skills and knowledge to safeguard their well-being. By raising awareness about sanitation and fire safety measures, to mitigate health risks and prevent avoidable tragedies.

Women's safety initiatives are poised to create safer environments and combat gender-based violence and discrimination. Through self-defense workshops and advocacy campaigns AEC's NSS seeks to empower women with tools and confidence to assert their rights and navigate public spaces with dignity and security.

Medical camps, aims to improve health outcomes and enhance quality of life for all, to offer essential healthcare services to underserved communities, bridging gaps in access and affordability. By bringing healthcare professionals and resources directly to the doorstep of public.

YOUTH RED CROSS

Youth Red Cross is an active unit in the institute, it organizes various society oriented programs like blood donation camps, road safety awareness programs, etc.,

OTHER REMARKABLE ACTIVITIES

The institute has organized other social activities for school students and general public as well. A motivational program for HSC students named 'Padikkarkkal' was organized in campus in which students from various schools brought to the campus for the program. Eminent speakers and academicians are called to interact with students on various topics which help the students to prepare and face their board exams confidently.

Drawing competition for school students named “Oviyaa” is being organized for the students up to 10th Standard. An open call is made to various school students and the competition is organized inside institute campus. Social-oriented themes are given as topics. A huge number of students participate in the event and portray their skills. Eminent persons in fine arts are called as chief guest to judge the drawings and prizes are given to students.

The blood donation camp, a cornerstone of the special camp. Through voluntary donations from students, faculty, and public, to make a tangible impact on healthcare accessibility and emergency response in the surrounding areas.

COMMUNITY SERVICES / PROJECTS ARUNAI ENGINEERING COLLEGE

Arunai Engineering College (AEC) has actively engaged in various community service projects and initiatives aimed at societal development and student involvement. Some of the notable activities are given below:

1. SOCIETY-ORIENTED PROJECTS: Online Car Parking Management System:

The Deepam festival of Tiruvannamalai district faces much chaos. The work of Arunai Engineering College students had a huge impact in managing the parking slots of the vehicles, during the festive. They digitalized the systematic approach of the car parking, with pre-booking sessions for their respective slots. This reduced the chaos and also reduced the unnecessary movements of the vehicles, which led to decreased traffic. The information regarding the system is published on Tamilnadu Police's official page.



CHILD CARE SYSTEM:

Reference link - WWW.TVMPPOURNAMI.IN/CHILDCARE

The crowd proliferation during the Deepam festive will somehow depart the children from their parents during the Girivalam. This will complicate the situation and panic the people and the children as well. The initiative has been taken by AEC faculty and students to overcome this situation. They made a web application to find and protect the lost children, and hand them safely to their parents. They created a network of people who assisted the Police department with their skills. More than 100 children have been found using this web application.



DRONE SURVEILLANCE:

The Technical team of the Police department in Tiruvannamalai district, had started using Drone for surveillance. The faculty of Arunai Engineering College, had learnt drone operation from the technical team and assisted the Police department for the surveillance of the crowd during the Deepam Festive, especially on 7th day(07-12-19) of the festive. They monitored and assisted the Police department to identify and track malicious activities in the region.

DRONE CAMP:

The faculty of Arunai Engineering College conducted a Drone Camp for school students. The aim is to educate Drone Technology, Assembly, modifications and drone driving experience to school Students. This camp has enriched knowledge about how to competently and correctly pilot a drone to the school students.

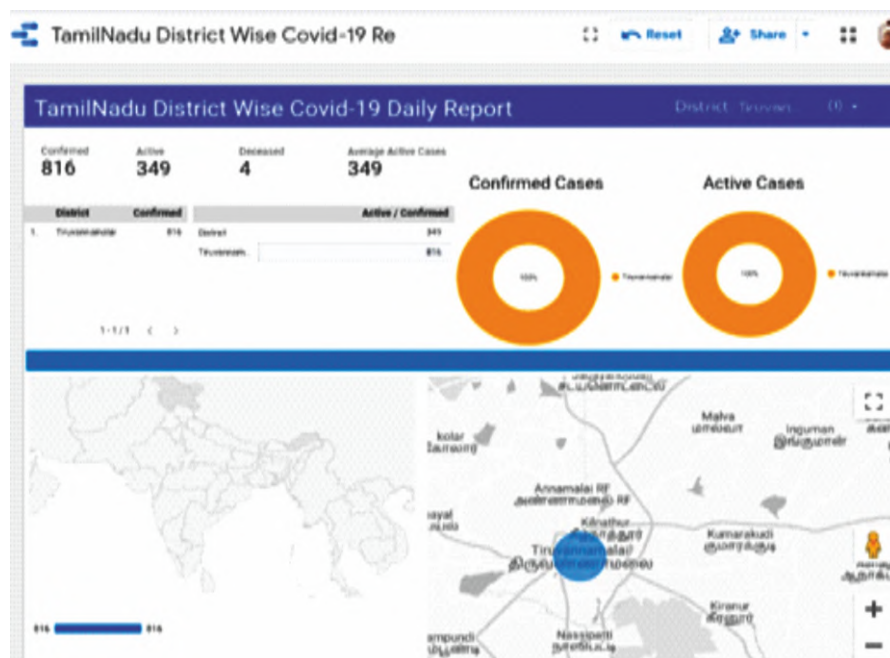


COVID LOCKDOWN & QUARANTINE MONITORING:

COVID pandemic has been hectic for all the regions. Drone surveillance had a huge impact in maintaining the lockdown and ensures safety of the people. The AEC students had learnt drone operation from the technical team and assisted the Police department for the surveillance of quarantined zones, during events like protest, and throughout the lockdown. This had been a great technical support for the Police department.

COVID REPORT GENERATION:

The COVID quarantine has been hectic to analyze and protect the entire district. The AEC students had provided technical support, by generating entire analysis report, stating the exact situation of COVID pandemic in our district. They used Google data studio, for analysis and used Google maps to locate and pinpoint the quarantined regions.



TVM POLICE APP:

Every work of the Police department entirely has been a manual work from beginning. The faculty and students of Arunai Engineering College, has taken the initiative to digitalize the processes and enlighten the communication between the cops and people. They created a pilot project, named "TVM Police", a mobile application, for various services such as women security, cops login and monitoring system, moral policing, spreading COVID awareness through chatbot and enormous information about the district.

TVM POLICE

POLICE

Travel Details:

Place of Board:

Place of Destination:

Vehicle Type:

☐ Car/cab/auto-riksha

☐ Train

☐ Bus

Vehicle No.:

NEXT

Description

BROWSE

UPLOAD

If you find something against your conscience and morality then collect some evidence and along with their description upload them here, to make our police aware about things and be a good citizen!!!

In the Covid-19 pandemic situation, the people should sanitize their hands and check their body temperature frequently is necessary. AEC students had provided the two in one solution, they came up with a prototype model called "SMART MED ASSIST" which measure the temperature and sanitize the hand simultaneously to the people additionally store the temperature in sheets and generate the receipt filled with Id no, Name column and temperature. This system helps the medical workers to maintain social distance with the patients which help in reduce the spread of corona virus. As a pilot project, the module is implemented in '**Sai Seva Yoga nature cure Hospital and Research Centre**' by **Dr. Ezhilmaran**.



TOUCHLESS SKETCHING BY GESTURE RECOGNITION:

With the tremendous advancements in science and technology over last few years, hand gesture recognition technology has emerged as an effective tool for reducing the risk of Covid-19 spread, such as touchless devices and to avoid touching equipment and screen. The faculty and students of AEC had provided solution, they came up with a prototype model called "Touchless user interface for sketching using hand gesture recognition" where actually human finger acts as a marker, it captures the movement of the finger and the drawing operation is performed. It creates an interface between user and the system. This system is based on socially connecting rural peoples and it creates digital artwork. We have implemented our project in Arunachala village school and Panchayat Government School and have received certification for our successful work.

SMART EDUCARE SYSTEM:

In the Covid-19 pandemic situation, Smart education makes students learning from anywhere at any place. The use of technology in classroom increases the understanding and interaction of the students. The students of Arunai Engineering College have designed the purpose of "Smart educate system for the development of rural educational welfare" which makes the students to feel enthusiastic and have fun while learning using Touch panels. This system provides a platform for students where kids can learn through Animated videos, mathematical games, puzzles and sounds and also can attend assessments based on their syllabus. By this means of Touch based sensitivity every child feels good while learning. It actually helps the children to visualize and to grasp concepts easily in their minds for long time. This system is based on socially connecting rural peoples by having interactive learning sessions. We have implemented our project in Arunachala village school and Panchayat Government School and have received certification for our successful work.



2. National Service Scheme (NSS) Activities:

Village Camps: NSS units conducted camps in nearby villages, focusing on sanitation, hygiene, women empowerment, and other social activities to uplift rural communities.

Blood Donation Drives: Regular blood donation camps were organized, encouraging students and staff to contribute to healthcare needs.

Tree Plantation Drives: Initiatives to plant trees were undertaken to promote environmental sustainability and awareness among students and the local community.

Road Safety Awareness Campaigns: Programs were conducted to educate the public on road safety measures, aiming to reduce accidents and enhance community well-being.

Swachh Bharat Abhiyan: In alignment with the national mission, cleanliness drives were organized to maintain and promote hygiene in and around the campus and local areas.



3. World Space Week Celebrations:

In coordination with the Indian Space Research Organization (ISRO), AEC organized events during World Space Week- 2019, attracting over 12,000 school students and members of the general public. This initiative aimed to foster interest in space science and technology.

These initiatives reflect Arunai Engineering College's commitment to societal development, environmental sustainability, and the holistic growth of its students through active community engagement.



4. Padikarkal:

Every year, Arunai Engineering College conducts a motivational program for school students, studying 12th which has helped them to plan their education after schooling. Around 5000 students participate in the program and get benefitted.

KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN

Kamban College of arts and science for women has actively engaged in various community service and initiatives aimed at societal development and student involvement. Some of the notable activities are given below

1.NATIONAL SERVICE SCHEME (NSS) ACTIVITIES:

Dental Camp:

NSS units from KCAS has organized dental camp. KCAS students and Staff members benefited by this camp. The main objective of the camp is to impart education on dental issues, oral hygiene and treatment.



FREE MEDICAL CAMP:

Free medical camp organized by NSS volunteers and given vaccination to students and staff members to prevent them from Covid – 19 and to improve healthy life.



EYE CAMP:

NSS units has organized Eye Camp in KCAS campus students and staff members of our college benefited by this program. NSS volunteers successfully participated and helped our students to prevent visual impairment and enrich their vision.

MENSTRUAL HEALTH ON WOMEN WEALTH:

RRC has organized an awareness program on the topic “MENSTRUAL HEALTH ON WOMEN WEALTH” nearly 200 students participated and benefitted.



ROAD SAFETY:

Road safety awareness is conducted to prevent ourselves and learn safety measures.

ANTI-DRUG CAMPAIGN:

Anti –drug campaign has organized by NSS units from KCAS and distributed pamphlets contains slogans against drug. NSS volunteers participated effectively to drive away drugs from our society and Nation.

NATTU NALAPANITHITTAM:

NSS volunteers donated stationery items for Govt primary school students and also provided basic needs, food items to orphanage students to inculcate helping tendency among people.

PLASTIC BAN AWARENESS PROGRAM:

Every day Plastic waste negatively affects the ecosystem, habitats, human health, and sustainable development across the world. So Kamban College of Arts and Science for Women NSS team and IQAC joined together to organize Plastic Ban Awareness at Government Higher Secondary School, That champattu on 06.12.2021 at 11 am. The aim is to increase public understanding and shape community perceptions of the danger of plastic pollution.



KARAN ARTS AND SCIENCE COLLEGE

Karan Arts and Science College (KASC), Tiruvannamalai, typically involves engaging students and faculty in activities that benefit the local community. These activities can range from educational initiatives, environmental projects, health awareness campaigns, and social welfare programs, to skill development workshops and support for underprivileged sections of society.

Some common types of community service activities that may be organized by the college include:



1. Health and Hygiene Campaigns: Organizing health check-up camps, blood donation drives, and promoting awareness about sanitation and personal hygiene.

2. Environmental Initiatives: Participating in tree planting drives, cleanliness campaigns, and raising awareness about waste management and environmental conservation.

3. Dental Camp: NSS units from KASC has organized dental camp. The main objective of the camp is to impart education on dental issues, oral hygiene and treatment.

Impact assessments or case studies demonstrating societal contributions.

Arunai Engineering College has won several awards time to time for its exemplary contribution to the society. The institute has won various prestigious awards for its outstanding contribution to the society. Exnora International has awarded the Institute with “Green Campus” award for having adequate environmental structure for protection of environment. National Service Scheme (NSS) is a key unit in carrying out extension activities to the nearby villages and needy people. Those services were well received by the beneficiaries and as an outcome several awards were won by the unit. The institute has bagged 35 National / State Awards from ISTE won under Students (UG +PG), Staff and Institution category. Ms.V. Ramya, third-year Information Technology student participated in the State-level NSS Republic Day Parade held in Chennai. Representing her institution under the affiliation of Anna University Chennai. Mr. Vasanth, a dedicated NSS volunteer, showcased exceptional talent and creativity as he participated in the Youth Festival organized by the State NSS Cell, Tamil Nadu. His passion for photography was



2.21. Global Outreach and Collaboration

Internationalization efforts, such as partnerships with global institutions for research, education, and student exchange.

NAME OF THE MOU/LINKAGE	NAME OF THE INSTITUTION/INDUSTRY WITH WHOM THE MOU / LINKAGE IS MADE, WITH CONTACT DETAILS	PURPOSE OF THE MOU/LINKAGE (INTERNSHIP, ON-THE-JOB TRAINING, PROJECT WORK, STUDENT/FACULTY EXCHANGE, AND COLLABORATIVE RESEARCH)
Collaborative research work for students and staff, Internships and visits for students and staff of both parties	National Kaohsiung University of Science and Technology, KAOHSIUNG, TAIWAN	Collaborative research work for students and staff, Internships and visits for students and staff of both parties
Collaborative research work for students and staff, Internships and visits for students and staff of both parties	University of Science & Computer Technology (STEKOM) Semarang Indonesia	Collaborative research work for students and staff, Internships and visits for students and staff of both parties
Abroad studies, exchange programs, Abroad internship, Abroad placements	MIT Square Group of Companies, United Kingdom	Abroad studies, exchange programs, Abroad internship, Abroad placements
IBM's provision of technology products and services for Customer;	IBM India Private Limited ("IBM")	Internship, on-the-job training
IBM's provision of technology products and services for Customer;	IBM India Private Limited ("IBM")	Internship, on-the-job training
AEC -EEE- SIMBA	POWERSYS, Les Jardins de Enterprise, 13610 Le Puy - Sainte- Reparade, France	Training SIMBA - Power Electronics & Drives Simulation Software for Students & Staff

2.22. OUTCOME-BASED METRICS

► Detailed data on academic outcomes, such as graduation rates, placements, higher studies, and competitive exam results.

► PLACEMENT RATE IN LAST THREE YEARS

2023-24	2022-23	2021-22
310	285	284

GRADUATES ENROLLED FOR HIGHER EDUCATION

2023-24	2022-23	2021-22
361	335	360

2.23. INCLUSIVITY AND DIVERSITY

Number of actual students admitted from the reserved categories year wise during last five years (Exclusive of supernumerary seats)

2023-24	2022-23	2021-22
340	280	238

Number of seats earmarked for reserved category as per GOI/ State Govt rule year wise during the last five years

2023-24	2022-23	2021-22
580	566	474

Details of policies and programs for inclusivity (support for underprivileged, differently-abled, and marginalized groups).

EDUCATION OF SCHEDULED CASTES & SCHEDULED TRIBES

Reservation of seats in favor of children belonging to SCs and STs is provided in proportion to their population in the concerned district provided that no such reservation will be less than the national average of 22.5 percent (15 percent for SCs and 7.50 percent for STs) and a maximum of 50 percent for both the categories (SCs & STs) taken together. These reservations are interchangeable and over and above the students selected under open merit.

CONCESSION IN FEE TO SC / ST / PHYSICALLY HANDICAPPED CANDIDATES

The SC/ST students are given concession in admission fees to the extent of 50% of course fees for Secondary and Higher Secondary Courses.

Tamilnadu Council for Open and Distance Learning focuses on the development of textbooks, teacher guides, supplementary reading materials, evaluation of textbooks, vocational education, educational technology, examination reforms, education of educationally disadvantaged groups.

Educational development of Scheduled Castes and Scheduled Tribes is an area of major concern of Tamilnadu Council. It carries out a number of studies relating to educational programmes and schemes for scheduled castes and scheduled tribes. It has also been generating material relating to educational institutions and the development of Scheduled Caste and Scheduled Tribe students.

RESERVATION POLICY OF UGC

As per the reservation policy, UGC has earmarked 15 percent and 7.50 percent reservations for SCs and STs respectively in appointments, both in teaching and non-teaching posts, admissions, hostel accommodation, etc., in universities/ colleges, professional and technical educational institutions administered by the Central Government. State universities follow reservation policies as prescribed by respective state governments. The commission has been issuing guidelines/directives/instructions from time to time for implementing the reservation policy of the Government of India.

Apart from the reservation, there is also a relaxation in the minimum qualifying marks for admission for SC/ST candidates. UGC has been implementing the program of Career Orientation to Education (vocationalisation of education) to ensure that the graduates have knowledge, skills, and attitudes for gainful employment in the wage sector in general, and self-employment in particular for all including SCs/STs. It also provides financial assistance for Remedial Coaching to SC/ST students. It provides financial assistance to the existing coaching centers to prepare SC/ST candidates for the National Eligibility Test (NET) conducted by UGC/CSIR. The Commission provides financial assistance for extension activities. Under the scheme, all groups of the society are covered including SCs/STs.

In order to contribute towards social equity and socioeconomic mobility of the under privileged sections of the society, UGC has introduced remedial coaching scheme at UG/PG level. The main objectives of the scheme are:

- i. To improve the academic skills and linguistic proficiency of the students in various subjects.
- ii. To raise the level of comprehension of basic subjects so as to provide a strong foundation for further academic work.
- iii. To strengthen their knowledge, skills and attitudes in the subjects where quantitative and qualitative techniques and laboratory work are involved
- iv. To improve the overall performance of these students in the examination.

Tamilnadu Council for Open and Distance Learning follows the rules and regulations laid down by UGC regarding education and reservation for SC/ST Candidates.

2.24. DIVERSITY AND INCLUSION

Year	Name of the scheme	Government/ non-government	Name of the individual/organization	Several students benefited	Amount (in INR)
18-19	First Graduate	Government	Government of Tamilnadu	814	20350000
	BC/MBC	Government	Government of Tamilnadu	310	1395000
	National Scholarship Portal	Government	-	NIL	NIL
	SC/ST	Government	Government of Tamilnadu	135	6750000
19-20	First Graduate	Government	Government of Tamilnadu	620	15500000
	BC/MBC	Government	Government of Tamilnadu	496	2322000
	National Scholarship Portal	Government	-	NIL	NIL
	SC/ST	Government	Government of Tamilnadu	116	6150000
20-21	First Graduate	Government	Government of Tamilnadu	753	18825000
	BC/MBC	Government	Government of Tamilnadu	499	2245500
	National Scholarship Portal	Government	-	NIL	NIL
	SC/ST	Government	Government of Tamilnadu	141	7050000
21-22	First Graduate	Government	Government of Tamilnadu	662	16550000
	BC/MBC	Government	Government of Tamilnadu	485	2182500
	National Scholarship Portal	Government	-	NIL	NIL
	SC/ST	Government	Government of Tamilnadu	202	10100000
	7.5% Reservation	Government	Government of Tamilnadu	17	850000
22-23	First Graduate	Government	Government of Tamilnadu	631	15775000
	BC/MBC	Government	Government of Tamilnadu	432	1944000
	National Scholarship Portal	Government	-	NIL	NIL
	SC/ST	Government	Government of Tamilnadu	199	9950000
	7.5% Reservation	Government	Government of Tamilnadu	26	1300000
	Institutional Scholarship (Mark Based)	Non-Government	Arunai Engineering College	71	781250

2.25. INDUSTRY COLLABORATION AND PARTNERSHIP OUTCOMES

Sl. No.	Name of the MoU / linkage	Name of the institution / industry with whom the MoU / linkage is made, with contact details	Year of signing MoU / linkage	Purpose of the MoU/Linkage (internship, on-the-job training, project work, student / faculty exchange and collaborative research)
1	Student training and Project support	JDNET Technologies, Cuddalore.	2023	Internship, Guest Lecture, Workshop
2	Internship and Project work	SSP Engineering Services Puducherry	2023	Internship & project work for students
3	Internship and Project work	Kineco Ltd (Defence & Industrial Division. Goa	2023	Internship & project work for students
4	Internship and Project work	Shri Sai Institute of NDT & Inspection Services, Pudhucherry	2023	Internship & project work for students
5	On-the-Job-Training	Ashok Layland, Hosur	2023	On-the-Job-Training
6	AEC -EEE- SIMBA	POWERSYS, Les Jardins de Entreprise, 13610 Le Puy - Sainte-Reparate, France	2023	Training SIMBA - Power Electronics & Drives Simulation Software for Students & Staff
7	Internship and Project work	M/s GREENTECH FERTILIZERS CORPORATION K.V NANDHAL-606755 TIRUVANNAMALAI	2022	Internship & project work for students
8	Placement and Internship	Ladder Survey Institute of Technology	2022	Internship, Training and Placement
9	Internship and Project work	JD Net Technologies, cuddalore	2022	Internship & project work for students
10	Internship	Gapcrud private Technologies	2022	Internship, Guest Lecture, Workshop,
11	Student training and Project support	Pantech PvtLtd,Chennai	2022	Internship, Guest Lecture, Workshop
12	Student training and Project support	FOXCONN	2022	Internship, Guest Lecture
13	Student training and Project support	JDNET Technologies, Cuddalore	2021	Internship, Guest Lecture,Placement,Wor kshop
14	Project and Training	Vector Solutions (P) Ltd, Puducherry	2021	Project and Training

15	Internship and Project work	M/s. Genprotic biopharma Pvt Ltd, No. 6 & 7, VGP Layout	2021	Internship & project work for students
		Part II, Gandhi Street, Chennai-600119.		
16	Student training and Project support	STU INNOVATION PVT LTD	2020	Internship, Guest Lecture
17	Internship and Project work	I Globe Soft Labs, Trichy	2020	Internship & project work for students
18	Internship and Project work	Venus Energy, Neyveli	2019	Internship & project work for students
19	Internship and Project work	Vaibhava Power Solutions, Kumbakonam	2019	Internship & project work for students
20	Internship and Project work	Alpha CADD Solutions, Tiruvannamalai	2019	Internship & project work for students
21	Internship	ONE ZERO Solution, Puducherry	2019	Internship, Guest Lecture
22	Internship	Sahyadri software technologies, Chennai	2019	Internship, Guest Lecture
23	Collaborative Research, R&D, Skill development programme, Industrial training & Visits	FluxGen Engineering Technologies (P) Limited	2018	Collaborative Research, R&D, Skill development programme, Industrial training & Visits
24	Design development & deployment of grid interfaced power conversion unit for	CPRI Bangalore	2016	Design development & deployment of grid interfaced power conversion unit for
	solar-wind power generation system			
25	On the Job Training about Microcontrollers	UTL Technologies Limited, Bengaluru	2015	On the Job Training about Microcontrollers

THE DETAILS OF INDUSTRY COLLABORATIONS: ENGINEERING COLLEGE

S No.	Name of the MoU/Linkage	Name of the institution / industry with whom the MoU / linkage is made, with contact details	Purpose of the MoU/Linkage (internship, on-the-job training, project work, student / faculty exchange and collaborative research)
1	Collaborative research work for students and staff, Internships and visits for students and staff of both parties	National Kaohsiung University of Science and Technology, KAOHSIUNG, TAIWAN	Collaborative research work for students and staff, Internships and visits for students and staff of both parties
2	Collaborative research work for students and staff, Internships and visits for students and staff of both parties	University of Science & Computer Technology (STEKOM) Semarang Indonesia	Collaborative research work for students and staff, Internships and visits for students and staff of both parties
3	Abroad studies, exchange programmes, Abroad internship, Abroad placements	MIT Square Group of Companies, UK	Abroad studies, exchange programmes, Abroad internship, Abroad placements
4	IBM's provision of technology products and services for Customer;	IBM India Private Limited ("IBM")	Internship, on-the-job training
5	IBM's provision of technology products and services for Customer;	IBM India Private Limited ("IBM")	Internship, on-the-job training
6	Value added course and intern	CIIC- Crescent Incubation and Innovation center	Value added course and intern
7	Intern value added course	Hebesec Technologies (P) Ltd	Intern value added course
8	Course access, customization, technical support, internship	Data Flair Indore	Course access, customization, technical support, internship
9	Student training and project support	Birdscale Technologies	Internship, Project Work
10	Student training and project support	Networking Consultants	Internship, Project Work
11	Student training and project support	SNY Auto Signing Ceremony	Internship, Project Work
12	Student training and project support	Saehan Stamping Private Limited	Internship, Project Work
13	Student training and project support	V3 Tech Serv	Internship, Project Work
14	Student training and Project support	JDNET Technologies, Cuddalore.	Internship, Guest Lecture, Workshop
15	Internship and Project work	SSP Engineering Services Puducherry	Internship & project work for students
16	Internship and Project work	Kineco Ltd (Defence & Industrial Division. Goa	Internship & project work for students

17	Internship and Project work	Shri Sai Institute of NDT & Inspection Services, Puducherry	Internship & project work for students
18	On-the-Job-Training	Ashok Leyland, Hosur	On-the-Job-Training
19	AEC -EEE- SIMBA	POWERSYS, Les Jardins de Enterprise, 13610 Le Puy - Sainte- Reparade, France	Training SIMBA - Power Electronics & Drives Simulation Software for Students & Staff
20	Internship, guest lecture, placement, workshop for students	JDNET Technologies, Cuddalore.	Internship, guest lecture, placement, workshop
21	Internship and Project work	M/s Greentech Fertilizers Corporation K.V Nandhal- 606755 Tiruvannamalai	Internship & project work for students
22	Placement and Internship	Ladder Survey Institute of Technology	Internship, Training and Placement
23	Internship and Project work	JD Net Technologies, Cuddalore	Internship & project work for students
24	Internship	Gap crud private Technologies	Internship, Guest Lecture, Workshop,
25	Student training and Project support	Pantech (P)Ltd, Chennai	Internship, Guest Lecture, Workshop
26	Student training and Project support	FOXCONN	Internship, Guest Lecture
27	Student training and Project support	JDNET Technologies, Cuddalore	Internship, Guest Lecture, Placement, Workshop
28	Project and Training	Vector Solutions (P) Ltd, Puducherry	Project and Training
29	Internship and Project work	M/s. Genprotic biopharma Pvt Ltd, No. 6 & 7, VGP Layout Part II, Gandhi Street, Chennai	Internship & project work for students
30	Student training and Project support	STU Innovation (P) Ltd	Internship, Guest Lecture
31	Internship and Project work	I Globe Soft Labs, Trichy	Internship & project work for students
32	Internship and Project work	Venus Energy, Neyveli	Internship & project work for students
33	Internship and Project work	Vaibhava Power Solutions, Kumbakonam	Internship & project work for students
34	Internship and Project work	Alpha CADD Solutions, Tiruvannamalai	Internship & project work for students
35	Internship	ONE ZERO Solution, Puducherry	Internship, Guest Lecture
36	Internship	Sahyadri software technologies, Chennai	Internship, Guest Lecture
37	Collaborative Research, R&D, Skill development programme, Industrial training & Visits	FluxGen Engineering Technologies (P) Ltd	Collaborative Research, R&D, Skill development programme, Industrial training & Visits
38	Design development & deployment of grid interfaced power conversion unit for solar-wind power generation system	CPRI Bangalore	Design development & deployment of grid interfaced power conversion unit for solar-wind power generation system
39	On the Job Training about Microcontrollers	UTL Technologies Limited, Bengaluru	On the Job Training about Microcontrollers

THE INDUSTRY COLLABORATIONS: KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN

S No	Name of the MoU/Linkage	Name of the institution / industry with whom the MoU / linkage is made, with contact details	Purpose of the MoU/Linkage (internship, on-the-job training, project work, student / faculty exchange and collaborative research)
1	Skill based training, Placement, Internship and industrial visit	LUCAS TVS Limited, Pondicherry	Skill based training, Placement, Internship and industrial visit
2	Project, Placement	FIIT to Lead Technology	Project, Placement
3	Career counselling, soft skill, technical training, workshops, Guidance for projects, Campus recruitments	SKAPARE CORPORATE SOLUTION PVT LIMT.,	Career counselling, soft skill, technical training, workshops, Guidance for projects, Campus recruitments
4	Standard curriculum and quality delivery of skill education and training	VIS Education and training company	Standard curriculum and quality delivery of skill education and training
5	For promoting entrepreneurship and career development	Entrepreneurship and Career development	For promoting entrepreneurship and career development
6	Placement training, skill development	NANDHI Foundation	Placement training, skill development
7	Placement training, skill development	STU Innovation Pvt Ltd.,	Placement training, skill development
8	Internship and skill development program	Intermediate English and Skill development	Internship and skill development program
9	To develop the skill	yavarum skill Development Association	To develop the skill
10	Skillfull preparation of students for various competitive Examination	Interface Academy of Skill and Research	Skillfull preparation of students for various competitive Examination
11	Skill development Training and placement	NASSCOM TO CONDUCT TRAINING AND PLACEMENT	Skill development Training and placement
12	Coaching for Bank exam	T.I.M.E - Competitive examination	Coaching for Bank exam
13	To develop the skill in accounts	Tally.ERP 9	To develop the skill in accounts

2.26. STATUTORY COMPLIANCE AND LEGAL FRAMEWORK

Sr.No	Name of the Awards	Name of Councils/Organizations
1	Certificate of Accreditation with NAAC A Grade	National Assessment and Accreditation Council
2	NBA (CSE, Biotech)	National Board of Accreditation
3	NBA (ECE, EEE, IT)	National Board of Accreditation
4	Certificate of Establishment for IIC	Innovation Cell, Ministry of Education, Govt of India
5	Certificate of Accreditation for National Rural Institutions Sustainability Grading (NRISG)	Govt of India
6	AZADI KA AMRIT MAGOTSAV	Govt. of India
7	Management System Certificate - ISO 9001:2015	DNV
8	Startup TN Pre-incubation Centre	TAMIL NADU STARTUP AND INNOVATION MISSION, Government of Tamil Nadu
9	Green Audit	EXNORA INTERNATIONAL, Tiruvannamalai
9	Certificate of Accreditation with NAAC Grade	National Assessment and Accreditation Council
10	Management System Certificate - ISO 9001:2015	Govt of India
11	All India Survey on Higher Education (AISHE)	Ministry of Education, Govt of India
12	2(f)	UGC, Govt of India
13	Permanent Affiliation (12 B)	UGC, Govt of India



2.27 INTERNAL QUALITY ASSURANCE CELL (IQAC) FUNCTIONING.

In pursuance of its Action Plan for performance evaluation, assessment & accreditation & quality up-gradation of institutions of higher education, NAAC proposes that every accredited institution should establish an Internal Quality Assurance Cell (IQAC) as a post-accreditation quality sustenance measure. Since quality enhancement is a continuous process, the IQAC will become a part of the institution's system & work towards realization of the goals of quality enhancement & sustenance. The prime task of the IQAC is to develop a system for conscious, consistent & catalytic improvement in the overall performance of institutions. For this, during the post-accreditation period, it will channelize all efforts & measures of the institution towards promoting its holistic academic excellence.

STRATEGIES

IQAC shall evolve mechanisms and procedures for:

- ▶ Ensuring timely, efficient and progressive performance of academic, administrative and financial tasks
- ▶ The relevance and quality of academic and research programmes
- ▶ Equitable access to and affordability of academic programmes for various sections of society
- ▶ Optimization and integration of modern methods of teaching and learning
- ▶ The credibility of evaluation procedures
- ▶ Ensuring the adequacy, maintenance and functioning of the support structure and services
- ▶ Research sharing and networking with other institutions in India and abroad

FUNCTIONS

Some of the functions expected of the IQAC are:

- ▶ Development and application of quality benchmarks/parameters for various academic and administrative activities of the institution
- ▶ Facilitating the creation of a learner-centric environment conducive to quality education and faculty maturation to adopt the required knowledge and technology for participatory teaching and learning process
- ▶ Arrangement for feedback response from students, parents and other stakeholders on quality-related institutional processes
- ▶ Dissemination of information on various quality parameters of higher education
- ▶ Organization of inter and intra institutional workshops, seminars on quality related themes and promotion of quality circles
- ▶ Documentation of the various programmes/activities leading to quality improvement
- ▶ Acting as a nodal agency of the Institution for coordinating quality-related activities, including adoption and dissemination of best practices
- ▶ Development and maintenance of institutional database through MIS for the purpose of maintaining/enhancing the institutional quality
- ▶ Development of Quality Culture in the institution
- ▶ Preparation of the Annual Quality Assurance Report (AQAR) as per guidelines and parameters of NAAC, to be submitted to NAAC

BENEFITS

IQAC will facilitate/contribute:

- ▶ Ensure heightened level of clarity and focus in institutional functioning towards quality enhancement
- ▶ Ensure internalization of the quality culture
- ▶ Ensure enhancement and coordination among various activities of the institution and institutionalize all good practices
- ▶ Provide a sound basis for decision-making to improve institutional functioning
- ▶ Act as a dynamic system for quality changes in HEIs
- ▶ Build an organized methodology of documentation and internal communication

COMPOSITION

IQAC may be constituted in every institution under the Chairmanship of the Head of the institution with heads of important academic and administrative units and a few teachers and a few distinguished educationists and representatives of local management and stakeholders. The composition of the IQAC may be as follows:

1. Chairperson: Head of the Institution
2. A few senior administrative officers
3. Three to eight teachers
4. One member from the Management
5. One/two nominees from local society, Students and Alumni
6. One/two nominees from Employers/Industrialists/stakeholders
7. One of the senior teachers as the coordinator/Director of the IQAC

2.28. ALUMNI CONTRIBUTIONS

- ▶ Alumni profiles showcasing contributions in national and global forums.
- ▶ Evidence of alumni engagement in mentoring, resource mobilization, and institution-building.

2.29. STUDENT-CENTRIC POLICIES

- ▶ Initiatives promoting mental health, well-being, and career guidance.
- ▶ Mechanisms to encourage innovation and entrepreneurship among students.

1. Integrated Mental Health and Career Services

- ▶ **Holistic Support Approach:** Recognize the intrinsic link between mental health and career development. Implement integrated services where mental health professionals and career advisors collaborate to provide comprehensive support.
- ▶ **Training for Career Services Staff:** Equip career services professionals with the skills to identify and address mental health concerns, enabling them to support students effectively during career planning and transitions.

2. Accessible Counseling and Support Services

- ▶ **On-Campus Counseling Centers:** Establish easily accessible mental health counseling centers staffed with qualified professionals to provide confidential support to students.
- ▶ **Peer Support Programs:** Develop peer counseling initiatives where trained students offer support, fostering a community of care and reducing stigma associated with seeking help.

3. Proactive Mental Health Education

- ▶ **Workshops and Seminars:** Organize regular sessions focusing on stress management, resilience building, and coping strategies to equip students with tools to maintain their mental well-being.

► **Integration into Curriculum:** Incorporate mental health education into the academic curriculum to raise awareness and promote a culture of well-being.

4. Comprehensive Career Guidance Services

► **Personalized Career Counseling:** Offer one-on-one career counseling sessions to help students identify their strengths, interests, and potential career paths.

► **Career Development Workshops:** Conduct workshops on resume building, interview skills, and job search strategies to prepare students for the job market.

5. Establishment of Innovation and Entrepreneurship Centers

► **Innovation Labs:** Create dedicated spaces equipped with resources and technology where students can experiment with ideas, collaborate on projects, and develop prototypes.

► **Entrepreneurship Programs:** Develop programs that provide mentorship, funding opportunities, and networking events to support student-led startups and entrepreneurial ventures.

6. Integration of Career Guidance and Mental Health Support

► **Collaborative Workshops:** Develop programs that address both career planning and mental well-being, acknowledging their interconnectedness and providing holistic support to students.

7. Promotion of Work-Life Balance

► **Flexible Academic Policies:** Implement policies that allow for academic flexibility, enabling students to balance their studies with personal well-being and extracurricular activities.

► **Supportive Campus Environment:** Foster a campus culture that values well-being, encouraging participation in recreational activities and providing spaces for relaxation.

8. Encouragement of Interdisciplinary Collaboration

► **Cross-Disciplinary Projects:** Encourage students from different academic backgrounds to collaborate on projects, fostering innovation through diverse perspectives.

► **Interdisciplinary Courses:** Offer courses that blend various fields of study, promoting a holistic educational experience and encouraging creative problem-solving.

9. Access to External Resources and Networks

► **Partnerships with Industry:** Establish collaborations with industry professionals and organizations to provide students with real-world insights, internships, and mentorship opportunities.

► **Alumni Networks:** Leverage alumni connections to offer guidance, support, and opportunities for current students interested in entrepreneurship and innovation.

10. Continuous Evaluation and Improvement

► **Feedback Mechanisms:** Implement regular assessments and feedback channels to evaluate the effectiveness of mental health services, career guidance, and entrepreneurship programs, ensuring they meet student needs.

► **Data-Driven Policy Adjustments:** Utilize collected data to make informed decisions and adjustments to programs, enhancing their impact and relevance.

These student-centric policies and initiatives, educational institutions can create a supportive environment that promotes mental health, well-being, career readiness, and entrepreneurial spirit among students.

2.30. STUDENT EMPOWERMENT AND LEADERSHIP

► Evidence of fostering student leadership through clubs, societies, and student council activities.

To provide a platform to exchange their knowledge, subject experts from both academia and industry were called to the campus for guest lectures, seminars, workshops, conferences and value-added courses.

The Institute is member in Institute of Electrical and Electronics Engineers (IEEE), Indian Society for Technical Education (ISTE), Computer Society of India (CSI), Indian Institute of Chemical Engineers (IICHE), Biotech Research Society of India (BRSI), Engineers Without Borders (EWB).

The Institute has active Institute Innovation Council which is articulated and monitored by the Ministry of Education, Government of India.

The Institute organizes project expo's in which innovative student's projects were exhibited to the general public to make them aware on technological developments. Society oriented projects by departments were taken and 2 of the projects Online Car Parking Management System and Missing Children Tracking System were implemented to benefit the District Police Department. Students and staff of the Institute consistently bag National awards for their innovative research work. More than 22 awards were bagged by them during last 5 years.

CLUB ACTIVITIES:

College club activities are integral part of the higher educational experience such club activities are a way for students to connect with others, explore their interest and develop new skills. We have created so many clubs such as National Service Scheme, Youth Red Cross, Red Ribbon Club, Eco Green Club, etc. Each club has their own motto and responsibilities and the governing body performed well. With the help of Eco - Green Club our institution undergoes several programs regarding environmental changes, planting trees, pollution, bio-degradable products and so on. The students eagerly participated in those programs and gave their best.

Our institution creates awareness among the next generation of Students Introducing urban students to rural living and advancing the country are the goals of the National Service Scheme's (NSS) Youth for Rural Reconstruction special camping program:

► **Recognize needs and issues:** Students ought to recognize the needs and issues of the community and participate in their resolution.

► **Foster civic and social duty:** Students ought to cultivate a sense of civic and social responsibility.

► **Encourage national integration:** Students ought to encourage social peace and national integration.

Our institution creates awareness among the Students to Providing basic needs to an orphanage can help ensure that children have access to food, clothing, shelter and sanitary napkins that can help educate people about menstrual cycles, reproductive health, and the natural and healthy process of menstruation.

Poverty is a major factor that leads children to be separated from their families and entered into orphanages. Families facing poverty may struggle to access essential services, such as health, education, and financial support.

The Youth Red Cross (YRC) club is one of the avenues available for the students indulge themselves in the social activities. The activities related with YRC includes conducting safety and first aid technique camp to the village people and school students, disaster management, scribe writing for visually challenged people, making safety survey on constructions. 'Assistance without discrimination' is the motto of the club.

The red ribbon is a symbol for both drug prevention and for the fight against AIDS. The Red Ribbon foundation is an organization founded in 1993 whose main purpose is the education about prevention of the Human Immunodeficiency Virus or HIV, Acquired Deficiency Syndrome Related Complex, ARC and AIDS.

The Red Ribbon Club (RRC) aims at harnessing the potential of the youth by equipping them with correct information on HIV/AIDS Prevention, Care, Support and Treatment.

The Entrepreneurship Development Cell (EDC) was started with the object of promoting specialized knowledge in the field of entrepreneurship development. In view of the worldwide shortage of jobs (both government and private sectors) leading to unemployment problems and lack of proper utilization of human resources; the cell strives to identify talented youth to entrepreneurial works. The cell plans to organize various programs regarding entrepreneurship development.

OBJECTIVES:

- (a) To create awareness among the students of the Institution regarding entrepreneurship as a career option.
- (b) To conduct training programme in the field of entrepreneurial skill development
- (c) To provide guidance and facilities to first time entrepreneurs during gestation.

The consumer club has been established to raise awareness among the students of their consumer rights and obligations, and also to inform students of their legal rights as consumers under the Consumer Protection Act. To raise awareness of the consumer rights and obligations outlined in the Indian Constitution, the Consumer Protection Act, and other Indian laws and to create knowledge about consumer rights.

The Women Empowerment Cell in our college is established to empower girl students and staffs and to enhance their understanding of issues related to women in the society and to face the challenges with great courage. The Goal of the cell is to bring about the advancement, development and empowerment of women. This cell creates a feel in them that our college campus is heaven and second home to them. The cell functions arduously to enhance the status and thereby empower them through Guest Lectures, Seminars. Workshops, various awareness programs and other welfare activities.

2.31. STAKEHOLDER FEEDBACK

Mechanisms for collecting and utilizing feedback from students, faculty, alumni, and employers and Structured feedback systems for curriculum, pedagogy, and facilities involving students, parents, and alumni.

Feedbacks are collected from the stakeholders to identify the curricular gaps and actions to rectify were taken accordingly. In order to strengthen the curricular outcome, the institute is organizing various co-curricular events such as Guest lectures, Seminars, conferences, symposiums, workshops, add-on courses. Feedback will be collected from the students and staff. Based on the feedback and end semester exam results an analytical report corrective and preventive actions will be prepared which will help to eradicate the shortfalls.

IMPROVEMENTS MADE BASED ON STAKEHOLDERS INPUT

Stakeholder feedback is vital to the continuous improvement process of any educational institution. By involving stakeholders such as students, faculty, staff, alumni, and parents, institutions can ensure that their development and growth align with the needs, expectations, and aspirations of the community they serve.

By actively listening to our stakeholders, taking decisive actions, and maintaining a cycle of evaluation and refinement, we are confident that we are building a stronger, more responsive, and dynamic institution. This commitment not only strengthens but also focused on providing the highest quality of education and services to those we serve.

The feedback gathered from stakeholders such as students, faculty, parents, alumni, and industry partners has been invaluable in guiding the improvements made at Kamban College of Arts and Science for Women. The actions taken based on this feedback have led to measurable improvements in campus infrastructure, academic support, career services, and stakeholder engagement.

2.32 CERTIFICATIONS AND ACHIEVEMENTS

Sr.No	Name of the Awards	Name of Councils/Organizations
1	Certificate of Accreditation with NAAC A Grade	National Assessment and Accreditation Council
2	NBA (CSE, Biotech)	National Board of Accreditation
3	NBA (ECE, EEE, IT)	National Board of Accreditation
4	Certificate of Establishment for IIC	Innovation Cell, Ministry of Education, Govt of India
5	Certificate of Accreditation for National Rural Institutions Sustainability Grading (NRISG)	Govt of India
6	AZADI KA AMRIT MAHOTSAV	Govt. of India
7	Management System Certificate - ISO 9001:2015	DNV
8	Startup TN Pre-incubation Centre	TAMIL NADU STARTUP AND INNOVATION MISSION, Government of Tamil Nadu
9	Green Audit	EXNORA INTERNATIONAL, Tiruvannamalai
9	Certificate of Accreditation with NAAC Grade	National Assessment and Accreditation Council
10	Management System Certificate - ISO 9001:2015	Govt of India
11	All India Survey on Higher Education (AISHE)	Ministry of Education, Govt of India
12	2(f)	UGC, Govt of India
13	Permanent Affiliation (12 B)	UGC, Govt of India

2.32 THE PROPOSAL OUTLINE

In alignment with its vision "to shape and transform youth with utmost care, innovative and dedicated service for morals and education," the project institutes are now poised to ascend to a new level of excellence by transforming into a "Distinct Deemed University." This transition is inspired by the goals set forth in the University Grants Commission (UGC) Regulations for Deemed Universities, 2023. By achieving this status, the institution aims to become a pioneering force in higher education, offering unique programs that seamlessly integrate technical excellence, ethical values, and holistic development, ultimately contributing to the nation's advancement and societal well-being.

This Detailed Project Report (DPR) encapsulates the strategic vision and operational framework that will guide AEC's transformation over the next fifteen years. The report is structured to provide a comprehensive overview of the strategic and rolling implementation plans that will underpin this journey.

These include:

- a) **Academic Plan:** Envisioning a curriculum that blends technical depth with interdisciplinary breadth, aiming to equip students with skills for emerging fields and societal challenges.
- b) **Faculty Recruitment Plan:** Building a team of accomplished, research-driven educators with international exposure, to create a knowledge ecosystem that promotes innovation and critical thinking.
- c) **Student Admission Plan:** Establishing an admission strategy that attracts diverse, high-caliber students from across the globe, enhancing the university's intellectual and cultural diversity.
- d) **Research Plan:** Creating an impactful research ecosystem, fostering collaboration with industry and academic institutions worldwide, and focusing on areas that address societal challenges and drive technological advancement.
- e) **Campus Information and Communication Technology Plan:** Adopting cutting-edge ICT infrastructure to create a digitally empowered campus, facilitating seamless teaching, learning, and administrative processes.
- f) **Infrastructure Development Plan:** Expanding and enhancing campus facilities to support a dynamic learning environment, inclusive of state-of-the-art laboratories, libraries, and recreational spaces.
- g) **Finance Plan:** Ensuring the financial sustainability and growth of the institution through diversified funding sources, including industry collaborations, alumni contributions, and government grants.
- h) **Administrative Plan:** Streamlining administrative processes to support efficient and effective operations, ensuring transparency, accountability, and alignment with institutional goals.
- i) **Governance Plan:** Establishing a governance framework that emphasizes strategic oversight, ethical leadership, and stakeholder engagement, ensuring the institution's integrity and commitment to its mission.

With clear annual milestones and action plans, this DPR details a transformative journey that aims to elevate Arunai College of Engineering into a leading institution recognized for its academic rigor, ethical standards, and contribution to societal progress. The report also includes an assessment of AEC's current academic and physical infrastructure, laying the groundwork for the envisioned development trajectory. By adhering to this roadmap, AEC aspires to achieve national and international distinction, nurturing future leaders who embody technical prowess, integrity, and a commitment to societal impact.



VISION:

"To be a pioneering university of distinction, renowned nationally and internationally for its extraordinary, unique programs that integrate technical excellence with ethical and holistic development. We are committed to cultivating globally competent professionals who are not only skilled and innovative but also deeply rooted in values and dedicated to social impact. By fostering transformative learning in a serene, inspiring environment, we empower individuals to excel in their fields, uplift communities, and serve as ambassadors of integrity and wisdom on the world stage."

This vision highlights AEC's commitment to unique, high-impact programs that elevate its reputation and reach, enhancing its appeal and influence both in India and globally.

MISSION:

"To advance knowledge and societal impact through distinctive programs that blend professional ability with moral education, enabling our graduates to become valued leaders who drive positive change both locally and internationally."

2.33 JUSTIFICATION FOR SEEKING "DISTINCT CATEGORY UNIVERSITY" STATUS.

The establishment of a 'Distinct University' under the deemed university category at Tiruvannamalai aligns with the University Grants Commission (UGC) Deemed University Regulations-2023, which emphasize academic excellence, innovation, and societal impact. This proposed university envisions offering unique programs integrating technical proficiency, moral education, and holistic development, thus addressing the evolving educational needs of the region and beyond. Located in the serene environs of Tiruvannamalai, the university aims to leverage its geographical advantage by attracting students and scholars globally to a multidisciplinary learning environment. Its focus on cutting-edge research, entrepreneurship, and strategic collaborations with industries and international institutions will foster a culture of innovation and provide a pathway for students to become global leaders. The university's emphasis on technical and ethical education uniquely positions it to contribute significantly to nation-building by creating competent professionals imbued with strong values.

The proposed 'Distinct University' also fulfills the objectives of the UGC Regulations by advancing societal development through education tailored to the region's cultural and economic context. It seeks to become a hub for innovative programs that address emerging challenges in science, technology, healthcare, and social sciences. By combining local relevance with global standards, the university will ensure inclusive growth and opportunities for rural and urban youth, particularly in Tamil Nadu and Pondicherry. Furthermore, its commitment to sustainability and ethical governance will serve as a model institution, meeting and exceeding UGC benchmarks for quality, autonomy, and innovation. The initiative to establish this university is a step towards fostering knowledge economies while ensuring equity and accessibility in higher education.

The establishment of a 'Distinct University' under the deemed university category at Tiruvannamalai aligns with the University Grants Commission (UGC) Deemed University Regulations-2023, which emphasize academic excellence, innovation, and societal impact. Leveraging the existing academic ecosystem of Arunai Engineering College and Kamban College of Arts and Science for Women, Karan Arts and Science College the proposed university envisions offering unique programs that integrate technical proficiency, moral education, and holistic development. Arunai Engineering College, with its legacy of quality technical education since 1993, provides a strong foundation for nurturing technical talent. Similarly, Kamban College's expertise in arts, commerce, and science contributes to the multidisciplinary fabric of this vision. Together, these institutions position the region as an emerging knowledge hub, and the establishment of the proposed university will amplify this potential by attracting global talent and fostering research and innovation. The serene and culturally rich environment of Tiruvannamalai offers a unique setting conducive to academic excellence and holistic growth.

The proposed 'Distinct University' fulfills the objectives of the UGC Regulations by advancing societal development through education tailored to the region's cultural and economic context. It will consolidate the strengths of Arunai Engineering College in technical disciplines and Kamban College's excellence in arts and sciences to offer a comprehensive, world-class education. By addressing regional and global challenges through innovative programs, the university will ensure inclusive growth and opportunities for rural and urban youth, particularly in Tamil Nadu and Pondicherry. Its strategic location, coupled with a commitment to sustainability and ethical governance, positions it as a model institution that aligns with UGC's benchmarks for quality, autonomy, and innovation. This endeavor is not just about creating another educational institution but about shaping a transformative ecosystem that inspires societal progress and global impact.

The background of the page is white, decorated with several large, overlapping geometric shapes in two shades of blue: a dark navy blue and a medium blue. These shapes, which include triangles and squares, are positioned in the corners and along the left and right edges, creating a modern, architectural feel. In the center of the page, the words "THE PROPOSAL" are written in a bold, dark blue, sans-serif typeface. The word "THE" is smaller and positioned directly above "PROPOSAL".

THE PROPOSAL

CHAPTER-III

3.1 VISION AND MISSION OF THE PROPOSED UNIVERSITY

VISION:

"To be a pioneering university of distinction, renowned nationally and internationally for its extraordinary, unique programs that integrate technical excellence with ethical and holistic development. We are committed to cultivating globally competent professionals who are not only skilled and innovative but also deeply rooted in values and dedicated to social impact. By fostering transformative learning in a serene, inspiring environment, we empower individuals to excel in their fields, uplift communities, and serve as ambassadors of integrity and wisdom on the world stage."

This vision highlights ARUNAI's (Deemed to be University) commitment to unique, high-impact programs that elevate its reputation and reach, enhancing its appeal and influence both in India and globally.

MISSION:

"To advance knowledge and societal impact through distinctive programs that blend professional ability with moral education, enabling our graduates to become valued leaders who drive positive change both locally and internationally."



3.2 Justification for seeking "Distinct Category University" status.

The establishment of a 'Distinct University' under the deemed university category at Tiruvannamalai aligns with the University Grants Commission (UGC) Deemed University Regulations-2023, which emphasize academic excellence, innovation, and societal impact. This proposed university envisions offering unique programs integrating technical proficiency, moral education, and holistic development, thus addressing the evolving educational needs of the region and beyond. Located in the serene environs of Tiruvannamalai, the university aims to leverage its geographical advantage by attracting students and scholars globally to a multidisciplinary learning environment. Its focus on cutting-edge research, entrepreneurship, and strategic collaborations with industries and international institutions will foster a culture of innovation and provide a pathway for students to become global leaders. The university's emphasis on technical and ethical education uniquely positions it to contribute significantly to nation-building by creating competent professionals imbued with strong values.

The proposed 'Distinct University' also fulfills the objectives of the UGC Regulations by advancing societal development through education tailored to the region's cultural and economic context. It seeks to become a hub for innovative programs that address emerging challenges in science, technology, healthcare, and social sciences. By combining local relevance with global standards, the university will ensure inclusive growth and opportunities for rural and urban youth, particularly in Tamil Nadu and Pondicherry. Furthermore, its commitment to sustainability and ethical governance will serve as a model institution, meeting and exceeding UGC benchmarks for quality, autonomy, and innovation. The initiative to establish this university is a step towards fostering knowledge economies while ensuring equity and accessibility in higher education.

The establishment of a 'Distinct University' under the deemed university category at Tiruvannamalai aligns with the University Grants Commission (UGC) Deemed University Regulations-2023, which emphasize academic excellence, innovation, and societal impact. Leveraging the existing academic ecosystem of Arunai Engineering College and Kamban College of Arts and Science for Women, Karan Arts and Science College, the proposed university envisions offering unique programs that integrate technical proficiency, moral education, and holistic development. Arunai Engineering College, with its legacy of quality technical education since 1993, provides a strong foundation for nurturing technical talent. Similarly, Kamban College's expertise in arts, commerce, and science contributes to the multidisciplinary fabric of this vision. Together, these institutions position the region as an emerging knowledge hub, and the establishment of the proposed university will amplify this potential by attracting global talent and fostering research and innovation. The serene and culturally rich environment of Tiruvannamalai offers a unique setting conducive to academic excellence and holistic growth.

The proposed 'Distinct University' fulfills the objectives of the UGC Regulations by advancing societal development through education tailored to the region's cultural and economic context. It will consolidate the strengths of Arunai Engineering College in technical disciplines and Kamban College and Karan College excellence in arts and sciences to offer a comprehensive, world-class education. By addressing regional and global challenges through innovative programs, the university will ensure inclusive growth and opportunities for rural and urban youth, particularly in Tamil Nadu and Pondicherry. Its strategic location, coupled with a commitment to sustainability and ethical governance, positions it as a model institution that aligns with UGC's benchmarks for quality, autonomy, and innovation. This endeavor is not just about creating another educational institution but about shaping a transformative ecosystem that inspires societal progress and global impact.

3.3. ACADEMIC INFORMATION- THE DISTINCT PROGRAMS

List of academic programs offered, including UG, PG, doctoral, and interdisciplinary programs are stated in Table NO.1 and Table No. 2 below.

TABLE NO.1: UNDERGRADUATE AND POST-GRADUATE PROGRAMS

STREAM	S.NO	COURSE NAME
Environment and Sustainability	1.	B.Tech. Climate Science & Engineering
	2.	M.Tech. Sustainability in Building Maintenance
Artificial Intelligence and Computers	3.	M.Tech. Natural Language Processing
	4.	M.Tech. AI in VLSI technology
Agriculture and Biotechnology	5.	M.Tech. Agro-Electronics
	6.	B.Sc. Organic Farming
	7.	M.Sc. Clinical Research and Health Informatics (Integrated Course)
Indian Knowledge System	8.	B.A. Tamil Culture and Civilization
	9.	M.A. Thozil Sar Kalaigal
Commerce and Modern Trade	10.	B.Com. Financial Risk Management and Taxation
Skill Development Courses	11.	PG Diploma Solar PV and Battery Management Systems
	12.	PG Diploma AIML in Health Care Management

THE FEE STRUCTURE

S.NO	COURSE	DURATION	INTAKE	FEE PER ANNUM
1.	B.Tech. Climate Science & Engineering	4 Years	60	50000
2.	M.Tech. Sustainability in Building Maintenance	2 Years	18	50000
3.	M.Tech. Natural Language Processing	2 Years	18	50000
4.	M.Tech. AI in VLSI technology	2 Years	18	50000
5.	M.Tech. Agro-Electronics	2 Years	18	50000
6.	B.Sc. Organic Farming	3 Years	50	30000
7.	M.Sc. Clinical Research and Health Informatics (Integrated Course)	2 Years	20	20000
8.	B.A. Tamil Culture and Civilization	3 Years	50	30000
9.	M.A. Thozil Sar Kalaigal	2 Years	20	20000
10.	B.Com. Financial Risk Management and Taxation	3 Years	50	30000

TABLE NO. 2: RESEARCH PROGRAMS/DOMAINS

S.NO	NAME OF THE PROGRAM	AVAILABLE VERTICAL SPECIALIZATION
1.	Agricultural Technologies	M.Tech (Agro Electronics) – Proposed Distinct Course B.Tech (Agricultural Engineering) – Existing Course
2.	VLSI	M.Tech. AI in VLSI technology – Proposed Distinct Course B. Tech (AIDS) – Existing Course B.E (AIML) – Existing Course B.E (ECE) – Existing Course
3.	Computer Science and Engineering (Specialization in AI)	M.Tech. Natural Language Processing – Proposed Distinct Course M.Tech. AI in VLSI technology – Proposed Distinct Course M.E Computer Science and Engineering B. Tech (AIDS) – Existing Course B.E (AIML) – Existing Course B.E Computer Science and Engineering – Existing Course
4.	Sustainable Technologies	M.Tech. Sustainability in Building Maintenance – Proposed Distinct Course B.E. Civil Engineering – Existing Course
5.	Health Informatics	M.Sc. Clinical Research and Health Informatics (Integrated Course) – Proposed Distinct Course M.Tech Biotechnology – Existing Course B.Sc. Microbiology – Existing Course B.Tech Biotechnology – Existing Course
6.	Tamil Literature and Tradition	M.A. Thozil Sar Kalaigal – Proposed Distinct Course M.A Tamil – Existing Course B.A. Tamil Culture and Civilization – Proposed Distinct Course B.A Tamil – Existing Course

3.4. DISTINCTIVENESS OF PROGRAMS

Comprehensive details on distinctive programs, courses, or areas of focus that set the institution apart.

The proposal includes three institute of the parent management, Arunai Engineering College, Kamban College of Arts and Science for Women and Karan Arts and Science College.

The proposal has a plan of establishing three major domains/faculties at the proposed university.

I. Science and Technology

II. Business Management and Commerce, and

III. Liberal Arts

The proposed university at Tiruvannamalai aims to offer a range of undergraduate (UG), postgraduate (PG), and research programs that reflect its commitment to distinctiveness, innovation, and integration of modern technology with traditional wisdom. These programs are designed to address the emerging needs of society and industries, blending theoretical foundations with practical applications to produce globally competent graduates. The university will emphasize inter-disciplinary and application-oriented learning across diverse fields like environment and sustainability, artificial intelligence, agriculture, biotechnology, Indian knowledge systems, commerce, and skill development.

Distinctiveness is a cornerstone of the proposed programs, evident in offerings like B. Tech in Climate Science and Engineering, which integrates advanced climate modeling with sustainable engineering practices. Similarly, M. Tech in AI for VLSI Technology introduces a confluence of artificial intelligence and semiconductor design, preparing students for cutting-edge technological advancements. Programs such as M.Sc. in Clinical Research and Health Informatics exemplify the integration of healthcare, biotechnology, and informatics to address modern healthcare challenges through data-driven solutions. Furthermore, the B.A. in Tamil Culture and Civilization and M.A. Thozil Sar Kalaigal celebrate India's rich heritage by fostering in-depth understanding and modern applications of traditional knowledge systems.

The programs leverage technology to enhance learning outcomes and research capabilities. Advanced laboratories, AI-powered tools, and online platforms will support experiential and blended learning methodologies. For instance, courses like M. Tech in Sustainability in Building Maintenance employ smart technologies for energy-efficient designs, aligning with global sustainability goals. In addition, the focus on research-oriented PG courses, such as Agro-Electronics, provides a seamless path for students to transition from advanced learning to impactful research.

The integration of skill development diplomas, such as Solar PV and Battery Management Systems, ensures that the university addresses industry demands for specialized skills while promoting green technologies. Research programs in specialized areas like sustainable technologies and health informatics will foster innovation by combining existing institutional strengths with cutting-edge advancements. This approach positions the proposed university as a hub of academic excellence, societal relevance, and technological leadership.

Evidence of integration of cutting-edge technology, such as AI, machine learning, or renewable energy, into academic and research programs.

The proposed university at Tiruvannamalai exemplifies the integration of cutting-edge technologies into its academic and research programs, fostering an environment of innovation and relevance to modern societal and industrial needs. Programs like M. Tech in Natural Language Processing and AI in VLSI Technology are prime examples of harnessing artificial intelligence and machine learning to address real-world challenges. These programs focus on developing systems that can process complex data and improve decision-making in various domains, from language processing to semiconductor design. The emphasis on AI ensures that students and researchers gain expertise in technologies shaping the future, providing them with competitive advantages in global markets. Renewable energy and sustainability are core areas where technological integration is evident. The B. Tech in Climate Science and Engineering incorporates advanced climate modeling and energy-efficient practices, preparing students to tackle global environmental challenges. Similarly, the M. Tech program in Sustainability in Building Maintenance utilizes smart building technologies to enhance energy efficiency and reduce environmental impact. The integration of these technologies in academic curricula reflects the university's commitment to sustainability and its alignment with global development goals, offering students hands-on experience with emerging green technologies that are critical to addressing climate change.

Evidence supporting the uniqueness and differentiation from other institutions. (Validation documents for the programs and curriculum)

The proposal for the 'Deemed University status under distinct category' of the three institutions (Arunai Engineering College, Kamban College of Arts and Science for Women, Karan Arts and Science College) submitted to UGC is unique and different due to the programmes designed. The following list of programmes was shortlisted to be distinct under various features and they are added in the proposal.

1. B.Tech. Climate Science & Engineering

Climate science encompasses a broad range of subjects, from atmospheric sciences and oceanography to the understanding of natural climate systems and anthropogenic influences on global warming. Engineering solutions are crucial in reducing carbon footprints, improving energy efficiency, and developing sustainable infrastructure to address the impacts of climate change. This program combines the scientific fundamentals with practical engineering applications, preparing graduates to tackle the complexities of climate-related challenges.

The increasing severity and unpredictability of extreme weather events, the rise in global temperatures, and the subsequent effects on agriculture, water resources, and biodiversity underscore the need for advanced studies in climate science and engineering. Additionally, the growing urgency for global cooperation and policy development in response to climate change makes this field highly relevant.

Why it is distinct?

The B.Tech. Climate Science and Engineering program stands out in its distinctiveness due to its interdisciplinary approach, blending climate science with engineering applications. Unlike traditional programs in environmental science or engineering, this degree provides students with a holistic perspective, enabling them to understand the complex interactions between climate, ecosystems, human society, and infrastructure.

2.M.Tech. Sustainability in Building Maintenance

Building maintenance consist of wide variety of tasks. It is important for a number of reasons such as safety, cost savings, sustainability and preventing various types of failures. So for design and construction of buildings there are lot of courses are available to learn but for maintenance there is some of the subjects only available.

This course is designed to equip the students with the technical know-how and management skills to be an effective leader in this specialist sector. Develop an understanding of the design, installation, operation and maintenance that make a building safe, comfortable, energy efficient and functional. This highly specialized skillset is always in demand. In fact, it's set to grow as buildings' designs evolve to become more sustainable.

Why it is distinct?

Building maintenance is often referred to as a "distinctive course" because it is a specialized field that requires a unique set of skills, knowledge, and expertise. The term emphasizes the specialized nature of the discipline within the broader construction, real estate, and facilities management industries. Building maintenance covers a wide range of tasks, from structural repairs to mechanical systems management. It requires knowledge in areas like plumbing, HVAC systems, electrical work, safety standards, and materials science, which makes it distinct from general construction or real estate management.

3. M.Tech. Natural Language Processing

This program focuses on equipping students with advanced skills in computational linguistics, machine learning, and AI for processing and understanding human language. It covers topics like text analytics, speech recognition, machine translation, and conversational AI. This program prepares graduates for careers in research, academia, and industries developing language-based AI applications such as chatbots, voice assistants, and translation systems.

NLP bridges the gap between human language and machines, enabling computers to understand, interpret, and generate human text. From chatbots and virtual assistants to sentiment analysis and machine translation, NLP fuels advancements across various industries. It facilitates more effective communication by enabling personalized experiences, improving accessibility, and breaking down language barriers. This program plays a crucial role in fields like linguistics, psychology, and social sciences, providing valuable insights into human language and cognition.

Why it is distinct?

This program stands out for its unique integration of linguistics, machine learning, and artificial intelligence, offering a deep focus on language-related AI technologies. Unlike general AI programs, it specializes in cutting-edge topics like sentiment analysis, conversational AI, and language modeling. The program bridges the gap between human communication and AI systems, offering innovative solutions for real-world applications. Its interdisciplinary approach ensures graduates are well-equipped to lead advancements in NLP research and industry.

4. M.Tech. AI in VLSI technology

The future of chip design is AI-driven, and this program prepares students to be at the forefront of this revolution. The integration of AI and VLSI design creates smarter, faster, and more energy-efficient semiconductor chips. It emphasizes automation, customization, and interdisciplinary learning, enabling students to master AI-driven tools for chip design, layout, verification, and testing. This program addresses the industry's growing need for custom AI chips used in smartphones, autonomous vehicles, wearable, and IoT devices.

Why it is distinct?

This program lies in its ability to produce next-generation engineers who are skilled in both AI and VLSI design. By focusing on AI-driven design automation, verification, optimization, and hardware-software co-design, the program prepares students for roles at the forefront of semiconductor innovation. Unlike traditional VLSI engineers, graduates of this program are equipped to design AI-specific hardware accelerators, support edge AI devices, and develop the next generation of neuromorphic chips. This unique blend of skills positions them to be industry leaders in fields like autonomous vehicles, IoT, robotics, and smart devices.

5. M.Tech. Agro-Electronics

Agro-Electronics is an exciting and impactful field, merging agriculture with advanced electrical and electronic solutions to promote sustainable farming practices. It combines engineering principles with agricultural sciences to develop smart, automated, and technologically-driven solutions tailored for the agricultural industry.

This program provides an opportunity for the graduates to support agricultural efficiency, environmental sustainability, and food security.

Why it is distinct?

The M. Tech Agro-Electronics is a distinct and interdisciplinary program that blends agriculture, electronics, and technology to address modern agricultural challenges through technological innovation. It integrates electronics, sensors, AI, and automation with agricultural sciences. It equips students to revolutionize traditional agriculture practices through technology-driven, sustainable, and precision-based solutions, making them future-ready for both agriculture and technology industries.

Graduates of this program possess skills that span agriculture, electronics, and IT industries, making them versatile for careers in Agri-tech start-ups, Government agricultural departments, Agro-electronics product development and Research & development in smart agriculture.

6. B.Sc. Organic Farming

This course enables the students to understanding and application of organic farming in environment, human health, animal welfare. It helps to maintain environment health by reducing the level of pollution, greenhouse gas emissions and soil erosion. The organic farming theory and practicals designing and evaluating policy solutions to a variety of environmental issue, soil health, climate change and to produce safe foods. explores the organic farming principles and policies related to environmental issues, Human and animal Health Welfare, Food safety and environmental sustainability

Why it is distinct?

This program stands out by combining traditional farming knowledge with modern ecological principles, helping students to become proficient in sustainable agricultural practices that align with global trends toward sustainability and healthy food production. It not only equips students with scientific knowledge but also fosters a deep understanding of how agriculture interacts with the environment, economy, and society.

7. M.Sc. Clinical Research and Health Informatics (Integrated Course)

A clinical research requires a systematic approach with diligent planning, execution and sampling in order to obtain reliable and validated results, as well as an understanding of each research methodology is essential for researchers. Indeed, selecting an inappropriate study type, an error that cannot be corrected after the beginning of a study, results in flawed methodology. Health informatics is a specialized field within the healthcare industry that combines communications, information and healthcare with the goal of improving patient care. It involves using resources such as health information systems in collecting, storing, retrieving, analyzing and utilizing healthcare information for a variety of purposes

Our course is designed is to produce high-quality graduates with significant communication, leadership, intellectual skills backed by ethics and social responsibility. Also this course is to introduce the principles of clinical research, including skills and knowledge on why and how to do research, how to build a good research environment, and where to find additional resources, tools, and reference packages to support a research project.

Why it is distinct?

M.Sc. Clinical research & health informatics is a distinctive course, since it includes co-curricular and extracurricular activities to make learning also it offers soft skills training programs to help students develop corporate manners and other skills needed for success in the global marketplace. Clinical research plays a pivotal role in advancing medical knowledge and improving patient outcomes. By investigating the safety and efficacy of new interventions, clinical trials provide valuable insights that guide treatment decisions and enhance the quality of care. Furthermore, research findings inform evidence-based guidelines, empowering healthcare providers to deliver personalized and effective therapies tailored to individual patient needs. As a result, clinical research serves as a catalyst for transformative breakthroughs that have the potential to revolutionize healthcare delivery and disease management. The integration of artificial intelligence in clinical research holds promise for streamlining data analysis and accelerating the discovery of novel therapeutic interventions. Health informatics plays a crucial role in healthcare by combining information technology, computer science, clinical knowledge, and management science to improve patient care, enhance efficiency, and reduce overall costs.

8. B.A. Tamil Culture and Civilization

The B.A. Tamil Culture and Civilization program explores the rich heritage, traditions, and advancements of Tamil society. It focuses on understanding the evolution of Tamil culture and its impact on various aspects of human life, including language, art, architecture, philosophy, and social practices.

This program provides students with a comprehensive understanding of Tamil civilization from its ancient origins to its modern developments. It emphasizes the unique contributions of Tamil culture to global civilization, encouraging students to appreciate their heritage while fostering critical thinking and cultural awareness.

Why it is distinct?

The B.A. in Tamil Culture and Civilization is a distinctive program that offers an enriching educational experience focused on the history, traditions, and contributions of Tamil civilization. With its multidisciplinary approach, it nurtures a deep understanding of the Tamil language, literature, arts, and cultural practices, preparing students to become experts in preserving and promoting Tamil heritage both in India and globally. This program serves not only as an academic pursuit but also as a means to celebrate and sustain the vibrant culture of the Tamil people.

9. B.A. Thozil Sar Kalaigal

B.A. in Thozil Sar Kalaikal is an undergraduate program designed to offer students a comprehensive understanding of various art forms, craftsmanship, and professional skills. This program combines the cultural richness of traditional arts with the modern applications of creativity, providing students with a unique opportunity to explore the practical and artistic aspects of different art disciplines. The course typically focuses on vocational skills, creative expression, and cultural heritage, empowering students to pursue careers in the arts, design, and cultural preservation fields.

Why it is distinct?

B.A. in Thozil Sar Kalaikal stands out by offering a blend of practical and theoretical knowledge about the arts, focusing on both traditional craftsmanship and contemporary creative expression. It is designed to prepare students for a range of creative professions while fostering a deeper understanding of cultural heritage and artistic innovation. The program emphasizes hands-on learning, community engagement, and the development of professional skills, making it an ideal choice for those looking to pursue a career in the vibrant world of applied and professional arts.

10. B.Com. Financial Risk Management and Taxation

This undergraduate program in Financial Risk Management and Taxation is designed to equip students with a comprehensive understanding of auditing and taxation principles, practices, and regulations. The program aims to develop professionals who can provide expert advice on taxation and auditing matters, and who can help organizations maintain transparency, accountability, and compliance with regulatory requirements.

11. PG Diploma in Solar PV and Battery Management Systems

The PG Diploma in Solar and Battery Management Systems aims to address the growing demand for skilled professionals in the renewable energy and energy storage sectors. Graduates of this program will emerge as skilled engineers, designers, and energy consultants, ready to contribute to the global transition toward renewable energy and sustainable power solutions.

12. PG Diploma in AIML in Health Care Management

The course emphasizes the importance of patient-centered care, where healthcare managers prioritize patients' needs, preferences, and values. It promotes evidence-based management practices, where healthcare managers make informed decisions based on research, data, and best practices. This course also highlights the importance of collaboration and teamwork among healthcare professionals, managers, and stakeholders to achieve common goals.

Additionally, interdisciplinary programs like M.Sc. Clinical Research and Health Informatics and M. Tech in Agro-Electronics demonstrate the university's focus on combining advanced technology with societal applications. Health informatics integrates AI-driven data analysis for improved healthcare delivery, while agro-electronics applies smart sensors and automation to enhance agricultural productivity. These programs utilize state-of-the-art laboratories, AI-powered research tools, and industry collaborations to ensure practical, real-world impact. By embedding technologies such as AI, machine learning, and renewable energy into both teaching and research, the proposed university fosters a transformative academic ecosystem that equips students with skills to lead in a technologically advanced world.

INTERDISCIPLINARY AND MULTIDISCIPLINARY APPROACHES TO CURRICULUM DESIGN.

The proposed university at Tiruvannamalai demonstrates a strong potential for adopting interdisciplinary and multidisciplinary approaches to curriculum design, blending diverse fields of study to create holistic and future-ready education programs. By integrating technology, sustainability, and cultural heritage into its curriculum, the university ensures that students acquire a broad knowledge base and the ability to tackle complex challenges through collaborative problem-solving. This approach aligns with global trends in higher education, emphasizing cross-disciplinary synergy to foster innovation and societal impact.

The shortlisted programs offer immense potential for interdisciplinary and multidisciplinary approaches in curriculum design, enabling students to explore cross-cutting themes and address complex societal challenges. Courses such as **M.Tech. Sustainability in Building Maintenance** and **B.Tech. Climate Science & Engineering** can integrate engineering principles with environmental science, policy studies, and social sciences to create a comprehensive understanding of sustainable development. This would involve collaborations between civil engineering, environmental science, and urban planning faculties, fostering innovative solutions for ecological challenges.

Programs in artificial intelligence and biotechnology, such as Additionally, interdisciplinary programs like **M.Sc. Clinical Research and Health Informatics** and **M. Tech in Agro-Electronics** demonstrate the university's focus on combining advanced technology with societal applications. Health informatics integrates AI-driven data analysis for improved healthcare delivery, while agro-electronics applies smart sensors and automation to enhance agricultural productivity. These programs utilize state-of-the-art laboratories, AI-powered research tools, and industry collaborations to ensure practical, real-world impact. By embedding technologies such as AI, machine learning, and renewable energy into both teaching and research, the proposed university fosters a transformative academic ecosystem that equips students with skills to lead in a technologically advanced world.

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Programs in artificial intelligence and biotechnology, such as **M.Tech. Natural Language Processing** and **M.Sc. Clinical Research and Health Informatics (Integrated Course)**, can benefit from a multidisciplinary approach combining computer science, data analytics, and healthcare. These programs can include collaborations across domains like linguistics, bioinformatics, and health informatics, enabling the development of AI-driven solutions in clinical decision-making and personalized medicine.

Furthermore, the integration of Indian knowledge systems, as seen in programs like **B.A. Tamil Culture and Civilization** and **M.A. Thozil Sar Kalaigal**, with modern disciplines such as commerce or skill development, opens doors to innovative pedagogy. For example, combining traditional arts and language studies with financial risk management or AI applications could foster a deeper cultural appreciation while equipping students with modern skills.

One notable example of interdisciplinary is the **M. Tech in Agro-Electronics**, which combines agricultural sciences, electronics, and automation technologies. This program equips students to develop precision farming tools using smart sensors, IoT, and AI to enhance agricultural productivity. It bridges the gap between traditional farming practices and modern technological advancements, addressing the growing need for sustainable agriculture solutions.

The **M.Sc. Clinical Research and Health Informatics** program is another excellent example. This integrated course merges healthcare, biotechnology, and informatics to improve medical research and data-driven decision-making in the healthcare industry. Students learn to apply AI and data analytics to clinical trials, patient care, and health management, reflecting the multidimensional nature of healthcare challenges and solutions.

The **B.Tech in Climate Science and Engineering highlights** the synergy between environmental science, engineering, and policy studies. This program trains students to address climate challenges through technological innovations like renewable energy systems and sustainable infrastructure. It also incorporates policy and regulatory frameworks, ensuring graduates are equipped to influence decision-making at both technical and administrative levels.

Similarly, the **B.A. in Tamil Culture and Civilization** integrates humanities, archaeology, and technology, offering a unique perspective on cultural preservation. By including modern tools like digital archiving and AI-based linguistic analysis, the program ensures that traditional knowledge is preserved and revitalized for contemporary relevance. This blend of heritage and innovation provides a model for multidisciplinary education that honors the past while addressing the future.

The **PG Diploma in Solar PV and Battery Management Systems** combines renewable energy technology, electrical engineering, and business strategies. Students are trained not only in the technical aspects of solar and battery systems but also in market dynamics, financial modeling, and sustainability practices. This comprehensive curriculum prepares graduates to contribute meaningfully to the global transition to clean energy.

The research programs such as those in **Agricultural Technologies and Sustainable Technologies** can adopt an interdisciplinary framework to address pressing challenges in food security and infrastructure resilience. Collaborations across agricultural engineering, biotechnology, and environmental sustainability can yield groundbreaking innovations, enhancing the relevance and impact of academic and research activities.

The designing such interdisciplinary and multidisciplinary programs, the university ensures its curriculum is aligned with industry demands, societal needs, and global challenges. These approaches encourage creativity, collaboration, and critical thinking, equipping students with the versatile skills needed for success in a rapidly changing world. The integration of technology, culture, and sustainability into diverse domains exemplifies the university's forward-looking vision and its commitment to producing well-rounded, competent graduates.

The research programs outlined in the list present vast opportunities for interdisciplinary and multidisciplinary approaches, enabling scholars to address real-world challenges through diverse academic perspectives. For instance, the Agricultural Technologies program, which includes M.Tech. Agro-Electronics as a proposed distinct course, can merge **agricultural engineering** with electronics and data science. By incorporating precision farming technologies, IoT-enabled devices, and sustainable agricultural practices, this program can drive innovations that improve crop yield, resource management, and environmental conservation.

Similarly, the **VLSI (Very Large-Scale Integration)** research program offers a fertile ground for collaboration between artificial intelligence, electronics, and materials science. The proposed M.Tech. AI in VLSI Technology can explore AI-driven chip design, energy-efficient computing, and smart hardware solutions. By bridging disciplines like computer engineering, applied mathematics, and physics, researchers can create next-generation semiconductor technologies that address computational demands while minimizing environmental impact.

The **Health Informatics** research program exemplifies the potential for interdisciplinary integration by combining biotechnology, clinical research, and information technology. The proposed M.Sc. Clinical Research and Health Informatics course can focus on developing AI-powered diagnostic tools, predictive health analytics, and personalized medicine platforms.

collaboration across bioinformatics, computer science, and public health ensures that these research initiatives have far-reaching implications for improving healthcare systems and patient outcomes. These examples highlight the transformative potential of interdisciplinary and multidisciplinary approaches in fostering cutting-edge research and innovation.

Explanation of the "distinct" nature of the university (e.g., innovative programs, societal impact, unique research areas, integration of technology, interdisciplinary focus).

The university's distinctiveness lies in its commitment to pioneering innovative programs that address critical global and regional challenges. Programs like **M.Tech. Sustainability in Building Maintenance** and **B.Tech. Climate Science & Engineering** reflect a dedication to solving environmental and sustainability issues, ensuring graduates are equipped with the knowledge and skills to drive meaningful change. By integrating technical, ethical, and societal perspectives, the university fosters a holistic educational experience that prepares students for leadership roles in their respective fields.

In terms of societal impact, the university's emphasis on unique research areas such as **Health Informatics**, **Agricultural Technologies**, and **Sustainable Technologies** ensures that its contributions extend beyond academia. These programs are tailored to address pressing concerns like healthcare accessibility, food security, and sustainable urban development. Collaborative partnerships with industries, government bodies, and NGOs further amplify the reach and relevance of its research, translating academic insights into tangible benefits for society.

The integration of technology across disciplines is another hallmark of the university's distinct approach. With programs like M.Tech. Natural Language Processing and **M.Tech. AI in VLSI Technology**, the institution positions itself at the forefront of the technological revolution. These courses emphasize cutting-edge applications of AI, machine learning, and data analytics, empowering students and researchers to push the boundaries of innovation. This technological focus is complemented by the university's interdisciplinary ethos, fostering a culture of collaboration across diverse domains to solve complex problems and drive progress. The university wish to stands out as a beacon of innovation and societal transformation, combining academic rigor with a visionary approach to education and research. Its unique blend of innovative programs, societal relevance, and interdisciplinary focus positions it as a leader in shaping future-ready graduates and impactful solutions for the challenges of tomorrow.

3.5 CURRICULUM DETAILS ALIGNED WITH NATIONAL EDUCATION POLICY (NEP) 2020.

The curriculum designed for the stated programs (table NO.1) above are **aligned with National Education Policy (NEP) 2020**

The National Education Policy (NEP) 2020 emphasizes a holistic and multidisciplinary education system with flexible curricular structures. This document outlines the guidelines for implementing the four-year undergraduate (UG) engineering curriculum in alignment with NEP 2020, focusing on flexibility, choice, and interdisciplinary learning.

KEY FEATURES OF THE CURRICULUM

1. Multidisciplinary Approach:

The curriculum integrates major and minor subjects, vocational skill courses, and value-based education. It encourages interdisciplinary learning with options for minors across different disciplines, fostering flexibility and diversity.

2. Multiple Entry and Exit Options:

Students can exit the program with qualifications at various stages:

- Certificate after one year (40 credits)
- Diploma after two years (80 credits)
- Bachelor's degree after three years (120 credits)
- Bachelor's degree with honors or research specialization after four years (160-194 credits).

3. Credit Distribution Framework:

- Core Courses: Minimum 50% of total credits, including mandatory, elective, and vocational subjects.
- Minors: 14 credits, completed between the second and fourth years.
- Open Electives: 8 credits, promoting inter-faculty learning.
- Skill Enhancement: 8 credits, emphasizing practical and industry-relevant skills.
- Ability and Value Education: 10 credits, including Indian Knowledge Systems (IKS) and environmental studies.
- Internships and Projects: Minimum of 12 credits for internships and field-based projects, ensuring hands-on learning.

4. Research and Honor's Options: Students pursuing honors degrees can earn additional credits through research projects or specialized courses, enhancing their academic and professional profiles.

5. Online and Hybrid Learning: Up to 40% of courses can be delivered online through platforms like SWAYAM, promoting accessibility and digital integration.

6. Industry-Academia Linkages: The framework mandates internships, apprenticeships, and community engagement projects to align academic learning with practical applications, increasing employability and societal impact.

3.6 IMPLEMENTATION AND SENSITIZATION

1. Institutional Development Plans (IDPs): Institutions must outline strategies for infrastructure upgrades, faculty training, and quality enhancement to support the curriculum.

2. Stakeholder Involvement: Universities and autonomous colleges must engage stakeholders through continuous sensitization programs to ensure effective implementation.

3. Action Plan: Establishing NEP implementation cells, adopting revised curricula, and finalizing subject combinations are key milestones. A timeline ensures readiness for the academic year 2025-26.

The NEP 2020 curriculum framework aims to foster interdisciplinary thinking, critical analysis, and societal engagement, preparing students for global challenges. By integrating flexibility, choice, and a focus on practical skills, the curriculum ensures a transformative educational experience.

3.7 INNOVATIONS IN TEACHING, LEARNING, AND EVALUATION METHODS.

The NEP 2020 curriculum framework emphasizes flexibility, interdisciplinary approaches, and real-world skill development. To deliver this framework effectively, institutions must adopt innovative teaching-learning strategies and assessment methods that align with its vision. The success of NEP 2020's curriculum depends on adopting dynamic teaching-learning strategies and inclusive, transparent assessment methods. These innovations ensure students are not only academically proficient but also equipped with real-world skills, critical thinking abilities, and ethical values. By aligning pedagogy and evaluation with NEP's principles, educational institutions can transform students into well-rounded professionals ready for global challenges.

INNOVATIVE TEACHING-LEARNING STRATEGIES

- 1. Blended Learning:** Integrating online and offline learning modes, leveraging platforms like SWAYAM for digital content delivery alongside traditional classroom interactions.
- 2. Problem-Based Learning (PBL):** Encouraging students to solve real-world problems by applying multidisciplinary knowledge and critical thinking.
- 3. Flipped Classroom:** Students access pre-recorded lectures and preparatory materials before class, reserving in-person sessions for discussions, problem-solving, and hands-on activities.
- 4. Experiential Learning:** Using internships, apprenticeships, and field projects to enhance hands-on skills and understanding of theoretical concepts.
- 5. Collaborative Learning:** Team-based projects and peer-to-peer teaching to foster teamwork, communication, and shared problem-solving skills.
- 6. Integration of Indian Knowledge Systems (IKS):** Incorporating traditional Indian knowledge, ethics, and values to provide a holistic educational experience.

EXAMPLES OF TEACHING STRATEGIES:

- 1. Simulation Exercises:** Using software to simulate engineering processes or business scenarios.
- 2. Industry Collaborations:** Guest lectures, live projects, or mentorship programs with industry experts.
- 3. Virtual Labs:** Online labs to conduct experiments in subjects like engineering or science.
- 4. Hackathons and Case Studies:** Encouraging innovative problem-solving and practical application of concepts.

ASSESSMENT AND EVALUATION METHODS

1. **Continuous Assessment:** Emphasis on frequent, low-stakes assessments to track progress and provide timely feedback.
2. **Competency-Based Assessment:** Evaluating students' mastery of specific skills or competencies rather than rote memorization.
3. **Portfolio-Based Assessment:** Students compile projects, essays, and practical assignments showcasing their learning journey and achievements.
4. **Rubric-Based Grading:** Clear criteria for grading complex tasks like research projects or presentations to ensure fairness and transparency.
5. **Gamified Assessments:** Quizzes and assignments using gamification elements to make evaluations engaging and interactive.
6. **Peer and Self-Assessment:** Involving students in evaluating their work and that of their peers to build critical reflection and accountability.

EXAMPLES OF ASSESSMENT METHODS:

1. **Capstone Projects:** End-of-program projects to integrate and apply all learned skills.
2. **Performance Tasks:** Assignments requiring practical application, such as designing prototypes or conducting field surveys.
3. **Mock Industry Assignments:** Students complete tasks mirroring real-world industry scenarios.
4. **Role-Play and Case Analysis:** Students assume roles in problem-solving exercises to demonstrate understanding.

3.8 DETAILS OF OUTCOME-BASED EDUCATION (OBE) PRACTICES.

Outcome-Based Education (OBE) is a student-centric education system that emphasizes achieving specific learning outcomes by the end of a course or program. OBE focuses on the measurable knowledge, skills, attitudes, and abilities students are expected to acquire. OBE practices are centered around designing, implementing, assessing, and improving the educational process to meet predefined outcomes.

OBE aims to ensure that education systems foster learning outcomes that are directly aligned with the needs of the society, the labor market, and the individual's professional and personal development. It encourages a shift from "teaching to testing" to a more holistic approach that emphasizes the application of knowledge in real-world contexts. This approach aligns with the rapidly evolving demands of modern educational environments, ensuring students are not just well-versed in theoretical concepts but are also capable of applying their learning in practical scenarios.

KEY PRINCIPLES OF OBE

1. Clarity of Learning Outcomes: OBE starts with the clear identification of specific learning outcomes. These outcomes are statements that define what students should be able to know, understand, and do by the end of a course or program. These outcomes are categorized into three domains:

- **Cognitive (Knowledge):** Focusing on the theoretical understanding and critical thinking skills.
- **Psychomotor (Skills):** Involving physical or practical skills and techniques.
- **Affective (Attitudes and Values):** Relating to values, ethics, and professional behaviors.

2. Curriculum Design and Alignment: The curriculum is designed around the desired learning outcomes. This ensures that all course activities, assessments, and teaching methods are aligned to help students achieve these outcomes. It is important that the learning experiences, such as lectures, workshops, projects, and fieldwork, facilitate the acquisition of knowledge and skills as per the outlined objectives. Faculty members need to ensure that their teaching methodologies (lectures, practical sessions, online learning) are consistent with the intended outcomes.

3. Student-Centered Approach: OBE is centered around the student's learning process, placing them at the heart of the educational system. The role of the teacher shifts from that of a mere instructor to that of a facilitator who guides students in achieving their desired outcomes. Students take more responsibility for their learning, engage in active learning, and are encouraged to reflect on their progress throughout the course. OBE fosters self-directed learning, where students set goals, track their progress, and understand their strengths and weaknesses.

4. Continuous Assessment and Feedback: Regular assessments are designed to evaluate students' progress toward achieving the defined learning outcomes. Assessments are not limited to end-of-semester exams but are continuous and diversified to cater to different learning styles. These may include quizzes, assignments, case studies, group projects, presentations, and practical tasks. Feedback is a critical element of OBE, helping students understand their learning gaps and strengths, motivating them to improve. Assessments must be designed to measure the achievement of specific outcomes rather than focusing solely on content recall.

5. Flexibility and Inclusivity: OBE accommodates various learning paces and styles. Some students may achieve outcomes quickly, while others may need more time or alternative learning resources. OBE practices are flexible and inclusive, allowing for adjustments in teaching methods, assessments, and support services. This ensures that all students, irrespective of their learning abilities or backgrounds, have the opportunity to achieve the intended outcomes.

OBE PRACTICES IN TEACHING AND LEARNING

1. Program and Course Design: The design of a program or course in an OBE system is done with identifying the program educational objectives (PEOs) and course outcomes (Cos). These objectives and outcomes are linked to the institution's mission, vision, and the needs of the industry and society. The course structure is designed in a way that allows each student to progressively develop the required competencies, starting from foundational knowledge to more complex skills and attitudes. Course outcomes are typically mapped to program outcomes (POs), which in turn align with the overall institutional goals.

2. Pedagogical Strategies: OBE encourages the use of diverse pedagogical strategies, which include:

- **Active Learning:** Students are engaged in tasks that require them to apply concepts in real-world contexts, such as problem-solving, case analysis, and project work.
- **Collaborative Learning:** Group activities such as team projects, discussions, and peer reviews foster teamwork and communication skills, which are essential in professional environments.
- **Blended Learning:** A mix of traditional face-to-face and online learning allows students to benefit from both direct interactions with faculty and the flexibility of digital resources.
- **Flipped Classroom:** In this model, students learn the theoretical content outside the classroom (through online videos or readings), while class time is used for interactive activities, discussions, and problem-solving.

3. Assessment and Evaluation: Assessment in OBE goes beyond traditional examinations. A combination of formative and summative assessments is used to measure how well students meet the learning outcomes. Formative assessments include quizzes, assignments, and projects, while summative assessments evaluate cumulative knowledge and skills at the end of the course. Rubrics are used to clearly outline the criteria for each assessment, ensuring transparency and fairness.

- **Direct Assessment:** Involves evaluating students based on their actual performance in tasks that reflect the learning outcomes (e.g., practical exams, projects, presentations).
- **Indirect Assessment:** Includes surveys, interviews, and feedback mechanisms to assess students' perception of their learning experience, and the effectiveness of teaching strategies.

4. Reflection and Continuous Improvement: OBE is a dynamic, iterative process. Regular reflection on the teaching-learning process helps institutions identify areas for improvement. Feedback from students, faculty, and industry stakeholders is used to continuously refine the curriculum, teaching methods, and assessment strategies. This feedback loop is integral to maintaining the relevance and quality of education.

5. Mapping and Accreditation: Mapping learning outcomes to accreditation standards ensures that the program meets both national and international quality standards. OBE practices are essential for institutional accreditation, as they demonstrate that the institution is systematically working towards achieving defined outcomes, and that it continuously assesses and improves the quality of education.

The proposed university will ensure the implementation of OBE practices significantly to enhance the educational experience, ensuring that students are not just passive recipients of information, but active participants in their learning journey. The clear articulation of learning outcomes is done in the design of the curriculum. At the same time, designing relevant curricula, employing diverse teaching methodologies, and engaging in continuous assessment and feedback, OBE fosters the holistic development of students. This approach prepares them to succeed in their professional and personal lives, equipped with the knowledge, skills, and values necessary for today's dynamic world. The integration of OBE practices ensures that educational institutions are responsive to the changing demands of the global workforce and society at large.

3.10 INTERNATIONAL COLLABORATIONS OR EXCHANGE PROGRAMS.

The proposed university have plans for International collaboration and exchange programs offer students, faculty, and institutions opportunities for cross-cultural learning, academic growth, and global networking. For an institution looking to develop global partnerships and programs, various strategies can be implemented to foster mutual cooperation, improve academic standards, and prepare students for the globalized workforce. Below are several strategies for collaboration and exchange programs, followed by potential international institutions for collaboration.

The following strategies will be followed by the applicant's university.

1. Collaborative Research Programs

- **Strategy:** Partner with internationally recognized universities to undertake joint research initiatives. These programs could focus on areas of mutual interest, such as sustainable development, technology, engineering, health sciences, or social innovation.
- **Implementation:** Develop joint research centers, organize collaborative conferences and workshops, and encourage faculty and student involvement in shared research projects. Also, consider applying for international grants and funding to support joint research endeavors.
- **Examples of Collaborations:** Faculty and students from both institutions can engage in cross-border research, attend international conferences, and publish research papers together.

2. Dual Degree Programs

- **Strategy:** Establish dual degree programs with international universities, allowing students to earn degrees from both institutions in a shorter time frame. This is especially beneficial for graduate (PG) and research-level programs.
- **Implementation:** Clearly define the requirements for both institutions, including courses to be taken, credit transfer systems, and faculty responsibilities. Dual degree programs can attract international students and foster cross-cultural academic exchanges.
- **Examples of Collaborations:** A partnership in a master's or PhD program could involve students completing part of their coursework at one institution and the rest at the collaborating university.

3. Faculty Exchange Programs

- **Strategy:** Promote faculty exchanges between partner universities to encourage knowledge sharing and best practices in teaching, research, and academic management. Faculty exchanges enable educators to bring innovative ideas, research collaborations, and educational methodologies to the institutions.
- **Implementation:** Set up short-term and long-term exchange programs where faculty members teach, conduct research, and attend workshops at the partner institution. This strengthens academic relationships and enriches the educational experience for both faculty and students.
- **Examples of Collaborations:** A faculty exchange program where a professor from one institution spends a semester or year at the partner university to offer courses, mentor students, and engage in joint research projects.

4. Student Exchange and Internship Programs

- **Strategy:** Develop student exchange programs to provide undergraduate (UG), postgraduate (PG), and research students opportunities to study abroad for a semester or year. Internships with international organizations, industries, and research labs can offer hands-on experience in a global setting.
- **Implementation:** Organize structured exchange programs where students study specific courses at the partner institution. Additionally, create internship partnerships with global organizations to place students in international settings.
- **Examples of Collaborations:** A summer internship exchange program with a well-established industry partner, such as multinational companies in technology or health sectors.

5. Online and Blended Learning Partnerships

- **Strategy:** With the rise of digital learning, establish online or blended learning programs in collaboration with international universities. These programs could be in the form of MOOCs (Massive Open Online Courses), online certifications, or joint degree programs with a combination of online and in-person classes.
- **Implementation:** Partner with international universities to offer online courses or certifications that provide access to global learning opportunities. These courses can cover emerging topics like AI, Data Science, and Digital Transformation.
- **Examples of Collaborations:** Collaboration with institutions offering digital learning platforms like Coursera or edX for global access to online courses and certifications.

6. Short-Term International Programs and Workshops

- **Strategy:** Organize short-term programs, workshops, and seminars that offer students and faculty the opportunity to engage in international academic exchange. These could include summer schools, specialized workshops, and international academic competitions.
- **Implementation:** Design a schedule of thematic workshops or specialized programs on contemporary topics. These events can bring students and faculty together to collaborate on global challenges.
- **Examples of Collaborations:** Partnering with top universities and research institutes to host a “Global Innovation Workshop” or a “Sustainability Summer School” on pressing international issues.

7. International Accreditation and Recognition

Strategy: Pursue international accreditation for certain degree programs to ensure global recognition. This can help attract international students and faculty and improve the overall credibility of the institution.

- **Implementation:** Work with international accreditation bodies such as ABET (for engineering and technology programs), AACSB (for business schools), or WFME (for medical programs) to gain accreditation.
- **Examples of Collaborations:** Achieving dual accreditation with institutions in the US or Europe will boost the international reputation of your academic programs.

8. International Industry and business Partnerships

- **Strategy:** Collaborate with international industries and organizations for industry-specific academic partnerships, such as industry internships, live projects, and research collaborations.
- **Implementation:** Set up partnerships with global firms for student internships, faculty consulting, joint research, and collaborative development of innovative products and services.
- **Examples of Collaborations:** Tie-ups with companies like Microsoft, Google, or Siemens to offer collaborative internships, innovation labs, and research funding for students.

9. Cross-Cultural and Language Programs

- **Strategy:** Facilitate cross-cultural exchange programs to help students and faculty develop a global perspective and cultural competence. These can include language exchange programs, global internships, and collaborative research projects on cultural diversity.
- **Implementation:** Promote global citizenship by encouraging students to immerse themselves in different cultures, learn new languages, and develop intercultural communication skills.
- **Examples of Collaborations:** Partner with universities in non-English speaking countries like Japan, South Korea, or France to promote language and culture exchange programs.

Based on the proposed list of UG, PG, and Research programs, below are some potential international institutions that could be excellent candidates for collaboration and exchange programs:

1. Massachusetts Institute of Technology (MIT), USA

- Strong focus on engineering, technology, and entrepreneurship.
- Potential collaborations in computer science, AI, and robotics.

2. University of Cambridge, UK

- Renowned for engineering, medical, and social sciences programs.
- Collaboration in areas such as sustainability, biomedical engineering, and economics.

3. ETH Zurich, Switzerland

- Expertise in engineering, technology, and environmental sciences.
- Collaboration in research and technology innovation.

4. National University of Singapore (NUS)

- Strong programs in engineering, business, and global governance.
- Potential for joint research in smart cities, AI, and global development.

5. University of Melbourne, Australia

- Offers programs in law, engineering, business, and health sciences.
- Possible student exchange programs and research collaborations in health technology and law.

1. Tsinghua University, China

- Known for its engineering, science, and technology programs.
- Collaborative research on AI, sustainable development, and infrastructure.

2. Technical University of Munich (TUM), Germany

- Leader in technical education and innovation.
- Opportunities for research collaboration in fields like renewable energy, engineering, and digital transformation.

3. Indian Institute of Technology (IIT) Bombay, India

- Strong programs in engineering, management, and technology.
- Strategic collaborations in the field of research and development.

By pursuing these strategies and establishing collaborations with these renowned institutions, the institution shall create a robust international network that benefits both students and faculty, enhancing the overall educational experience and expanding global opportunities.

3.11. INTERDISCIPLINARY RESEARCH AND PROGRAMS

Integration of multiple disciplines to address global and regional challenges.

TABLE NO.1: UNDERGRADUATE AND POST-GRADUATE PROGRAMS

STREAM	S. No	COURSE NAME
Environment and Sustainability	1.	B.Tech. Climate Science & Engineering
	2.	M.Tech. Sustainability in Building Maintenance
Artificial Intelligence and Computers	3.	M.Tech. Natural Language Processing
	4.	M.Tech. AI in VLSI technology
Agriculture and Biotechnology	5.	M.Tech. Agro-Electronics
	6.	B.Sc. Organic Farming
	7.	M.Sc. Clinical Research and Health Informatics (Integrated Course)
Indian Knowledge System	8.	B.A. Tamil Culture and Civilization
	9.	M.A. Thozil Sar Kalaigal
Commerce and Modern Trade	10.	B.Com. Financial Risk Management and Taxation
Skill Development Courses	11.	PG Diploma Solar PV and Battery Management Systems
	12.	PG Diploma AIML in Health Care Management

Special research projects that intersect technology, health, environment, and social sciences.

TABLE NO. 2: RESEARCH PROGRAMS/DOMAINS

Sr.No	Name of the Program	Available Vertical Specialization
1.	Agricultural Technologies	M.Tech (Agro Electronics) – Proposed Distinct Course B.Tech (Agricultural Engineering) – Existing Course
2.	VLSI	M.Tech. AI in VLSI technology – Proposed Distinct Course B. Tech (AIDS) – Existing Course B.E (AIML) – Existing Course B.E (ECE) – Existing Course
3.	Computer Science and Engineering (Specialization in AI)	M.Tech. Natural Language Processing – Proposed Distinct Course M.Tech. AI in VLSI technology – Proposed Distinct Course M.E Computer Science and Engineering B. Tech (AIDS) – Existing Course B.E (AIML) – Existing Course B.E Computer Science and Engineering – Existing Course
4.	Sustainable Technologies	M.Tech. Sustainability in Building Maintenance – Proposed Distinct Course B.E. Civil Engineering – Existing Course
5.	Health Informatics	M.Sc. Clinical Research and Health Informatics (Integrated Course) – Proposed Distinct Course M.Tech Biotechnology – Existing Course B.Sc Microbiology – Existing Course B.Tech Biotechnology – Existing Course
6.	Tamil Literature and Tradition	M.A. Thozil Sar Kalaigal – Proposed Distinct Course M.A Tamil – Existing Course B.A. Tamil Culture and Civilization – Proposed Distinct Course B.A Tamil – Existing Course

3.11 PROPOSED NAME FOR THE RESEARCH CENTER:

3.11.1 “AGRITECH NEXUS: ADVANCING SUSTAINABLE FARMING INNOVATIONS”

1. Parameters Under the New Research Center Title

1. Precision Agriculture Using IoT and AI

► Scope:

- Real-time monitoring of soil and crop health.
- Optimizing irrigation and fertilization.
- Predictive analytics for pest and disease management.

► Key Technologies:

- IoT-enabled sensors.
- Drones with imaging capabilities.
- AI-powered analytics tools.
- GPS-based precision equipment.

2. PROPOSED NAME FOR THE RESEARCH CENTER:

- Enhanced crop yield.
- Reduced water, fertilizer, and pesticide usage.
- Improved profitability.

Proposed Name for the Research Center:

► Scope:

- Cultivation in limited spaces with minimal resources.
- Year-round production of pesticide-free produce.
- Reduced dependency on arable land.

► Key Technologies:

- Hydroponic systems with nutrient reservoirs.
- LED grow lights and automated control systems.

► Outcomes:

- Sustainable farming practices.
- High-quality crop output.
- Efficient resource utilization.

3. Solar-Powered Irrigation Systems

► Scope:

- Promoting renewable energy in irrigation.
- Reducing dependency on conventional energy.
- Ensuring sustainable water management.

► Key Technologies:

- Solar panels and pumps.
- Smart controllers for water distribution.

► Outcomes:

- Lower energy costs.
- Improved irrigation efficiency.
- Increased crop productivity.

4. Automated Greenhouse Systems

► Scope:

- Controlled cultivation of high-value crops.
- Climate-resilient farming with extended growing seasons.
- Promotion of export-quality produce.

► Key Technologies:

- Climate control and environmental sensors.
- Automated irrigation and misting systems.

► Outcomes:

- Consistent and higher-quality yields.
- Reduced labor and water costs.
- Increased farmer income.

5. Blockchain for Agricultural Supply Chain Management

► Scope:

- Enhancing transparency and traceability in the supply chain.
- Reducing inefficiencies and post-harvest losses.
- Enabling direct farmer-to-consumer transactions.

► Key Technologies:

- Blockchain platforms.
- IoT tracking devices.
- Mobile applications for seamless interfaces.

► Outcomes:

- Fair pricing for farmers.
- Increased consumer trust.
- Reduced supply chain fraud.

The suggested technology research proposal addresses key challenges in agriculture while promoting sustainable and innovative practices. Each technology is designed to boost productivity, improve sustainability, and empower farmers.

"Centre for AI-Driven Innovations in VLSI and Emerging Technologies"

1. Design and Optimization of Energy-Efficient VLSI Circuits for AI Applications

► Scope:

- Develop low-power architectures for AI workloads.
- Optimize circuits for real-time inference and training.
- Improve energy efficiency in edge devices for IoT and AI systems.

► Key Technologies:

- Advanced VLSI techniques, novel transistor technologies.
- AI-based optimization for hardware accelerators.

► Outcomes:

- Reduced power consumption for AI hardware.
- Improved battery life for edge devices.
- Scalable and energy-efficient solutions for AI.

2. AI-Driven VLSI Design Automation

► Scope:

- Automate VLSI design processes using AI.
- Minimize errors and reduce design lifecycle time.

► Key Technologies:

- Machine learning and reinforcement learning in design.
- AI-powered Electronic Design Automation (EDA) tools.

► Outcomes:

- Faster and more accurate design workflows.
- Reduced development costs.
- High-quality and optimized chip designs.

3. Neuromorphic Computing and Brain-Inspired VLSI Architectures

► Scope:

- Develop brain-inspired VLSI architectures for AI.
- Enhance efficiency in deep learning and cognitive computing.

► Key Technologies:

- Memristors and spiking neural network accelerators.
- Neuromorphic circuits for neuron and synapse modeling.

► Outcomes:

- Breakthroughs in low-power, high-performance AI chips.
- Advanced neuromorphic solutions for robotics and AI systems.

4. VLSI-Based Security Solutions for AI Systems

► Scope:

- Address vulnerabilities in AI systems through secure VLSI.
- Enhance robustness against hardware attacks.

► Key Technologies:

- Cryptographic hardware accelerators and Physically Unclonable Functions (PUFs).
- Secure boot architectures and tamper-proof designs.

► Outcomes:

- Increased trust in AI-powered devices.
- Improved security in defense, healthcare, and finance applications.

5. Integration of AI with VLSI for Advanced 5G and IoT Networks

► Scope:

- Develop AI-enabled VLSI solutions for 5G and IoT.
- Optimize AI algorithms for low-power chip designs.
- Key Technologies:
- AI-driven signal processing and adaptive modulation.
- Real-time analytics for communication hardware.

► Outcomes:

- High-performance 5G and IoT hardware.
- Intelligent and low-power devices for next-generation networks.

This R&D initiative aims to position the center as a leader in cutting-edge VLSI and AI technologies, supporting academic excellence, industry collaboration, and innovation.

"CENTRE FOR AI-POWERED NATURAL LANGUAGE INNOVATIONS AND APPLICATIONS"

1. Development of Multilingual Machine Translation Systems

► Scope:

- Seamless communication across diverse languages, with a focus on low-resource Indian languages.
- Enhanced accuracy and efficiency in translation systems.

► Key Technologies:

- Neural Machine Translation (NMT) techniques using transfer learning and transformer architectures.
- Pre-trained models tailored for multilingual corpora.

► Outcomes:

- High-quality translation tools for Indian languages.
- Bridging communication gaps in governance, education, and commerce.

► Requirements:

- Software: TensorFlow, PyTorch, Hugging Face Transformers.
- Hardware: High-performance GPU clusters and multilingual corpora storage.

2. Sentiment Analysis and Emotion Recognition in Regional Texts

► Scope:

- Sentiment and emotion detection in local languages.
- Tools to identify misinformation and hate speech in regional contexts.

► Key Technologies:

- Transformer-based models like BERT and annotated datasets for regional dialects.
- Natural Language Processing (NLP) libraries such as spaCy and NLTK.

► Outcomes:

- o Advanced sentiment analysis tools for businesses.
- o Social media monitoring for policy enforcement and misinformation detection.

► Requirements:

- Software: TensorFlow, NLP libraries.
- Hardware: GPUs and storage for annotated datasets.

3. Building Speech-to-Text and Text-to-Speech Systems for Regional Languages

► Scope:

- Enhancing accessibility for visually and hearing-impaired users.
- Enabling voice-based real-time interactions in local languages.

► Key Technologies:

- Deep learning models for speech processing, optimized for mobile applications.
- Advanced tools like Kaldi, Mozilla DeepSpeech, and Google TensorFlow.

► Outcomes:

- High-quality voice-enabled tools and virtual assistants.
- Accessibility improvements for education and governance.

► Requirements:

- Software: Audio processing tools like Praat.
- Hardware: GPUs and large-scale storage for speech datasets.

4. Creation of Domain-Specific Question-Answering Systems

► Scope:

- Customized question-answering systems for industries such as healthcare, education, and agriculture.
- Enhanced conversational AI for domain-specific applications.

► Key Technologies:

- Fine-tuned NLP models like GPT and BERT for domain-specific knowledge bases.
- Tools like Elasticsearch for efficient data retrieval.

► Outcomes:

- Improved customer service automation.
- Advanced information retrieval systems for specialized industries.

• Requirements:

- Software: Hugging Face Transformers.
- Hardware: GPUs and storage for large knowledge bases.

5. Development of NLP-Based Knowledge Graphs for Semantic Understanding

► Scope:

- Semantic networks for domains like healthcare and legal.
- Advanced recommendation systems and enhanced data discovery.

► Key Technologies:

- NLP techniques for unstructured data extraction and graph building.
- Tools like Neo4j and TensorFlow for knowledge graph development.

► Outcomes:

- Semantic search engines and enhanced analytics tools.
- Improved decision-making and data integration capabilities.

► Requirements:

- Software: Graph databases and spaCy.
- Hardware: GPU clusters and large-scale storage.

The R&D Center for NLP technology aims to position itself as a leader in multilingual and domain-specific AI solutions, fostering innovation and accessibility across diverse industries.

"CENTRE FOR SUSTAINABLE CONSTRUCTION TECHNOLOGIES AND CLIMATE-RESILIENT INFRASTRUCTURE"

1. Development of Green Concrete for Sustainable Infrastructure in India

► Scope:

- Utilize industrial by-products like fly ash and recycled aggregates for eco-friendly concrete.
- Address CO₂ emissions from traditional concrete production.

► Key Technologies:

- Advanced mixing and curing techniques.
- Industrial by-product utilization.

► Outcomes:

- Reduced carbon footprint in construction.
- Enhanced durability and mechanical properties of concrete.

► Laboratory Requirements:

- Universal Testing Machine (UTM), X-ray Diffraction (XRD), and SEM.

2. Energy-Efficient Building Materials for Smart Cities

► Scope:

- Develop materials with low thermal conductivity to minimize energy usage.
- Align innovations with India's smart city initiatives.

► Key Technologies:

- Thermal insulation material development.
- Simulation-based energy efficiency testing.

► Outcomes:

- Lower operational energy costs for buildings.
- Contribution to India's Net Zero Emissions goal.

► Laboratory Requirements:

- Thermal conductivity testing apparatus and environmental simulation chambers.

3. Smart Water Management Systems in Urban Infrastructure

► Scope:

- Design IoT-enabled systems for efficient urban water management.
- Prevent urban flooding and reduce water wastage.

► Key Technologies:

- IoT-based water monitoring.
- Hydraulic simulation and rainwater harvesting systems.

► Outcomes:

- Improved water management and sustainability.
- Reduction in urban flooding incidents.

► Laboratory Requirements:

- IoT development kits and hydraulic simulation setups.

4. Recycling and Reuse of Construction and Demolition Waste

► Scope:

- Process construction waste to produce cost-effective recycled materials.
- Reduce environmental pollution and landfill usage.

► Key Technologies:

- Waste sorting, crushing, and aggregate production.

► Outcomes:

- Promoted circular economy in construction.
- Cost-effective recycled building materials.

► Laboratory Requirements:

- Crushing and screening equipment, tensile testing machines.

5. Carbon Capture and Utilization Technologies in Construction

► Scope:

- Capture and utilize CO₂ emissions during cement production.
- Enhance concrete properties using CO₂ curing methods.

Key Technologies:

- CO₂ capture reactors and curing chambers.
- Gas analysis for carbon integration.

► Outcomes:

- Reduced emissions and improved concrete durability.
- Incentivized carbon capture adoption.

► **Laboratory Requirements:**

- CO₂ capture reactors and gas testing equipment.

6. Design and Development of Climate-Resilient Infrastructure

► **Scope:**

- Develop resilient construction techniques for cyclone- and flood-prone regions.
- Study soil stability and climatic patterns to create adaptive infrastructure.

► **Key Technologies:**

- Wind and storm impact simulation.
- Geotechnical testing and GIS-based planning.

► **Outcomes:**

- Improved disaster resilience and reduced recovery costs.
- Guidelines for climate-resilient policies.

► **Laboratory Requirements:**

- Flood simulation facility, geotechnical testing lab, GIS, and remote sensing systems.

Significance and Benefits

- Supports India's sustainability goals and climate resilience strategies.
- Promotes eco-friendly practices and circular economy in construction.
- Advances disaster management and adaptive infrastructure planning.
- Establishes collaboration between academia, government, and industry.

The proposal integrates emerging research areas relevant to India's construction challenges, aligning with global sustainability goals.

"CENTRE FOR ADVANCED HEALTH INFORMATICS AND TECHNOLOGY-DRIVEN HEALTHCARE SOLUTIONS"

1. AI-Powered Diagnostics and Predictive Analytics for Rural Healthcare

► **Scope:**

- Develop AI tools for early detection and diagnosis of prevalent diseases in rural areas.
- Predictive analytics for timely healthcare interventions.

► **Key Technologies:**

- Machine learning algorithms for disease prediction.
- AI-driven diagnostic tools using medical imaging data.

► **Outcomes:**

- Improved diagnostic accuracy and healthcare equity in underserved areas.
- Cost-effective and scalable diagnostic solutions.

► **Laboratory Requirements:**

- High-performance computing systems.
- Repositories for medical imaging and diagnostic data.

2. Telemedicine Solutions for Chronic Disease Management

► **Scope:**

- Build telemedicine platforms integrated with health informatics for remote monitoring.
- Support continuous management of chronic diseases like diabetes and hypertension.

► **Key Technologies:**

- IoT-enabled wearable devices.
- Cloud-based telemedicine platforms for data sharing and consultations.

► **Outcomes:**

- Reduced hospital visits and patient costs.
- Enhanced doctor-patient engagement in underserved regions.

► **Laboratory Requirements:**

- Remote monitoring systems and wearable device integration.
- Cloud computing infrastructure.

3. Blockchain for Secure Electronic Health Records (EHR) Management

► **Scope:**

- Design blockchain systems for secure and interoperable electronic health records.
- Ensure compliance with data security and privacy regulations in India.

► **Key Technologies:**

- Decentralized ledgers for secure health data sharing.
- Encryption tools for patient confidentiality.

► **Outcomes:**

- Enhanced trust in digital health systems.
- Seamless data sharing across healthcare providers.

► **Laboratory Requirements:**

- Blockchain development tools and secure servers.
- Software for encryption and data validation.

4. Personalized Medicine Using Genomic Data Analytics

► **Scope:**

- Utilize genomic data to provide tailored treatments for diseases like cancer and genetic disorders.
- Develop bioinformatics platforms for analyzing sequencing data.

► **Key Technologies:**

- Next-generation sequencing (NGS) machines.
- AI models for genomic data analysis and treatment optimization.

► **Outcomes:**

- Precision in treatment plans and improved patient outcomes.
- Reduction in treatment side effects and costs.

► **Laboratory Requirements:**

- NGS machines, bioinformatics software.
- High-capacity data storage and processing systems.

5. Health Informatics for Epidemic and Pandemic Preparedness

► **Scope:**

- Build real-time data collection and visualization systems for outbreak prediction and management.
- Leverage big data analytics to guide decision-making during health crises.

► **Key Technologies:**

- Real-time data dashboards and epidemiological models.
- Big data platforms for health surveillance and resource allocation.

► **Outcomes:**

- Early detection and management of outbreaks.
- Enhanced preparedness and resource allocation during health crises.

► **Laboratory Requirements:**

- Epidemiological data collection tools and big data platforms.
- Dashboards for visualization and real-time reporting.

Significance and Benefits

- Revolutionizes healthcare delivery through innovative technology solutions.
- Bridges the gap in rural and underserved healthcare access.
- Strengthens India's capabilities in personalized medicine and epidemic management.
- Enhances patient trust and security in digital health systems.

"CENTRE FOR TAMIL LITERATURE, ARTS, AND CULTURAL HERITAGE STUDIES"

1. Decoding Ancient Tamil Manuscripts and Inscriptions: Unveiling Historical Narratives

► **Scope of Research:**

- Digitize, translate, and interpret ancient Tamil manuscripts and inscriptions.
- Derive historical, cultural, and literary insights.

► **Outcomes:**

- Digital repository of translated manuscripts and inscriptions.
- Insights into Tamil Nadu's socio-economic and cultural history.
- Creation of educational materials for academic use.

2. Exploring Sangam Literature's Socio-Economic and Ethical Paradigms

► **Scope of Research:**

- Analyze Sangam literature for its socio-economic structures, ethical codes, and governance models.

► **Outcomes:**

- Research on socio-economic insights from Sangam literature.
- Policy recommendations inspired by ancient Tamil ethics.
- Enhanced understanding of Tamil Nadu's intellectual heritage.

3. Reviving Tamil Folk Arts: Documentation and Socio-Cultural Impact

► **Scope of Research:**

- Document and analyze Tamil folk arts for historical relevance & cultural preservation.
- Propose strategies for their revival.

► **Outcomes:**

- Digital archive of folk arts and performances.
- Strategies for preserving and revitalizing folk traditions.
- Boost to cultural tourism and local economies.

4. Contribution of Tamil Literature to Feminist Thought and Gender Studies

► **Scope of Research:**

- Study feminist themes and gender dynamics in classical and modern Tamil literature.

► **Outcomes:**

- Academic papers on gender studies in Tamil literature.
- Integration of Tamil feminist perspectives into global discourse.
- Awareness campaigns on women's contributions to Tamil literature.

5. Tamil Contributions to Global Maritime Trade and Cultural Exchange

► **Scope of Research:**

- Study Tamil Nadu's role in historical maritime trade and cultural exchanges.

► **Outcomes:**

- Detailed analysis of Tamil Nadu's maritime history.
- Enhanced understanding of global cultural exchanges.
- Resources for academic and cultural institutions.

6. Ethnobotany in Tamil Literature: Insights into Traditional Medicine and Agriculture

► **Scope of Research:**

- Document references to traditional medicine, agriculture, and ethnobotany in Tamil literature.

► **Outcomes:**

- Comprehensive repository of ethnobotanical knowledge.
- Integration of traditional practices into modern healthcare and agriculture.
- Sustainable development models based on Tamil literature.

7. Tamil Music and Poetry: Evolution and Global Influence

► **Scope of Research:**

- Study the evolution of Tamil music and poetry and their global influence.

► **Outcomes:**

- Research papers and exhibitions showcasing Tamil music and poetry.
- Increased awareness of Tamil Nadu's cultural heritage.
- Enhanced cultural tourism and international collaborations.

SIGNIFICANCE AND BENEFITS

- Preserves and promotes Tamil heritage and knowledge systems.
- Enhances academic and cultural understanding of Tamil literature.
- Contributes to socio-economic development through cultural tourism and policy frameworks.
- Promotes Tamil Nadu as a hub for global cultural and historical research. This R&D centre will serve as a hub for scholars, policymakers, and cultural enthusiasts, ensuring Tamil Nadu's rich tradition and knowledge system continue to thrive and inspire future generations.

PROJECT-WISE COSTING

PROPOSED NAME OF RESEARCH CENTRE	RESEARCH PROJECT	Cost (INR Crores)
AgriTech Nexus: Advancing Sustainable Farming Innovations	Precision Agriculture Using IoT and AI	1.8
	Hydroponics and Vertical Farming	2.2
	Solar-Powered Irrigation Systems	1.4
	Automated Greenhouse Systems	2.5
	Blockchain for Agricultural Supply Chain Management	1.0
Centre for AI-Driven Innovations in VLSI and Emerging Technologies	Energy-Efficient VLSI Circuits	3.0
	AI-Driven VLSI Design Automation	2.5
	Neuromorphic Computing	3.5
	Security Solutions for AI Systems	2.8
	Integration of AI with 5G/IoT Networks	4.0
Centre for AI-Powered Natural Language Innovations and Applications	Multilingual Machine Translation Systems	3.2
	Sentiment Analysis & Emotion Recognition	2.5
	Speech-to-Text & Text-to-Speech Systems	3.5
	Domain-Specific Question-Answering Systems	3.0
	NLP-Based Knowledge Graphs	3.5
Centre for Sustainable Construction Technologies and Climate-Resilient Infrastructure	Green Concrete	0.35
	Energy-Efficient Materials	0.3
	Smart Water Management	0.2
	Recycling Construction Waste	0.25
	Carbon Capture Technologies	0.4
	Climate-Resilient Infrastructure	0.8
Centre for Advanced Health Informatics and Technology-Driven Healthcare Solutions	AI-Powered Diagnostics	0.35
	Telemedicine Solutions	0.3
	Blockchain for Secure EHR	0.3
	Personalized Medicine with Genomic Analytics	0.5
	Health Informatics for Epidemic Preparedness	0.4
Centre for Tamil Literature, Arts, and Cultural Heritage Studies	Tamil Manuscripts & Inscriptions	0.15
	Sangam Literature's Socio-Economic Paradigms	0.07
	Reviving Tamil Folk Arts	0.1
	Tamil Literature & Feminist Thought	0.08
	Tamil Maritime Trade Contributions	0.12
	Ethnobotany in Tamil Literature	0.18
	Tamil Music & Poetry Evolution	0.1
	TOTAL	35.00

3.12. PROVISIONS FOR MULTIDISCIPLINARY EDUCATION

The proposed university at Tiruvannamalai demonstrates a strong potential for adopting interdisciplinary and multidisciplinary approaches to curriculum design, blending diverse fields of study to create holistic and future-ready education programs. By integrating technology, sustainability, and cultural heritage into its curriculum, the university ensures that students acquire a broad knowledge base and the ability to tackle complex challenges through collaborative problem-solving. This approach aligns with global trends in higher education, emphasizing cross-disciplinary synergy to foster innovation and societal impact.

The shortlisted programs offer immense potential for interdisciplinary and multidisciplinary approaches in curriculum design, enabling students to explore cross-cutting themes and address complex societal challenges. Courses such as **M.Tech. Sustainability in Building Maintenance** and **B.Tech. Climate Science & Engineering** can integrate engineering principles with environmental science, policy studies, and social sciences to create a comprehensive understanding of sustainable development. This would involve collaborations between civil engineering, environmental science, and urban planning faculties, fostering innovative solutions for ecological challenges.

Programs in artificial intelligence and biotechnology, such as **M.Tech. Natural Language Processing** and **M.Sc. Clinical Research and Health Informatics (Integrated Course)**, can benefit from a multidisciplinary approach combining computer science, data analytics, and healthcare. These programs can include collaborations across domains like linguistics, bioinformatics, and health informatics, enabling the development of AI-driven solutions in clinical decision-making and personalized medicine.

Furthermore, the integration of Indian knowledge systems, as seen in programs like **B.A. Tamil Culture and Civilization** and **M.A. Thozil Sar Kalaigal**, with modern disciplines such as commerce or skill development, opens doors to innovative pedagogy. For example, combining traditional arts and language studies with financial risk management or AI applications could foster a deeper cultural appreciation while equipping students with modern skills.

One notable example of interdisciplinary is the **M. Tech in Agro-Electronics**, which combines agricultural sciences, electronics, and automation technologies. This program equips students to develop precision farming tools using smart sensors, IoT, and AI to enhance agricultural productivity. It bridges the gap between traditional farming practices and modern technological advancements, addressing the growing need for sustainable agriculture solutions.

The **M.Sc. Clinical Research and Health Informatics** program is another excellent example. This integrated course merges healthcare, biotechnology, and informatics to improve medical research and data-driven decision-making in the healthcare industry. Students learn to apply AI and data analytics to clinical trials, patient care, and health management, reflecting the multidimensional nature of healthcare challenges and solutions.

The **B. Tech in Climate Science and Engineering** highlights the synergy between environmental science, engineering, and policy studies. This program trains students to address climate challenges through technological innovations like renewable energy systems and sustainable infrastructure. It also incorporates policy and regulatory frameworks, ensuring graduates are equipped to influence decision-making at both technical and administrative levels.

Similarly, the **B.A. in Tamil Culture and Civilization** integrates humanities, archaeology, and technology, offering a unique perspective on cultural preservation. By including modern tools like digital archiving and AI-based linguistic analysis, the program ensures that traditional knowledge is preserved and revitalized for contemporary relevance. This blend of heritage and innovation provides a model for multidisciplinary education that honors the past while addressing the future.

The **PG Diploma in Solar PV and Battery Management Systems** combines renewable energy technology, electrical engineering, and business strategies. Students are trained not only in the technical aspects of solar and battery systems but also in market dynamics, financial modeling, and sustainability practices. This comprehensive curriculum prepares graduates to contribute meaningfully to the global transition to clean energy.

Integration of Arts, Humanities, and Sciences in the Curriculum

The integration of arts, humanities, and sciences within a university curriculum fosters a holistic approach to education, ensuring that students develop a well-rounded perspective that combines technical expertise with critical thinking, creativity, and ethical reasoning. By blending these disciplines, the curriculum prepares students not only for professional success but also for active and informed participation in societal issues. This approach emphasizes the interconnectedness of knowledge and the importance of interdisciplinary collaboration in addressing real-world challenges.

Few examples illustrating the integration of arts, humanities, and sciences within the curriculum are stated here.

1. B.A. Tamil Culture and Civilization with Digital Archiving and AI-Based Linguistic Analysis

► **Disciplines Integrated:** Humanities (Tamil culture, language), Arts (cultural preservation), and Technology (digital archiving, AI).

► **Description:** This program combines traditional studies in Tamil culture and civilization with modern technological approaches such as AI-based linguistic analysis and digital archiving. Students learn about the rich heritage of Tamil culture while using contemporary digital tools to preserve and revitalize ancient texts, artifacts, and linguistic features. This integration enhances the relevance of cultural studies by connecting the past with future technologies.

► **Outcome:** Students develop skills in both cultural preservation and modern technology, making them well-equipped to work in fields related to cultural heritage management, digital preservation, and even AI applications in the humanities.

2. M.Tech. Sustainability in Building Maintenance with Environmental Science and Social Sciences

► **Disciplines Integrated:** Engineering (building maintenance), Environmental Science, and Social Sciences (policy studies, community development).

► **Description:** This program integrates principles from civil engineering, environmental science, and social sciences to address sustainability in building maintenance. Students explore eco-friendly building materials, energy-efficient technologies, and urban planning, while also studying the social and policy aspects of sustainable development. The inclusion of social sciences ensures that graduates understand the human impact of environmental decisions and can advocate for policies that promote both ecological and social sustainability.

► **Outcome:** Graduates are capable of designing and maintaining sustainable buildings while considering the social and environmental implications of their work, making them leaders in the green building and urban development sectors.

3. M.Sc. Clinical Research and Health Informatics with Biotechnology and Data Science

► **Disciplines Integrated:** Healthcare (clinical research), Biotechnology, Data Science, and Computer Science (health informatics, AI).

► **Description:** This integrated course brings together clinical research, biotechnology, and health informatics to improve decision-making in healthcare. Students gain knowledge in biotechnology and data analytics, with a focus on applying AI and data-driven solutions in clinical settings. For example, they may work on AI algorithms for personalized medicine, predictive healthcare analytics, or developing better data systems for clinical trials.

► **Outcome:** This program equips students to tackle complex healthcare challenges using both the scientific understanding of biology and the technical capabilities of data science and informatics.

4. B.Tech. Climate Science and Engineering with Environmental Engineering and Policy Studies

► **Disciplines Integrated:** Engineering (climate science, renewable energy), Environmental Science, and Policy Studies.

► **Description:** The B.Tech. program in Climate Science and Engineering merges the scientific study of climate change with the technological innovations needed to address it. Students engage with engineering solutions such as renewable energy technologies, energy-efficient building designs, and climate modeling. In addition, they study policy frameworks that can influence the development of sustainable infrastructure and climate change mitigation strategies, integrating science, engineering, and social responsibility.

► **Outcome:** Graduates are prepared to work in the development of sustainable infrastructure, renewable energy systems, and in policy-making for climate resilience and environmental justice.

5. PG Diploma in Solar PV and Battery Management Systems with Electrical Engineering and Business Strategies

► **Disciplines Integrated:** Engineering (solar PV and battery systems), Business (market dynamics, financial modeling), and Environmental Science (sustainability).

► **Description:** This program offers a unique fusion of electrical engineering, business strategy, and sustainability. Students are taught the technical aspects of solar energy systems and

battery technologies, while also gaining an understanding of market trends, financial models, and business strategies. This ensures that they are not only capable of designing and implementing renewable energy solutions but also have the expertise to navigate the economic and business aspects of the clean energy industry.

► **Outcome:** Graduates are well-equipped to drive the global transition to renewable energy, both in technical and entrepreneurial roles, contributing to the clean energy movement from multiple angles.

The integration of arts, humanities, and sciences in the curriculum allows students to approach complex problems from multiple perspectives, enriching their learning experience and fostering innovation. These interdisciplinary programs create well-rounded individuals who are not only experts in their fields but also socially and ethically conscious, prepared to contribute to global challenges in innovative and thoughtful ways. By offering such programs, universities can cultivate the next generation of leaders who understand the interconnectedness of disciplines and the importance of a holistic approach to problem-solving.

3.13 Plans for implementing interdisciplinary and multidisciplinary programs aligned with NEP 2020.

1. Curriculum Integration and Flexibility

► Design curricula that facilitate the integration of multiple disciplines, enabling students to choose courses across various fields (e.g., engineering, humanities, social sciences, and sciences). The programs will be structured to provide flexibility in course selection, allowing students to create a personalized academic journey aligned with their interests and career aspirations.

2. Collaborative Teaching and Learning

► Encourage collaboration between faculty members from different departments to design and teach interdisciplinary courses. Cross-departmental teams will be formed to create programs that blend theory and practical applications from multiple disciplines, such as AI in healthcare, sustainability in engineering, or digital humanities.

3. Project-Based Learning and Real-World Problem Solving

► Implement project-based learning where students work on real-world challenges that require interdisciplinary approaches. For example, projects on climate change can involve engineering, environmental science, policy studies, and social sciences. Students will work in teams to provide holistic, innovative solutions to these challenges.

4. Flexible Degree Programs and Credit Transfer System

► Introduce flexible undergraduate and postgraduate degree programs where students can earn credits from various departments. These degrees could include a combination of disciplines like B.Tech. in Climate Science and Engineering or M.Tech. in Sustainability, integrating courses from engineering, environmental science, and policy studies. Establish a credit transfer system for students to switch between programs and disciplines within the university.

5. Incorporation of Technology and Emerging Fields

► Promote the incorporation of technology and emerging fields, such as Artificial Intelligence, Machine Learning, Biotechnology, and Data Science, into traditional programs. Students will

be exposed to technical and non-technical subjects to encourage holistic learning and innovation. For instance, incorporating AI into healthcare programs for personalized medicine and digital healthcare solutions.

6. Interdisciplinary Research Centers

► Establish interdisciplinary research centers that focus on pressing global issues like climate change, healthcare, sustainable agriculture, and digital humanities. These centers will promote collaborative research among students, faculty, and industry experts, fostering innovation and knowledge generation that spans multiple fields.

7. Industry and Global Collaborations

► Form strategic partnerships with industry leaders, academic institutions, and research organizations globally. This will enable students and faculty to engage in joint research, exchange programs, and internships that are interdisciplinary. Collaborations will help bridge the gap between theory and practice while providing a global perspective on local challenges.

8. Cross-Disciplinary Skill Development

► Offer workshops, seminars, and short-term courses that focus on skill development across disciplines. For example, a course on "Ethics in Technology" could be a collaboration between philosophy and computer science departments. This will enhance students' capabilities to work in diverse, multidisciplinary teams and address multifaceted challenges in the workplace.

9. Integration of Indian Knowledge Systems

► Infuse Indian traditional knowledge systems with modern disciplines, especially in programs like cultural studies, medicine, and environmental sciences. For example, integrating traditional agricultural practices with modern technology in agro-sciences, or exploring Ayurveda alongside biotechnology and health informatics, will provide students with unique insights into sustainable practices rooted in India's cultural heritage.

10. Promoting Holistic Development through Multidisciplinary Programs

► Ensure that interdisciplinary programs support not only academic excellence but also holistic personal development. These programs will integrate soft skills, leadership, ethics, and social responsibility, helping students understand the broader impact of their work on society. A focus on emotional intelligence and ethical decision-making will be incorporated across all programs.

11. Encouraging Student Mobility and Exchange Programs

► Establish student mobility programs that allow students to take courses or participate in research projects at partner institutions, both domestically and internationally. These exchange programs will provide exposure to different disciplines, perspectives, and global challenges, further enriching their educational experience. Institutions will also collaborate on joint degree programs that encourage cross-border interdisciplinary learning.

These plans, aligned with the National Education Policy (NEP) 2020, aim to foster an inclusive, flexible, and holistic approach to education that equips students with the skills, knowledge, and interdisciplinary perspective required to thrive in a rapidly evolving world.

3.14. EVALUATION AND CONTINUOUS IMPROVEMENT

Institutional mechanisms for self-evaluation and quality enhancement and Examples of corrective measures taken from previous assessments.

Assessment and Evaluation Methods

- 1. Continuous Assessment:** Emphasis on frequent, low-stakes assessments to track progress and provide timely feedback.
- 2. Competency-Based Assessment:** Evaluating students' mastery of specific skills or competencies rather than rote memorization.
- 3. Portfolio-Based Assessment:** Students compile projects, essays, and practical assignments showcasing their learning journey and achievements.
- 4. Rubric-Based Grading:** Clear criteria for grading complex tasks like research projects or presentations to ensure fairness and transparency.
- 5. Gamified Assessments:** Quizzes and assignments using gamification elements to make evaluations engaging and interactive.
- 6. Peer and Self-Assessment:** Involving students in evaluating their work and that of their peers to build critical reflection and accountability.

EXAMPLES OF ASSESSMENT METHODS:

- 1. Capstone Projects:** End-of-program projects to integrate and apply all learned skills.
- 2. Performance Tasks:** Assignments requiring practical application, such as designing prototypes or conducting field surveys.
- 3. Mock Industry Assignments:** Students complete tasks mirroring real-world industry scenarios.
- 4. Role-Play and Case Analysis:** Students assume roles in problem-solving exercises to demonstrate understanding.

3.15. Compliance with the NEP 2020

Steps taken to implement NEP guidelines
(e.g., multidisciplinary education, academic credit banks, holistic development).

1. Curriculum Restructuring for Multidisciplinary Education

► **Action:** Restructure the curriculum to offer multidisciplinary and interdisciplinary courses, allowing students to select subjects from diverse fields such as humanities, engineering, social sciences, and natural sciences. This ensures that students can pursue knowledge from various disciplines, promoting cross-disciplinary learning.

► **Outcome:** A flexible and dynamic curriculum that encourages students to combine their primary field of study with elective courses from other departments, fostering a well-rounded educational experience.

2. Introduction of the Academic Credit Bank System (ACBS)

► **Action:** Implement an Academic Credit Bank that allows students to accumulate and transfer academic credits across different institutions and programs. This system provides greater flexibility for students to pursue education at their own pace and take courses from various disciplines without the constraints of traditional degree structures.

► **Outcome:** Increased flexibility and mobility for students, enabling them to enhance their skills, switch between programs, and accumulate credits at multiple institutions, thus promoting lifelong learning.

3. Focus on Skill Development and Vocational Education

► **Action:** Integrate skill development programs, vocational training, and industry-specific certifications within the academic curriculum. Offer courses that develop both technical and soft skills such as communication, leadership, and ethical decision-making, ensuring that students are workforce-ready.

► **Outcome:** Graduates with a combination of academic knowledge and practical, marketable skills, making them more employable and adaptable to industry demands.

4. Promotion of Holistic and Sustainable Development

► **Action:** Encourage programs that focus not only on academic excellence but also on the personal development of students. Introduce courses on ethics, mental health, sustainability, and social responsibility, which contribute to the overall growth of the individual and their contribution to society.

► **Outcome:** A generation of students who are not only academically skilled but also socially conscious, responsible citizens with a balanced and sustainable approach to life and work.

5. Emphasis on Technology Integration

► **Action:** Introduce courses and programs focused on emerging technologies like Artificial Intelligence, Data Science, Biotechnology, and Renewable Energy, ensuring that students acquire skills relevant to the future job market. Additionally, embed digital literacy and online learning tools across all disciplines to enhance teaching and learning.

► **Outcome:** A future-ready workforce that is proficient in modern technologies and able to solve complex, global challenges using cutting-edge tools and techniques.

6. Establishment of Multidisciplinary Research and Innovation Centers

► **Action:** Set up research centers that encourage collaboration between different faculties (e.g., engineering, arts, and sciences) to work on interdisciplinary projects. These centers focus on solving societal problems, such as climate change, healthcare innovation, and sustainable development, through cross-disciplinary research.

► **Outcome:** Breakthrough innovations and solutions derived from collaborative efforts that address pressing issues while fostering a culture of research excellence.

7. Student-Centric Pedagogy and Active Learning Strategies

► **Action:** Transition from traditional lecture-based learning to more student-centric approaches such as project-based learning, flipped classrooms, and experiential learning. Promote collaborative activities, real-world problem-solving, and peer-to-peer learning to foster deeper engagement with the subject matter.

► **Outcome:** Enhanced student engagement, critical thinking, and collaborative skills that are crucial for solving complex problems in interdisciplinary settings.

8. Encouraging Student Mobility and International Exchange Programs

► **Action:** Facilitate student exchange programs with national and international universities to allow students to experience different educational systems and cultural contexts. Encourage the development of joint degree programs with partner institutions globally.

► **Outcome:** Greater exposure to diverse academic methodologies and global perspectives, enriching students' educational journeys and broadening their horizons.

9. Faculty Development and Interdisciplinary Training

► **Action:** Provide training and professional development opportunities for faculty members to enhance their expertise in interdisciplinary teaching and research. Encourage faculty from

different departments to collaborate on curriculum design, research projects, and cross-disciplinary workshops.

► **Outcome:** A highly skilled faculty that can teach across disciplines, fostering an environment of knowledge-sharing and innovative teaching practices.

10. Incorporation of Indian Knowledge Systems (IKS)

► **Action:** Introduce courses that integrate Indian traditional knowledge with modern disciplines. For example, courses on Ayurveda, Indian philosophy, and ancient sciences can be combined with modern health sciences, biotechnology, and environmental studies.

► **Outcome:** Students gain a broader perspective by understanding the value of Indian knowledge systems alongside global scientific and technological advancements.

11. Holistic Student Support Systems

► **Action:** Establish comprehensive student support systems that include career counseling, mental health services, academic advising, and extracurricular activities. This ensures that students receive a well-rounded educational experience, promoting their intellectual, emotional, and social well-being.

► **Outcome:** A nurturing environment that helps students balance academic pressure with personal development, leading to better overall performance and mental health.

The above steps are incorporated in the strategic plan of the institutions which can align the educational programs with the guidelines of NEP 2020, creating a more flexible, inclusive, and holistic learning environment. This will empower students to develop into well-rounded individuals with the skills and knowledge to tackle complex global challenges in an increasingly interconnected world.

3.16 PATHWAYS FOR FLEXIBLE AND LIFE-LONG LEARNING OPPORTUNITIES.

The National Education Policy (NEP) 2020 envisions a holistic and inclusive education system, emphasizing flexibility, lifelong learning, and accessibility for all. The policy emphasizes providing learners with opportunities to continuously upgrade their skills, knowledge, and competencies throughout their lives. Below are key pathways to facilitate flexible and lifelong learning opportunities:

1. Credit-Based System and Academic Credit Bank (ACB)

► **Pathway:** NEP 2020 proposes the introduction of a Credit-Based System, including an Academic Credit Bank (ACB), where students can accumulate and transfer academic credits over time, even from different institutions. Students can earn credits from various learning experiences, such as formal education, MOOCs (Massive Open Online Courses), industry certifications, and work experience.

► **Impact:** This enables learners to take courses from different fields and institutions, giving them the flexibility to pursue education at their own pace and from various providers, without being tied to a specific location or program. The system supports modular learning, allowing students to exit and re-enter education at different points, ensuring continuity of learning.

2. Multiple Entry and Exit Points

► **Pathway:** NEP 2020 advocates for a flexible academic structure with multiple entry and exit points in degree programs. For instance, students can exit after completing one year with a Certificate, after two years with a Diploma, or after three or four years with a Degree. Further, those wishing to return to complete their studies can do so without losing their previous credits.

► **Impact:** This system empowers learners to make informed decisions based on their personal, professional, or financial situations. They can return to complete their education at a later stage in life, accommodating those balancing work and family commitments, thus supporting lifelong learning.

3. Blended and Online Learning Opportunities

► **Pathway:** The policy strongly promotes online education, blended learning, and the use of technology to enhance the accessibility of education. The introduction of digital platforms, like SWAYAM, enables learners to access high-quality courses across various disciplines, anytime and anywhere. Institutions will offer more hybrid and fully online courses to cater to diverse learning needs.

► **Impact:** This pathway provides opportunities for individuals to continue learning without the constraint of geographic location or time, creating opportunities for people in rural areas or working professionals to enhance their skills and knowledge.

4. Lifelong Learning Centers and Community-Based Education

► **Pathway:** NEP 2020 emphasizes the establishment of Lifelong Learning Centers in universities, colleges, and community spaces to offer continuous learning opportunities. These centers will provide education for all age groups, from school dropouts to working professionals, fostering the development of vocational skills, entrepreneurial skills, and personal growth.

► **Impact:** These centers will cater to learners of all ages, enabling people to acquire new skills or upgrade existing ones throughout their lives. It creates pathways for individuals to pursue education beyond traditional schooling and formal university degrees, promoting social mobility and individual empowerment.

5. Integration of Vocational Education

► **Pathway:** NEP 2020 advocates for vocational education to be integrated into mainstream education from the school level, ensuring students have access to skill-based learning alongside academic learning. By the 2030s, every student should have the opportunity to develop vocational skills as part of their formal education.

► **Impact:** This approach offers practical skills training that makes learning relevant to the workforce. Students can pursue lifelong learning through vocational programs that enhance their employability, ensure career growth, and adapt to emerging market needs, such as digital skills or green technologies.

6. Recognition of Prior Learning (RPL) and Prior Work Experience

► **Pathway:** Recognition of Prior Learning (RPL) will be used to recognize the skills, knowledge, and experience gained through informal learning or work. This allows individuals to get academic credit for skills and knowledge they have acquired outside formal educational systems, such as through work, volunteering, or self-learning.

► **Impact:** This pathway opens up opportunities for adult learners and professionals to formalize their competencies and gain recognition for their experience, thus enhancing career prospects, career advancement, and fostering lifelong learning.

7. Micro-Credentials and Modular Learning

► **Pathway:** NEP 2020 encourages the development of micro-credentials and modular learning, which are small units of learning that focus on specific skills or knowledge areas. These can be pursued independently and stacked together over time to earn a larger credential, such as a certificate or degree.

► **Impact:** This approach allows learners to acquire focused, bite-sized knowledge relevant to their careers and industries, making education more flexible and aligned with evolving industry needs. Micro-credentials will allow learners to continuously update and diversify their skills in a rapidly changing job market.

8. Credit Transfer Between Formal and Non-Formal Education Systems

► **Pathway:** The establishment of mechanisms for credit transfer between formal education institutions and non-formal learning platforms will facilitate lifelong learning. For example, credits earned from online courses, industry certifications, and community-based education can be transferred and recognized by formal education institutions.

► **Impact:** This system provides a seamless pathway for individuals to continue learning and upgrading their qualifications, irrespective of whether they choose formal education routes or non-formal, skill-based learning options.

9. Work-Integrated Learning and Industry Linkages

► **Pathway:** Work-integrated learning (WIL) is strongly encouraged by NEP 2020. This involves internships, apprenticeships, and industry-based projects, where students and professionals can gain practical experience alongside their academic education.

► **Impact:** WIL programs foster hands-on learning, helping individuals acquire real-world skills, while making them more competitive in the job market. These experiences, combined with academic learning, enhance career outcomes and prepare individuals for the future of work.

10. Personalized Learning Pathways

► **Pathway:** NEP 2020 supports personalized learning pathways, where learners can tailor their education to their interests, learning pace, and career goals. The flexibility to choose between a wide range of courses and interdisciplinary subjects allows students to chart their own educational journey.

► **Impact:** Personalized learning ensures that learners have control over their education and are motivated to continue learning throughout their lives. It allows individuals to engage in continuous self-improvement, making them adaptable to changing personal or professional circumstances.

11. Collaborations with International Institutions for Lifelong Learning

► **Pathway:** NEP 2020 encourages the establishment of international collaborations for lifelong learning, enabling learners to engage in global learning opportunities. Partnerships with foreign universities, online platforms, and industry experts will facilitate exposure to different educational systems, knowledge, and innovations.

► **Impact:** This promotes the global mobility of learners, helping them expand their skills and perspectives through international exposure. It enhances cultural awareness and prepares learners to function in a globalized world.

The pathways for flexible and lifelong learning in NEP 2020 ensure that individuals can continuously evolve throughout their lives, adapting to new challenges, skills, and knowledge needs. By integrating technology, enhancing accessibility, and recognizing informal learning, NEP 2020 empowers individuals to pursue education at any stage of life, helping them stay relevant in an ever-changing world.

3.17 POLICY FOR COURSE CORRECTIONS AND UPDATES BASED ON FEEDBACK

1. Establishing Feedback Mechanisms

- ▶ Implement multiple channels for collecting feedback from students, alumni, faculty, industry partners, and accreditation bodies.
- ▶ Use tools like online surveys, focus group discussions, course evaluations, and alumni interactions.

2. Formation of a Curriculum Review Committee

- ▶ Constitute a standing committee responsible for reviewing feedback, analyzing curriculum gaps, and recommending changes.
- ▶ Include representatives from academia, industry, and student bodies to ensure diverse perspectives.

3. Annual Academic/Curriculum Audit

- ▶ Conduct an annual review of all courses to assess alignment with program objectives, emerging technologies, and industry requirements.
- ▶ Use feedback as a primary input for the audit process.

4. Structured Feedback Integration

- ▶ Establish clear guidelines to integrate feedback into course corrections, prioritizing inputs based on relevance and impact.
- ▶ Create a feedback mapping framework to link feedback directly with course objectives and program outcomes.

5. Continuous Update Policy

- ▶ Mandate biannual updates to the curriculum for minor changes (e.g., adding a new topic) and major revisions every two years.
- ▶ Incorporate changes in consultation with academic boards and regulatory bodies.

6. Transparent Approval Process

- ▶ Define a streamlined process for approval of course updates, involving department heads, curriculum review committees, and academic councils.
- ▶ Publish changes with rationale and expected outcomes for transparency.

7. Monitoring and Assessment

- ▶ Regularly monitor the impact of implemented course corrections through student performance, placement records, and stakeholder satisfaction.
- ▶ Use data-driven insights to refine the process further.

8. Faculty Training and Orientation

- ▶ Organize workshops and training sessions to familiarize faculty with new course content and pedagogical approaches.
- ▶ Provide resources and support to ensure effective delivery of updated courses.

9. Documentation and Communication

- ▶ Maintain comprehensive records of feedback, proposed changes, approvals, and implementation timelines.
- ▶ Communicate updates effectively to all stakeholders through institutional portals, newsletters, and meetings.

EXAMPLES OF COURSE CORRECTIONS/UPDATES BASED ON FEEDBACK.

1. Enhanced Practical Learning Opportunities

- ▶ **Feedback:** Students and faculty observed a gap between theoretical knowledge and real-world applications.
- ▶ **Correction:** Introduction of additional lab sessions, workshops, and real-life project-based assignments in core subjects like Strength of Materials and Digital Signal Processing to bridge the gap.
- ▶ **Policy Update:** Mandating at least two industry-sponsored mini-projects before graduation.

2. Interdisciplinary Electives

- ▶ **Feedback:** Students expressed interest in interdisciplinary fields like AI in civil engineering or IoT applications in mechanical systems.
- ▶ **Correction:** Introduction of electives such as Artificial Intelligence for Smart Infrastructure and IoT for Mechanical Automation.
- ▶ **Policy Update:** Offering a flexible credit system allowing students to choose courses from other departments.

3. Inclusion of Emerging Technologies

- ▶ **Feedback:** Alumni highlighted the lack of exposure to contemporary technologies such as block chain and robotics.
- ▶ **Correction:** Adding new courses like Block chain in Engineering and Introduction to Robotics and Automation to the curriculum.
- ▶ **Policy Update:** Revising the curriculum every two years to incorporate rapidly evolving technologies.

4. Stronger Industry-Academia Linkages

- ▶ **Feedback:** Limited interaction with industry professionals and inadequate preparation for the job market.
- ▶ **Correction:** Incorporating mandatory internships, industry webinars, and co-designed courses with industry partners.
- ▶ **Policy Update:** Making internships a credit-based activity with specific evaluation metrics.

5. Focus on Outcome-Based Education (OBE)

- ▶ **Feedback:** NAAC and NBA assessments indicated a lack of clear mapping between course objectives and program outcomes.

- ▶ **Correction:** Redesigning course objectives and outcomes to align explicitly with OBE frameworks, ensuring measurable assessments.
- ▶ **Policy Update:** Establishing a review committee to evaluate OBE compliance annually.

6. Soft Skills and Ethical Training

- ▶ **Feedback:** Employers reported that students lack communication skills and ethical awareness in professional settings.
- ▶ **Correction:** Embedding courses such as Professional Communication and Engineering Ethics and Sustainability in the first and final years.
- ▶ **Policy Update:** Including a soft skills assessment as part of the final year evaluation.

7. Assessment Reforms

- ▶ **Feedback:** Students expressed concerns about traditional examination methods focusing on rote memorization rather than conceptual understanding.
- ▶ **Correction:** Introducing open-book exams, group assessments, and design-oriented evaluation methods.
- ▶ **Policy Update:** Adopting a 40-30-30 weightage system: 40% continuous assessment, 30% practical/project work, and 30% final exams.

8. Environmental Sustainability Integration

- ▶ **Feedback:** Stakeholders emphasized the growing importance of sustainability in engineering practice.
- ▶ **Correction:** Adding courses like Green Engineering and incorporating sustainability modules into existing courses such as civil and mechanical engineering.
- ▶ **Policy Update:** Mandating a sustainability project for all final-year students.

9. Language Flexibility in Curriculum

- ▶ **Feedback:** Regional students struggled with technical English.
- ▶ **Correction:** Introducing regional language support in foundational courses and multilingual resources for technical subjects.
- ▶ **Policy Update:** Developing dual-medium (English and regional language) instructional materials for core subjects.

10. Flexible Learning Pathways

- ▶ **Feedback:** Students requested more control over their academic journey.
- ▶ **Correction:** Incorporating choice-based credit systems (CBCS), allowing students to select electives and focus areas across disciplines.
- ▶ **Policy Update:** Adopting a policy that enables lateral entry students to pick up missing foundational courses through bridge programs.

These course corrections /updates reflect a proactive approach to integrating feedback from students, alumni, industry, and accreditation bodies, ensuring that the engineering curriculum remains relevant and impactful.

3.18. CULTURAL AND ETHICAL ENRICHMENT

Based on the NEP-2020 Curriculum Design Guidelines, the following components are incorporated into the curriculum:

1. Efforts to Preserve and Promote Regional or National Culture and Heritage

The curriculum emphasizes the **Indian Knowledge System (IKS)**, which includes modules that promote understanding of India's cultural, literary, and historical heritage. These modules encourage students to engage with local traditions, art forms, and historical practices. For example, a **2-credit course on IKS** is integrated into the first year of UG programs. Through community engagement projects, students are involved in activities that explore and document regional heritage. This fosters pride in India's cultural legacy while supporting its preservation and relevance for future generations.

2. Integration of Ethical and Moral Education in the Curriculum

Courses on **Value Education (VEC)**, such as **Understanding India, Environmental Science, and Digital and Technological Solutions**, provide students with a foundation in ethical practices and moral reasoning. These courses focus on sustainability, empathy, and integrity, addressing societal challenges from a moral perspective. The curriculum's design ensures students develop a holistic understanding of ethics applicable to their professional and personal lives. This prepares them for responsible citizenship and ethical decision-making in their fields.

3. Initiatives for Promoting Inclusivity, Gender Equity, and Social Justice

The NEP-2020 guidelines include **Community Engagement Projects (CEP) and Field Projects** that expose students to socio-economic challenges, enabling them to design and implement inclusive solutions. Furthermore, co-curricular activities like NSS/NCC and initiatives focusing on gender equity and marginalized communities are integrated into the learning experience. Students gain a deeper understanding of societal inequalities and develop solutions-oriented thinking. These initiatives prepare them to advocate for and implement policies promoting inclusivity, gender equity, and social justice.

3.19. ETHICS AND SOCIAL RESPONSIBILITY

University Policy on Ethical Practices in Academic, Administrative, and Research Activities & Contributions to Fostering a Socially Responsible Academic Community

1. ETHICAL PRACTICES IN ACADEMIC ACTIVITIES

Objective:

To ensure that all academic activities are conducted in a manner that is fair, transparent, and promotes integrity, respect, and accountability.

POLICY:

•Academic Integrity:

All students and faculty members are expected to uphold high standards of honesty and integrity. This includes adhering to guidelines on plagiarism, cheating, and falsification of data or results. Violations will be met with disciplinary actions ranging from warnings to expulsion, depending on the severity.

•Assessment and Evaluation:

Assessments should be conducted transparently and fairly, ensuring that grading is based on merit. Faculty members must avoid any bias based on race, religion, gender, socioeconomic status, or personal beliefs.

•Respect for Intellectual Property:

The university promotes respect for intellectual property rights. Faculty and students must acknowledge the work and ideas of others in academic writing, research, and projects.

•Inclusive Education:

Academic programs and teaching materials will be designed to ensure inclusivity, promoting equal access for all students, including those with disabilities.

2. ETHICAL PRACTICES IN ADMINISTRATIVE ACTIVITIES**Objective:**

To create a work environment that is fair, just, transparent, and free from corruption or favoritism in all administrative functions.

POLICY:**•Transparency in Decision-Making:**

Administrative decisions, including faculty recruitment, promotions, and resource allocation, must be transparent and based on objective criteria. Regular audits of administrative processes should be carried out to ensure compliance with established procedures.

•Fairness in Employee Treatment:

All university employees, including administrative staff, will be treated with fairness, respect, and dignity. The university is committed to fostering an environment of equal opportunity, free from discrimination based on gender, ethnicity, age, or any other bias.

•Conflict of Interest:

All administrators and faculty members must disclose any potential conflicts of interest that may arise in their professional capacity. Any decisions made with conflicts of interest should be reported and handled according to the university's conflict-of-interest policy.

•Whistleblower Protection:

Employees and students who report unethical practices or violations of university policies will be protected from retaliation. A confidential and secure reporting system will be made available.

3. ETHICAL PRACTICES IN RESEARCH ACTIVITIES**Objective:**

To promote ethical conduct in all research activities, ensuring that research outcomes are reliable, reproducible, and contribute positively to society.

POLICY:**•Research Integrity:**

Researchers must ensure the honesty, accuracy, and integrity of their work. Data must be reported truthfully, and any discrepancies or errors discovered in research findings must be corrected promptly. Fabrication or falsification of data will result in serious disciplinary actions.

•Ethical Review of Research:

All research projects involving human participants, animals, or sensitive data must undergo ethical review by the university's Institutional Review Board/Ethics Committee (IRB) to ensure they comply with ethical standards, legal requirements, and societal considerations.

Environmental Responsibility:

Research activities should prioritize sustainability and minimize harm to the environment. The university will encourage the use of green technologies and support research that addresses environmental issues.

Collaboration and Sharing of Knowledge:

Researchers must maintain ethical standards when collaborating with external institutions and stakeholders, ensuring fairness and respect for intellectual property. Research outcomes should be published in a manner that is accessible and contributes to the advancement of knowledge.

4. CONTRIBUTIONS TO FOSTERING A SOCIALLY RESPONSIBLE ACADEMIC COMMUNITY**Objective:**

To create a university environment where academic, administrative, and research activities contribute to the betterment of society and promote social responsibility.

POLICY:**Community Engagement:**

The university will foster a culture of community engagement, encouraging students, faculty, and staff to participate in social service, volunteer work, and community-based research. Activities should address local, national, and global issues such as poverty, healthcare, education, and the environment.

Service Learning:

The university will incorporate service-learning projects into academic programs, allowing students to apply their academic knowledge to solve real-world problems in communities, particularly underserved areas.

Ethics and Social Responsibility Education:

The curriculum will include ethics courses or modules that address social responsibility, corporate social responsibility (CSR), and sustainability. Faculty members will be encouraged to integrate these themes into their teaching.

Promoting Diversity and Inclusivity:

The university will actively promote diversity in student and faculty recruitment, ensuring that all individuals, regardless of background, have equal opportunities for education and development.

Sustainable Practices:

The university will adopt green practices and sustainable technologies, aiming to reduce its carbon footprint and promote environmental responsibility among students, faculty, and staff.

MONITORING AND ENFORCEMENT

The university will establish a **Code of Ethics and Conduct Committee** to ensure the enforcement of these policies. This committee will review allegations of unethical conduct, recommend actions, and oversee the resolution of complaints. Additionally, regular training and awareness programs on ethical practices will be conducted for faculty, staff, and students.

Annual Reporting:

The university will prepare an annual report on its ethical practices and social responsibility initiatives, highlighting progress, challenges, and future goals. This report will be made available to all stakeholders, including students, faculty, alumni, and the public.

3.20. PREPAREDNESS FOR ACCREDITATION AND RANKING

Strategies and Steps for Alignment with Frameworks like NAAC, NBA, NIRF, and International Rankings (QS, THE) & Evidence of Preparation for External Quality Assessments

1. Alignment with NAAC (National Assessment and Accreditation Council)

Strategy: To align with NAAC accreditation, focus on fulfilling the criteria across seven key areas-Curriculum, Teaching-Learning, Research & Innovation, Infrastructure, Student Support, Governance, and Institutional Values.

STEPS:

Preparation of Self-Study Report (SSR):

- ▶ Collect data and evidence related to the seven NAAC criteria.
- ▶ Engage departments to ensure all academic, administrative, and support services align with NAAC's requirements.
- ▶ Draft the SSR, ensuring it reflects the institution's strengths and areas for improvement.

Internal Audit and Gap Analysis:

- ▶ Conduct a self-assessment and identify gaps in academic delivery, research, governance, infrastructure, and student support.
- ▶ Initiate improvements and alignments to meet NAAC standards.

Student and Faculty Feedback:

- ▶ Regularly collect feedback on curriculum, teaching, infrastructure, and support services from students and faculty.
- ▶ Implement the feedback to improve educational delivery and engagement.

Training and Awareness:

- ▶ Organize workshops for faculty and staff on NAAC's assessment process and criteria.
- ▶ Ensure that all stakeholders are aware of their role in the accreditation process.

Documentation and Evidence Collection:

- ▶ Maintain and update records of academic programs, research outputs, student activities, infrastructure developments, and administrative practices.
- ▶ Ensure that evidence for all activities is easily accessible for the peer review process.

Engagement with Stakeholders:

- ▶ Actively engage with students, alumni, and employers to gather testimonials and feedback for the SSR.

2. Alignment with NBA (National Board of Accreditation)

Strategy: Align academic programs with NBA's accreditation standards for engineering and technical programs, focusing on program outcomes, curriculum design, faculty qualifications, and student learning assessments.

STEPS:**Program Outcome Mapping:**

- ▶ Review and map the program outcomes (Pos), program-specific outcomes (PSOs), and course outcomes (COs) with NBA's prescribed standards.
- ▶ Ensure that the curriculum facilitates the achievement of these outcomes.

Internal Reviews and Faculty Involvement:

- ▶ Organize internal reviews of the curriculum, pedagogy, and teaching resources to ensure alignment with NBA's requirements.
- ▶ Involve faculty in mapping their courses to POs and PSOs to demonstrate how the curriculum contributes to these outcomes.

Infrastructure and Facilities Evaluation:

- ▶ Evaluate and upgrade laboratories, classrooms, and technological infrastructure to meet NBA's standards for equipment, safety, and access.

Continuous Monitoring:

- ▶ Implement regular assessments and quality checks to monitor the progress of students toward achieving defined learning outcomes.

Industry and Alumni Feedback:

- ▶ Gather and analyze feedback from industry partners and alumni to ensure that the curriculum is relevant and aligned with industry standards.
- o Use the feedback to update the curriculum and enhance teaching practices.

Documentation for NBA Review:

- ▶ Compile all the documentation, including course materials, faculty qualifications, assessment reports, and student outcomes.
- o Submit a well-organized accreditation report with clear evidence of adherence to NBA criteria.

3. Alignment with NIRF (National Institutional Ranking Framework)

Strategy: Enhance institutional performance by focusing on key NIRF metrics—Teaching, Learning & Resources, Research & Professional Practices, Graduation Outcomes, Outreach & Inclusivity, and Perception.

Steps:

Data Collection and Reporting:

- ▶ Ensure accurate and timely collection of data on teaching resources, student-faculty ratio, research output, industry collaborations, and infrastructure.
- ▶ Regularly update the NIRF submission with current statistics on faculty qualifications, research publications, student outcomes, and infrastructure.

Focus on Research & Innovation:

- ▶ Increase the volume and quality of research output by encouraging faculty and students to publish in peer-reviewed journals and participate in international conferences.

- ▶ Strengthen partnerships with industries and research organizations for collaborative projects and funding.

Faculty Development Programs:

- ▶ Implement regular faculty development programs to enhance teaching quality, research skills, and pedagogical techniques.
- ▶ Encourage faculty to undertake training programs that align with national and international educational standards.

Student Engagement and Success:

- ▶ Focus on improving student graduation rates, job placements, and skills development.
- ▶ Strengthen mentoring, career services, and internships to improve employability and practical exposure.

Outreach & Inclusivity Initiatives:

- ▶ Enhance inclusivity through scholarships, financial aid, and support systems for marginalized and economically disadvantaged groups.
- Develop community engagement projects, promote social responsibility, and create awareness of the institution's impact on society.

Perception Management:

- ▶ Foster positive perceptions through active alumni networks, public relations, and social media presence.
- ▶ Ensure strong industry relationships to enhance the university's reputation and employer trust.

4. Alignment with International Rankings (QS, THE)

Strategy: Improve the institution's global reputation by focusing on international standards of academic performance, research output, faculty quality, student experience, and global engagement.

Steps:

Enhance Research Output and Citations:

- ▶ Focus on increasing research publications in high-impact journals and expanding citation networks globally.
- ▶ Develop strategic research collaborations with international universities and research institutes.

Global Faculty Recruitment:

- ▶ Attract faculty members with international experience, strong academic backgrounds, and global research networks.
- ▶ Offer international faculty exchange programs to increase cross-border collaborations.

Improve International Student Diversity:

- ▶ Expand international student recruitment through targeted marketing and collaboration with global institutions.
- ▶ Provide a comprehensive support system to international students, ensuring a welcoming and conducive learning environment.

Global Partnerships and Collaborations:

- ▶ Strengthen partnerships with universities and research organizations abroad.
- ▶ Create exchange programs, joint degrees, and international research opportunities to enhance global visibility.

International Accreditation:

- ▶ Seek accreditation from global agencies relevant to the institution's programs, such as ABET (for engineering) and AACSB (for business schools).

Internationalization of Curriculum:

- ▶ Integrate global perspectives into the curriculum by offering courses on international issues, global challenges, and cross-cultural understanding.
- ▶ Foster collaboration between international faculty and local faculty for curriculum development.

Branding and Visibility:

- ▶ Participate in global rankings, conferences, and symposiums to enhance the institution's visibility on the world stage.
- ▶ Promote institutional achievements on international platforms to improve the university's global reputation.

5. Evidence of Preparation for External Quality Assessments

Strategy: Systematically prepare for external assessments by ensuring the institution is aligned with the prescribed frameworks, has all the required documentation in place, and can demonstrate ongoing improvement.

Steps:

I. Internal Quality Assurance Systems:

- Develop a robust internal quality assurance (IQA) mechanism to continuously assess and improve academic, administrative, and research practices.
- Regularly conduct internal audits to identify areas that need strengthening before external assessments.

II. Regular Data Collection and Reporting:

- Maintain a central repository for all required documentation, including curriculum development records, faculty qualifications, research outputs, and student performance metrics.
- Prepare and update all data to ensure it is accurate, comprehensive, and easily accessible for external reviewers.

III. Engagement with Accreditation Bodies:

- Maintain regular communication with accreditation bodies and international ranking agencies to stay updated on new guidelines and processes.
- Attend workshops, webinars, and training sessions offered by accreditation bodies to understand best practices.

IV. Stakeholder Involvement:

- a. Involve faculty, staff, and students in preparing for external assessments by sharing the importance of these frameworks and the role everyone plays in achieving accreditation.
- b. Encourage faculty and staff to participate in institutional improvement initiatives aligned with these frameworks.

V. Mock Assessments and Feedback Loops:

- a. Conduct mock assessments or internal reviews mimicking the external assessment process to identify areas of improvement.
- b. Use feedback from these reviews to make final adjustments before the actual assessment.

3.21. SCALABILITY AND REPLICABILITY

Approach and Steps for Establishing a Deemed University under Distinct Category

To establish a **Deemed University** under the **Distinct category**, the institution has planned to develop unique and scalable aspects of its educational model, ensuring it has a broader, transformative impact both locally and globally. This involves building upon the institution's distinctiveness and aligning with the broader higher education landscape. The following approach outlines steps that emphasize **scaling the institution's distinctiveness model** and **leveraging case studies or models** as benchmarks for other universities.

1. Scaling the Institution's Distinctiveness Model for Wider Impact**Step 1:**

Define the Institution's Core Distinctiveness

The first step is to formalize what sets the institution apart. This could include:

- **Curriculum Innovation:** Emphasize interdisciplinary programs, problem-based learning, and student-centric pedagogies.
- **Research Excellence:** Build on distinctive research themes, such as applied sciences, sustainability, social innovation, or industry-specific research.
- **Community Engagement:** Demonstrate the institution's commitment to social responsibility through community service, extension programs, or rural outreach.
 - Industry Partnerships: Showcase collaborations with industry leaders for real-world exposure, internships, and placements.

Actionable Steps:

- Host workshops with faculty, industry experts, and alumni to define the core distinctiveness.
- Conduct a SWOT (Strengths, Weaknesses, Opportunities, and Threats) analysis to ensure scalability and sustainability.
- Identify specific areas where the institution can lead in terms of innovation, academic offerings, research, and social outreach.

Step 2: Design a Scalable Model for Institutional Growth

To scale the distinctiveness model, the institution must create a flexible, adaptable model that can expand across multiple campuses, regions, or even international borders.

Actionable Steps:

- **Standardize Core Components:** Define the pillars of the distinctiveness model (curriculum, research, community engagement) and create standardized frameworks for replication.

•**Technological Infrastructure:** Implement an integrated learning management system (LMS) and ERP for seamless management across multiple locations. Include e-learning platforms to expand access globally.

•**Collaborations and Partnerships:** Foster partnerships with international universities, research institutions, and industries to scale research initiatives and globalize the curriculum.

Step 3: Build Institutional Capacity

For scalability, the institution must ensure that its infrastructure, faculty, and governance systems are capable of supporting the expansion of its distinctiveness model.

Actionable Steps:

- Invest in faculty development programs and leadership training to ensure high-quality academic delivery.
- Strengthen governance structures to support scaling and adapt policies to manage expansion smoothly.
- Ensure infrastructure—both physical (campuses, classrooms) and digital (online platforms)—is designed to support large-scale operations.

Step 4: Marketing and Communication for Global Reach

For scaling, the institution needs to effectively communicate its distinctiveness and its scalability potential to a broader audience, including students, faculty, industry, and the public.

Actionable Steps:

- Develop marketing materials that highlight the institution's unique approach to education and its potential for scaling.
- Create case studies and success stories from current programs to demonstrate how the institution's model has impacted students, faculty, and the community.
- Attend national and international educational conferences to share the institution's distinctiveness and model for scaling.

2. Case Studies or Models That Could Serve as Benchmarks for the proposed Universities

By identifying successful global models that exemplify scaling of educational distinctiveness, the institution can benchmark its approach. Below are examples that can guide the institution in establishing itself as a leader in the higher education space.

Case Study 1: Massachusetts Institute of Technology (MIT) – Open Courseware (OCW)

Overview: MIT's Open Courseware (OCW) has revolutionized access to world-class education by making their course materials freely available to learners worldwide. This openness reflects MIT's commitment to education, research, and the democratization of knowledge.

Relevance to the Institution: The model of OCW could be adapted by the institution to expand its academic reach globally. By making select courses or program materials accessible online, the institution can scale its academic model, attracting students and faculty from around the world.

Key Elements for Scaling:

- Open access to academic content
- Integration of technology for global reach
- Reputation building through educational innovation

Actionable Steps:

- Create an Open Courseware initiative for select courses to demonstrate commitment to global education access.
- Collaborate with industry and international partners to enhance online education platforms.

Case Study 2: Stanford University – Entrepreneurial Ecosystem

Overview: Stanford University's ecosystem has been instrumental in fostering entrepreneurship and innovation. It blends academic resources with industry partnerships, creating a fertile ground for startups, research commercialization, and innovation.

Relevance to the Institution: The institution can replicate Stanford's approach by establishing a university-wide entrepreneurial ecosystem, creating accelerators, incubators, and industry partnerships. The focus should be on building a research-to-market pipeline and fostering entrepreneurial thinking across all disciplines.

Key Elements for Scaling:

- Strong industry-university collaborations
- Support for student startups and research commercialization
- Creation of an innovation ecosystem

Actionable Steps:

- Develop an entrepreneurship and innovation hub that serves students, faculty, and the community.
- Build stronger industry partnerships to ensure the commercialization of research outputs.
- Provide seed funding and mentoring for student startups.

Case Study 3: University of California – Community Engagement Model

Overview: The University of California system integrates strong community engagement into its educational and research missions. Its focus on local communities, service learning, and public health has enabled it to expand its impact beyond the academic sphere.

Relevance to the Institution: This community-focused model can serve as a benchmark for the institution to scale its impact through outreach programs, service learning, and social responsibility initiatives. Focusing on social causes will allow the institution to play a vital role in societal development.

Key Elements for Scaling:

- Commitment to social impact and community service
- Extension programs that engage students and faculty in real-world challenges
- Focus on sustainability and local development

Actionable Steps:

- Expand community engagement and service-learning programs, particularly in rural and underserved areas.
- Collaborate with government agencies and NGOs to implement scalable social impact projects.
- Align academic programs with real-world challenges, emphasizing sustainability, social justice, and public health.

Case Study 4: University of Cambridge – Global Research Collaboration

Overview: Cambridge University has built a strong reputation for its research output, particularly in science and technology. Its collaborative approach, with global academic and industrial partners, has enabled it to scale its research and educational influence worldwide.

Relevance to the Institution: The institution can adopt a similar research collaboration model, focusing on global partnerships and interdisciplinary research. This will scale its impact through high-quality research that addresses global challenges.

Key Elements for Scaling:

- Interdisciplinary research and global partnerships
- Centers of excellence and specialization in cutting-edge research fields
- Strong global academic collaborations and research consortia

Actionable Steps:

- Build research consortia with international universities and research organizations.
- Focus on interdisciplinary research in areas with global relevance (e.g., climate change, AI, healthcare).
- Position the institution as a global leader in solving pressing global challenges.

In establishing a Deemed University under the Distinct category, the institution must develop a scalable model that highlights its unique strengths and potential for wider impact. By drawing on successful models like MIT's OpenCourseWare, Stanford's entrepreneurial ecosystem, the University of California's community engagement, and Cambridge's global research collaborations, the institution can create a distinctive and impactful educational ecosystem. The steps outlined in this approach focus on leveraging these best practices while adapting them to the institution's local context to achieve broader recognition and influence.

3.22. MONITORING AND EVALUATION FRAMEWORK**Approach and Steps for Establishing a Deemed University under Distinct Category**

To successfully establish a Deemed University under the Distinct category, the applicant has planned a robust system in place to monitor both academic and administrative performance, along with effective mechanisms to evaluate and refine the institution's distinctiveness. This will ensure that the university maintains high standards and continuously improves its unique elements to meet evolving academic and societal needs. The following steps outline the approach to achieve this, integrating internal and external mechanisms, and aligning with international best practices for accreditation and rankings.

1. Institutional Systems for Monitoring Academic and Administrative Performance

To ensure continuous quality improvement, the institution shall have comprehensive systems for monitoring and evaluating both academic and administrative performance. These systems should be aligned with best practices in quality assurance, institutional governance, and international accreditation standards.

Step 1: Establish an Internal Monitoring Framework**Academic Performance Monitoring:**

- Outcome-Based Education (OBE) System: Implement an Outcome-Based Education framework for monitoring student learning outcomes. This includes clearly defined program outcomes (Pos), course outcomes (COs), and program-specific outcomes (PSOs).

► **Regular Course and Faculty Reviews:** Establish periodic reviews of course syllabi, teaching methods, and faculty performance. Faculty should be evaluated using student feedback, peer reviews, and self-assessments.

► **Learning Management System (LMS):** Use an LMS to track student engagement, progress, and performance across all courses. Analytics from the LMS will allow for real-time assessment and intervention if needed.

► **Research Monitoring:** Develop systems for tracking research output, including publications, patents, grants, and industry collaborations. Establish key performance indicators (KPIs) to evaluate research quality and impact.

Administrative Performance Monitoring:

-Key Performance Indicators (KPIs) for Administration: Define KPIs for various administrative departments (admissions, HR, finance, student services) to track efficiency, effectiveness, and student satisfaction.

► **Internal Audits:** Conduct regular internal audits to assess the efficiency of administrative processes such as admissions, financial management, and resource allocation.

► **Feedback Mechanisms:** Collect feedback from students, faculty, staff, and external stakeholders (industry, alumni) through surveys and focus groups. Analyze feedback regularly to ensure continuous improvement.

Step 2: Implement External Monitoring and Evaluation Mechanisms Accreditation:

► **National Accreditation:** Regularly apply for accreditation from national bodies like NAAC and NBA. These agencies provide detailed criteria for evaluating academic and administrative performance, and their feedback will ensure alignment with national standards.

► **International Accreditation:** Seek accreditation from international agencies like AACSB, ABET, or EQUIS. These accreditations provide global validation of the university's academic quality, administrative efficiency, and research capabilities.

► **Annual Academic and Administrative Audits:** Engage third-party evaluators for annual academic and administrative audits to assess the effectiveness of current systems and processes.

Benchmarking Against Global Standards:

-Participation in Global Rankings: Actively participate in global university rankings such as QS, THE, and NIRF. These rankings provide external benchmarks for academic performance, research output, administrative practices, and international collaboration.

-International Peer Reviews: Regularly invite international experts for peer reviews of academic programs, research initiatives, and administrative processes. Their feedback will help refine strategies and improve institutional performance.

Step 3: Establish Continuous Feedback and Improvement Cycles

-Annual Review Meetings: Conduct annual meetings with faculty, administrators, and governing bodies to review academic and administrative performance. Use data from internal monitoring systems, accreditation reports, and external audits to make informed decisions.

-Quality Circles: Establish quality circles involving faculty, students, and staff to discuss issues related to academic and administrative performance. These groups can propose improvements and ensure the system remains dynamic and responsive.

2. Mechanisms to Evaluate and Refine the Distinctiveness Elements

The institution's distinctiveness model shall be continually evaluated and refined to ensure that it evolves in line with global trends in higher education, remains relevant to the community it serves, and delivers high-quality outcomes for students, faculty, and society at large.

Step 1: Establish Internal Mechanisms to Assess Distinctiveness

Self-Assessment Framework:

- Develop an internal self-assessment framework based on the institution's unique characteristics (e.g., curriculum innovation, research themes, social responsibility).
- Evaluate the distinctiveness elements through the lens of impact, sustainability, and scalability. This framework should assess whether these elements are achieving their intended outcomes and are adaptable for future growth.

Student and Stakeholder Feedback:

- Surveys and Feedback Loops: Regularly survey students, faculty, alumni, and industry partners on the impact of the distinctiveness elements. Use structured surveys and focus groups to collect both qualitative and quantitative feedback.
- Engagement in Social Responsibility Programs: Evaluate the effectiveness of community outreach programs and their alignment with the institution's vision of social impact.

Step 2: External Evaluation of Distinctiveness

Accreditation and Ranking Feedback:

- NAAC and NBA Reports: The feedback from national accreditation bodies like NAAC and NBA will provide insights into the effectiveness of the institution's distinctiveness model. These bodies assess criteria like innovation, industry collaborations, and societal impact, which are critical for refining distinctiveness.
- International Accreditation and Rankings: The institution should aim for recognition from QS, THE, and other global ranking systems, which will evaluate distinctiveness elements such as research quality, industry partnerships, and international outlook.

Case Study and Peer Comparison:

- Regularly benchmark the institution's distinctiveness elements against other universities with similar missions. Identify best practices, innovative models, and successful programs from global institutions that can be adapted or scaled.
- Study institutions that have successfully scaled their distinctiveness models, such as Stanford for entrepreneurship, MIT for innovation, or University of California for community engagement.

Step 3: Mechanisms for Refining Distinctiveness Elements

Data-Driven Refinement:

- Use data analytics from internal audits, feedback systems, and external evaluations to refine the distinctiveness elements. This could include modifying curricula, enhancing research initiatives, or adapting community outreach programs.
- Establish a dedicated Innovation and Strategy Office responsible for tracking the evolution of distinctiveness elements and ensuring that they remain relevant and impactful.

Global Collaborations:

- Engage in international collaborations with universities, research organizations, and industry to refine the distinctiveness model based on global trends and emerging needs.
- Foster global student and faculty exchanges to bring new perspectives that contribute to refining the institution's educational offerings and community impact.

Annual Strategic Review:

- Conduct an annual strategic review to assess the relevance and effectiveness of the institution's distinctiveness elements. This review should include input from stakeholders across the institution, as well as external peers.
- Use the review to fine-tune the institution's programs, research, and community outreach, ensuring they continue to meet global standards and respond to societal needs.

To successfully establish a Deemed University under the Distinct category, the institution shall develop robust systems for monitoring academic and administrative performance and effective mechanisms for evaluating and refining its distinctiveness elements. These systems should align with best practices in national and international accreditation and university rankings. By adopting a dynamic, data-driven approach to performance evaluation and leveraging global best practices, the institution will be well-positioned to scale its impact, maintain its uniqueness, and contribute meaningfully to the global higher education landscape.

3.23. BENCHMARKING WITH GLOBAL INSTITUTIONS

1. Massachusetts Institute of Technology (MIT), USA
 - ▶ Strong focus on engineering, technology, and entrepreneurship.
 - ▶ Potential collaborations in computer science, AI, and robotics.
2. University of Cambridge, UK
 - ▶ Renowned for engineering, medical, and social sciences programs.
 - ▶ Collaboration in areas such as sustainability, biomedical engineering, and economics.
3. ETH Zurich, Switzerland
 - ▶ Expertise in engineering, technology, and environmental sciences.
 - ▶ Collaboration in research and technology innovation.
4. National University of Singapore (NUS)
 - ▶ Strong programs in engineering, business, and global governance.
 - ▶ Potential for joint research in smart cities, AI, and global development.
5. University of Melbourne, Australia
 - ▶ Offers programs in law, engineering, business, and health sciences.
 - ▶ Possible student exchange programs and research collaborations in health technology and law.
6. Tsinghua University, China
 - ▶ Known for its engineering, science, and technology programs.
 - ▶ Collaborative research on AI, sustainable development, and infrastructure.

7. Technical University of Munich (TUM), Germany

- ▶ Leader in technical education and innovation.
- ▶ Opportunities for research collaboration in fields like renewable energy, engineering, and digital transformation.

8. Indian Institute of Technology (IIT) Bombay, India

- ▶ Strong programs in engineering, management, and technology.
- ▶ Strategic collaborations in the field of research and development.

3.24. GOVERNANCE AND POLICY FRAMEWORK

- ▶ A comprehensive governance structure reflecting autonomy and accountability.
- ▶ Visionary leadership practices driving distinctiveness.
- ▶ Institutional policies on academic freedom, ethical practices, and conflict resolution.

3.25. DATA-DRIVEN DECISION-MAKING

Mechanism and Structure for Data-Driven Decision-Making for Deemed University Status

In the process of applying for Deemed University status, an effective Data-Driven Decision-Making (DDDM) framework is essential for ensuring that academic, operational, and strategic decisions are based on robust data analytics and metrics. The following framework will help the institution align with modern higher education standards, improve efficiency, and ensure continuous improvement in academic and operational areas.

1. Institutional Efforts in Adopting Analytics and Metrics for Decision-Making

Step 1: Establish a Data-Driven Decision-Making Culture

▶ **Vision & Leadership:** The institution's leadership must clearly communicate the importance of data-driven decision-making in achieving academic excellence and operational efficiency. Create awareness among faculty, administrators, and staff about the benefits of data analytics for informed decision-making.

▶ **Data Governance Framework:** Establish a Data Governance Committee consisting of faculty, administrators, and IT experts to oversee data management, privacy, security, and quality standards. This committee will ensure that data is collected, analyzed, and used in accordance with ethical standards and institutional goals.

Step 2: Set Up Data Infrastructure and Analytics Tools

▶ **Data Collection Mechanisms:** Implement integrated systems like Learning Management Systems (LMS), Enterprise Resource Planning (ERP), Student Information Systems (SIS), and Human Resources Management Systems (HRMS) to gather data on academic, administrative, and operational activities.

▶ **Analytics Tools:** Invest in modern analytics tools like Power BI, Tableau, or Google Analytics to analyze data across various domains. These tools will enable decision-makers to visualize trends, track key performance indicators (KPIs), and make data-informed decisions.

► **Data Warehousing:** Set up a centralized data warehouse that stores academic, financial, operational, and research-related data. This will provide a single source of truth for all data-related queries, ensuring consistency and reliability in decision-making.

Step 3: Develop Key Metrics and Performance Indicators

► **Academic Performance Metrics:** Define metrics for evaluating student performance (e.g., graduation rate, course completion rate, dropout rate, exam results), faculty performance (e.g., student feedback, teaching quality, research output), and program effectiveness (e.g., learning outcomes, employability).

► **Operational Efficiency Metrics:** Set up KPIs for measuring administrative performance, such as response time for student queries, faculty recruitment time, financial health (e.g., cost per student, budget utilization), and resource utilization (e.g., classroom occupancy rate, lab usage).

► **Research and Innovation Metrics:** Track research output, publications, patents, industry partnerships, and grants to assess the institution's contribution to research and innovation.

Step 4: Data Literacy and Capacity Building

► **Training and Workshops:** Conduct workshops and training for faculty, administrators, and staff to build their capacity in using data analytics tools. Focus on helping them understand how to interpret data, spot trends, and make decisions based on evidence.

► **Data Champions:** Identify data champions within each department who can act as points of contact for data-related queries and who can help drive the culture of data usage in decision-making.

2. Use of Data to Enhance Academic and Operational Efficiency

Step 1: Enhancing Academic Efficiency with Data

• **Personalized Learning:** Use data from LMS to personalize the learning experience for students. Track student engagement, participation, and performance to identify struggling students and intervene with personalized academic support (e.g., tutoring, additional resources).

• **Curriculum Improvement:** Analyze performance data at the course and program level to identify areas where students consistently underperform. Use this data to inform curriculum adjustments, teaching methods, and course content updates.

• **Faculty Development:** Use data from student feedback, peer reviews, and faculty performance metrics to identify areas for faculty development. This can inform professional development programs that focus on improving teaching quality and enhancing faculty skills.

• **Assessment and Feedback:** Implement continuous assessment mechanisms using data from quizzes, assignments, and projects to provide timely feedback to students. This allows for adaptive teaching techniques and ensures a high level of academic performance.

Step 2: Optimizing Operational Efficiency with Data

• **Resource Allocation:** Use data on classroom occupancy, lab utilization, and student enrollment to optimize the allocation of physical and human resources. This can help improve efficiency in scheduling, reduce underutilized resources, and ensure cost-effective use of space and infrastructure.

• **Admissions and Enrollment:** Leverage data from admissions and student demographics to predict enrollment patterns and manage capacity. Use predictive analytics to forecast student demand and optimize recruitment efforts, thereby improving the efficiency of the admission process.

- **Financial Management:** Utilize financial data to track expenditure, revenue, and budget utilization. Use data analytics to identify cost-saving opportunities and ensure financial sustainability. This could include analyzing spending patterns, identifying cost overrun areas, and planning future budgets based on historical data.
- **Student Retention:** Use data on student behavior (e.g., attendance, participation, grades) to develop predictive models for student retention. These models can help identify at-risk students and trigger early intervention strategies to improve retention rates.
- **Staffing and Recruitment:** Track HR data, including faculty and staff performance, hiring rates, and retention data. Analyze this to optimize staffing strategies, reduce turnover, and ensure that the institution has the right talent in place.

Step 3: Continuous Improvement Using Data

- **Data-Driven Strategic Planning:** Use data analytics to drive strategic planning. Historical data can inform future planning for program expansion, research initiatives, campus development, and faculty recruitment.
- **Dashboards for Real-Time Monitoring:** Create real-time dashboards to monitor academic, administrative, and financial performance. Decision-makers should be able to access up-to-date information on key metrics at any time.
- **Benchmarking and Peer Comparison:** Use external data from national and international rankings (e.g., NIRF, QS, THE) to benchmark the institution's performance against peer universities. This will help identify areas for improvement and inform decision-making to enhance competitive positioning.
- **Alumni and Stakeholder Feedback:** Regularly collect and analyze alumni feedback to measure the long-term success of graduates and ensure academic programs align with industry needs. Use this data to adjust programs, research initiatives, and curricula.

3. Governance and Oversight for Data-Driven Decision-Making

- **Data Analytics Steering Committee:** Form a steering committee consisting of key academic and administrative leaders, data scientists, and IT experts. This committee will be responsible for overseeing the institution's data strategy, ensuring data quality, and fostering a data-driven culture.
- **Regular Performance Reviews:** Hold quarterly or bi-annual meetings to review the performance data and assess whether institutional goals are being met. These meetings will focus on interpreting data and setting new goals or strategies based on the insights gained. Under the proposed Deemed University status, adopting data-driven decision-making is crucial for improving academic and operational efficiency. By setting up a robust data infrastructure, developing a culture of data literacy, and leveraging analytics tools, the institution can make informed decisions, optimize resource utilization, and ensure continuous improvement across all functions. These efforts will not only improve institutional performance but also align with national and international best practices, making the institution a strong candidate for **Deemed University** status.

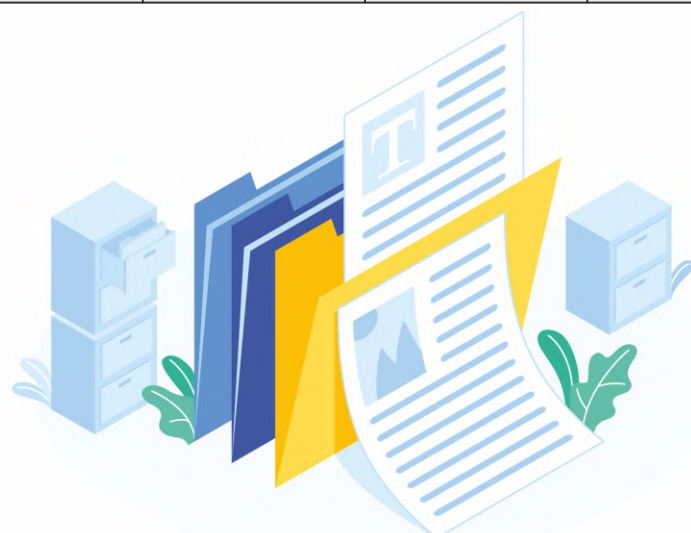
3.26. INTEGRATED DEVELOPMENT PLAN (IDP): A SEPARATE DOCUMENT IS ATTACHED

- A long-term roadmap that outlines the university's goals, timelines, and measurable outcomes.
- Focus on distinctiveness, innovation, societal impact, and alignment with Sustainable Development Goals (SDGs).

3.27. FINANCE AND RESOURCE MOBILIZATION

DETAILS OF THE ASSESTS

PARTICULARS	AEC	KCASW	KASC	TOTAL
A) Immovable Properties				
Building	605751025	46229792	58062740	710043557
Total - A	605751025	46229792	58062740	710043557
B) Movable Properties				
Computers & Accessories	24527425	13107619	6249295	43884339
Electrical Installation	7980037	7185626	4391216	19556879
Furniture Account	722486	520190	346793	1589469
Kitchen Equipment	9643229	1343218	1028749	12015196
Lab. Equipment	14925461	12364702	6298719	33588882
Library Books Account	4375905	3150653	2100435	9626993
Office Equipment	8941280	6437220	4291814	19670314
Sports Goods Account	420163	302517	201678	924358
Vehicles	155725130	26090710	14060473	195876313
Total - B	227261116	70502455	38969172	336732743
Total - A+B	833012141	116732247	97031912	1046776300



INCOME AND EXPENDITURE (LAST THREE YEARS DATA)

ARUNAI ENGINEERING COLLEGE

YEAR	2021 - 22	2022 - 23	2023 - 24
INCOME	Rs.	Rs.	Rs.
Tuition Fees	129271000	70706707	129762475
Hostel, Lunch & Transport Fees	6257106	67197602	21905650
Total	135528106	137904309	151668125
Expenditure	452653	337300	900000
Advertisement Expenses			
Affiliation & Approval fees	5652842	5926583	6008325
Book & Periodicals	2568456	271928	561900
Text Books for Students	68541	84523	365000
College Maintenance	405658	488954	512138
Electricity Charges	4856523	6982379	80,80,169
Function & Seminar Expenses	284514	382891	11,14,664
Laboratory Consumables & Services A/c	2,56,238	2,75,015	3,02,725
Mess Expenses	8452523	9654295	16658643
Miscellaneous Expenses	35846	152360	585638
Printing & Stationery Expenses	1845252	360497	421098
Professional Charges	1534000	1980000	1862989
Rent, Rate & Taxes	415268	398100	348787
Repairs & Maintenance	685236	40900	894292
Salaries and Allowances	42587563	58554121	58589222
EPF Paid	0	1350750	994266
Subscription	315276	203358	66134
Telephone, Post & Internet exp	1455863	1228487	1441015
Travelling & Conveyance	548567	69000	405646
Vehicle Maintenance Expenses	18525741	24892492	24225852
Bank Interest & Charges	265823	232478	258867
Total	91212383	113866411	124597370
	67.3014518	82.5691466	82.1513222

KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN

PARTICULARS	2021-2022	2022-2023	2023-2024
<u>INCOME</u>	Rs.	Rs.	Rs.
Tuition Fees, Hostel & Transport Fee	102161100	105689261	115313317
Total	102161100	105689261	115313317
<u>Expenditure</u>	284523	298526	461172
Advertisement Expenses			
Affiliation & Approval fees	285450	315270	495700
Book & Periodicals	89563	94523	117660
College Maintenance	105623	49527	53029
Electricity Charges	985623	1003250	11,33,358
Function & Seminar Expenses	78563	284574	4,12,000
Laboratory Consumables & Services A/c	1,85,636	1,95,287	65,440
Printing & Stationery Expenses	365823	185522	128000
Repairs & Maintenance	189523	106538	148900
Salaries and Allowances	24248563	25575852	28537935
EPF Paid	0	0	102972
Subscription	165723	105336	187640
Telephone, Post & Internet exp	685673	542536	737985
Travelling & Conveyance	150537	75850	45263
Vehicle Maintenance Expenses	8765236	19854254	27026229
Bank Interest & Charges	986596	2398587	2377010
Total	37572655	51085432	62030293

KARAN ARTS AND SCIENCE COLLEGE

PARTICULARS	2021-2022	2022-2023	2023-2024
<u>INCOME</u>	Rs.	Rs.	Rs.
Tuition Fees, Hostel & Transport Fee	20010852	23304300	39979856
Total	20010852	23304300	39979856
<u>Expenditure</u>			
Advertisement Expenses	184520	385620	397520
Affiliation & Approval fees	352674	1743640	2052404
Book & Periodicals	98563	105238	108410
College Maintenance	67436	45653	48247
Electricity Charges	735234	832573	11,23,423
Function & Seminar Expenses	285630	385640	5,01,256
Laboratory Consumables & Services A/c	48,756	56,234	59,308
Printing & Stationery Expenses	114520	119248	123470
Repairs & Maintenance	138563	141536	146320
Salaries and Allowances	10785623	11856423	19013280
Subscription	136458	165247	167842
Telephone, Post & Internet exp	548546	652410	682360
Travelling & Conveyance	45868	52364	57423
Vehicle Maintenance Expenses	4051360	4353260	9652380
Bank Interest & Charges	545476	741371	742530
Total		21636457	34876173
18139227			
	90.64694996	92.84319632	87.23436373

THE CUMULATIVE INCOME AND EXPENDITURE OF PROJECT INSTITUTES (Last Three-Year Data)

Head Income/ Expenditure	Arunai Engineering College	Kamban College of Arts and Science for Women	Karan Arts and Science College
Income	257700058	266897870	306961298
Expenditure	146924265	186588300	221503836
Surplus	110775793	80309570	85457462

STATEMENT OF INCOME AND EXPENDITURE PROJECTION FOR 2025 - 2026 TO 2035 - 2036

ARUNAI ENGINEERING COLLEGE

ARUNAI ENGINEERING COLLEGE	1	2	3	4	5	6	7	8	9	10
PARTICULARS	2025-2026	2026-2027	2027-2028	2028-2029	2029-2030	2030-2031	2031-2032	2032-2033	2033-2034	2034-2035
INCOME			Rs.	Rs.	Rs.	Rs.	Rs.	Rs.	Rs.	Rs.
Tuition Fees	206665100	246131700	288361000	333546400	381894700	433627400	488981400	548210100	611584800	679395800
Hostel, Lunch & Transport Fees	4779800	18114400	31382500	46579400	62840200	80239100	98856000	118776000	140090400	162896800
Total	211444900	264246100	319743500	380125800	444734900	513866500	587837400	666986100	751675200	842292600
Expenditure										
Advertisement Expenses	1190300	1368900	1574300	1810500	2082100	2394500	2753700	3166800	3641900	4188300
Affiliation & Approval Fees	7946100	9138100	10508900	12085300	13898100	15982900	18380400	21137500	24308200	27954500
Book & Periodicals	743200	854700	982900	1130400	1300000	1495000	1719300	1977200	2273800	2614900
Text Books for Students	482800	555300	638600	734400	844600	971300	1117000	1284600	1477400	1699100
College Maintenance	677400	779000	895900	1030300	1184900	1362700	1567100	1802200	2072600	2383500
Electricity Charges	12411100	14272800	16413699	18875800	21707200	24963300	28707800	33014000	37966200	43661200
Function & Seminar Expenses	1474200	1695400	1949700	2242200	2578600	2965400	3410200	3921800	4510200	5186800
Laboratory Consumables & Services	40500	46600	53600	61700	71000	81700	94000	108100	124400	143100
Mess Expenses	29756200	37719700	43377700	49884400	57367100	65972200	75868100	87248400	100335700	115386100
Miscellaneous Expenses	774600	890800	1024500	1178200	1355000	1558300	1792100	2061000	2370300	2726000
Printing & Stationery Expenses	557000	640600	736700	847200	974300	1120500	1288600	1481900	1704200	1959900
Professional Charges	2463900	2833500	3258600	3747400	4309500	4956000	5699400	6554300	7537500	8668100
Rent, Rate & Taxes	461300	530500	610100	701700	807000	928100	1067300	1227400	1411600	1623400
Repairs & Maintenance	1182800	1360300	1564400	1799100	2069000	2379400	2736300	3146800	3618900	4161800
Salaries and Allowances	97034300	121089500	147753000	176416000	206378500	237335400	272935800	313876300	360957800	415101600
EPF Paid	1314900	1512200	1739100	2000000	2300000	2645000	3041800	3498100	4022900	4626400
Subscription	87600	100800	116000	133400	153400	176400	202900	233400	268500	308900
Telephone, Post & Internet exp	1905800	2191700	2520500	2898600	3333400	3833400	4408400	5069700	5830200	6704800
Travelling & Conveyance	536500	617000	709600	816100	938600	1079400	1241300	1427500	1641700	1888000
Vehicle Maintenance Expenses	29288800	35182200	42959600	49403700	56814300	65336500	75137100	86407700	99368900	114274300
Bank Interest & Charges	342400	393800	452900	520900	599100	689000	792400	911300	1048100	1205400
Total	190671700	233773400	279840300	328317300	381065700	438226400	503961000	579556000	666491000	766466100

KAMBAN COLLEGE OF ARTS AND SCIENCE FOR WOMEN

KARAN ARTS & SCIENCE COLLEGE	1	2	3	4	5	6	7	8	9	10
PARTICULARS	2025 2026	2026 2027	2027 2028	2028 2029	2029 2030	2030-2031	2031-2032	2032-2033	2033-2034	2034-2035
INCOME	Rs.	Rs.	Rs.	Rs.	Rs.	Rs.	Rs.	Rs.	Rs.	Rs.
Tuition Fees	43777500	53842000	65611000	78203900	91678300	106095900	121522700	138029400	155691600	167390100
Hostel, Lunch & Transport Fees	5246000	5613300	6006300	6426900	6876900	7358400	7873600	8424900	9014800	9645900
Total	49023500	59455300	71617300	84630800	98555200	113454300	129396300	146454300	164706400	177036000
Expenditure	6,45,300	690500	738900	790700	846100	905400	968800	1036600	11 09200	1186900
Advertisement Expenses										
Affiliation & Approval fees	33,98,700	3636600	3891200	4163600	4455100	4767000	5100700	5457800	5839900	6248700
Book & Periodicals	2,21,500	237000	253600	271400	290400	310800	332600	355900	380900	407600
Text Books for Students	4,15,800	444900	476100	509500	545200	583400	624300	668000	714800	764900
College Maintenance	8,54,100	913900	977900	1046400	1119700	1198100	1282000	1371800	1467900	1570700
Electricity Charges	15,08,300	1613900	1726900	1847800	1977200	2115600	2263700	2422200	2591800	2773300
Function & Seminar Expenses	5,61,400	600700	642800	687800	736000	787600	842800	901800	965000	1032600
Laboratory Consumables & Services	66,500	71200	76200	81600	87400	93600	100200	107200	114700	122800
Mess Expenses	6,50,300	695900	744700	796900	852700	912400	976300	1044700	1117900	1196200
Miscellaneous Expenses	47,800	51200	54800	58700	62800	67200	71900	77000	82400	88200
Printing & Stationery Expenses	1,38,300	148000	158400	169500	181400	194100	207700	222300	237900	254600
Rent, Rate & Taxes	60,300	64600	69200	74100	79300	84900	90900	97300	104100	111400
Repairs & Maintenance	1,63,900	175400	187700	200900	215000	230100	246200	263500	282000	301800
Salaries and Allowances	2,27,94,900	25890600	32203000	37457300	47079400	57375100	63641500	77096500	91493300	99398000
Subscription	1,88,000	201200	215300	230400	246600	263900	282400	302200	323400	346100
Telephone, Post & Internet exp	7,64,300	817800	875100	936400	1002000	1072200	1147300	1227600	1313600	1405600
Travelling & Conveyance	64,300	68800	73700	78900	84500	90500	96900	103700	111000	118800
Vehicle Maintenance Expenses	1,10,46,300	11819600	15347100	19121500	20460100	21892400	24924900	26669700	28536700	31234400
Bank Interest & Charges	831700	890000	952300	1019000	1090400	1166800	1248500	1335900	1429400	1529500
Total	44421700	49031800	59664900	69542400	81411300	94111100	104449600	120761700	138215900	150092100

KARAN ARTS AND SCIENCE COLLEGE
TOTAL SURPLUS FROM THE PROJECT INSTITUTES (2025-2026 TO 2034-2035)

TOTAL SURPLUS FROM THE PROJECT INSTITUTES (2025-2026 TO 2034-2035)												
Project Institute	Particulars	2025-2026	2026-2027	2027-2028	2028-2029	2029-2030	2030-2031	2031-2032	2032-2033	2033-2034	2034-2035	
ARUNAI	Total income	211444900	264246100	319743500	380125800	444734900	513866500	587837400	666986100	751675200	842229600	
KAMBAN	Total income	145409900	165313700	185110900	206293900	228959600	253212000	279162200	306928900	336639000	368429200	
KARAN	Total income	49023500	59455300	71617300	84630800	98555200	113454300	129396300	146454300	164706400	177036000	
Grand Total income		405878300	489015100	576471700	671050500	772249700	880532800	996395900	1120369300	1253020600	1387757800	
ARUNAI	Total Expenditure	190671700	233773400	279840300	328317300	381065700	438226400	503961000	579556000	666491000	766466100	
KAMBAN	Total Expenditure	92841400	103841000	118511000	132307600	148170200	164242700	181790600	201316400	221909200	244643700	
KARAN	Total Expenditure	44421700	49031800	59664900	69542400	81411300	94111100	104449600	120761700	138215900	150092100	
Grand Total Expenditure		327934800	386646200	458016200	530167300	610647200	696580200	790201200	901634100	1026616100	1161201900	
Total Surplus		77943500	102368900	118455500	140883200	161602500	183952600	206194700	218735200	226404500	226555900	
TOTAL SURPLUS IN TEN YEARS (2025-26 TO 2034-2035): 166.31 Cr												

TOTAL SURPLUS IN TEN YEARS (2025-26 TO 2034-2035): 166.31 Cr
NOTE: THE RESOURCES REQUIRED LIKE MANPOWER, EQUIPMENT AND BOOKS IS ATTACHED AS PART OF THE ANNEXURE.

3.28. FUTURE PLANS (STRATEGIC PLAN) AND INSTITUTE DEVELOPMENT PLAN (IDP):

A separate Document is attached herewith

- ▶ Strategic goals and action plans for the next 10 years is the PART-II of the DPR.
- ▶ The Institute Development Plan is the PART-III of the DPR.

3.29. FUTURE ASPIRATIONS

- ▶ Aspirations for becoming a national/international hub for specific domains of excellence.
- ▶ Vision for leveraging the "Distinct Category" status to make transformational contributions.

Aspirations for Becoming a National/International Hub for Specified Domains of Excellence

The institution's shortlisted programs across diverse domains lay the groundwork for establishing itself as a leading national and international hub of excellence. By aligning educational objectives with societal and global needs, we aim to foster a culture of innovation, inclusivity, and impact. The following dimensions deepen our approach to achieving recognition:

1. Academic Leadership and Curriculum Innovation

- ▶ **Interdisciplinary Programs:** Programs such as M. Tech in Agro-Electronics and M.Sc. in Clinical Research and Health Informatics bridge multiple disciplines, preparing students for future-ready professions.
- ▶ **Outcome-Based Learning:** Curricula will focus on skills that meet industry standards and global expectations, ensuring graduates excel in their chosen fields.
- ▶ **Global Standards:** The institution will incorporate international benchmarks and accreditation frameworks like ABET, AACSB, and EQUIS, ensuring recognition for the quality of programs.

2. Research and Development (R&D) Leadership

- ▶ **Centers of Excellence (CoEs):** Dedicated CoEs in areas like AI in VLSI Technology, Sustainability, and Health Informatics will foster cutting-edge research, encouraging global collaborations with academia and industry.
- ▶ **Innovation Ecosystem:** Establishing incubators and funding mechanisms for startups in emerging fields will amplify the institution's research-to-market pipeline.
- ▶ **Global Conferences and Symposia:** Hosting international events to discuss challenges and advancements in domains like climate science and AI will attract thought leaders and increase visibility.

3. Internationalization and Global Partnerships

- ▶ **Strategic Collaborations:** MoUs with top-tier institutions and organizations worldwide will facilitate knowledge exchange, dual-degree programs, and joint research projects.
- ▶ **Student Mobility:** Scholarships for international students and exchange programs for domestic students will enhance global diversity on campus.
- ▶ **Global Rankings:** A targeted approach to improve rankings in QS, THE, and ARWU through impactful research, academic excellence, and outreach.

4. National Recognition and Policy Alignment ●

- ▶ **National Missions:** Programs aligned with initiatives such as Skill India, Digital India, and Make in India will contribute to national development.
- ▶ **Policy Advocacy:** Engaging in thought leadership by contributing research and white papers to policymakers on sustainability, AI ethics, and agricultural innovations.
- ▶ **Capacity Building:** Partnerships with state and central governments for workforce training and capacity development programs in areas like solar energy and organic farming.

5. Cultural Diplomacy and Indian Knowledge Systems ●

- ▶ **Global Outreach in Culture:** Courses like B.A. Tamil Culture and Civilization and M.A. Thozil Sar Kalaigal will position the institution as a global center for Indian heritage studies.
 - ▶ **Cultural Exchange Programs:** Hosting international students and researchers in Indian knowledge systems will promote cultural diplomacy and strengthen global ties.
- The shortlisted programs outlined emphasize a forward-thinking approach to education and research, aligning with global trends and national priorities. With these programs, we aspire to position ourselves as a hub of excellence in interdisciplinary and multidisciplinary domains. By integrating technical innovation, cultural heritage, and emerging research areas, the institution aims to address societal needs and global challenges.

1. Environment and Sustainability: ●

Programs like B. Tech in Climate Science and Engineering and M. Tech in Sustainability in Building Maintenance aim to prepare professionals capable of addressing critical environmental challenges. These courses position the institution as a leader in sustainable technology, fostering innovation in eco-friendly practices and contributing to national policies on sustainability.

2. Artificial Intelligence and Computing: ●

Distinct programs such as M. Tech in AI in VLSI Technology and M. Tech in Natural Language Processing leverage advancements in AI to enhance computational solutions and semiconductor technologies. These areas align with global industry demands and establish the institution as a hub for AI-driven innovation.

3. Agriculture and Biotechnology: ●

By offering specialized programs like M. Tech in Agro-Electronics and B.Sc. in Organic Farming, we aim to advance sustainable agricultural practices and integrate technology with farming. These initiatives ensure food security while promoting innovative agricultural research, making the institution a key player in AgriTech.

4. Indian Knowledge Systems: ●

Programs like B.A. in Tamil Culture and Civilization reflect a commitment to preserving and promoting indigenous knowledge systems. These courses attract international scholars and contribute to global academic dialogue on cultural heritage and its relevance in contemporary society.

5. Health Informatics: ●

Integrated courses such as M.Sc. in Clinical Research and Health Informatics address the intersection of healthcare and data science, positioning the institution as a pioneer in digital health research and solutions. These offerings aim to create a comprehensive ecosystem for research, innovation, and education, attracting collaborations with industries, governments, and academic institutions worldwide.

3.30 VISION FOR LEVERAGING THE 'DISTINCT CATEGORY' STATUS

The recognition as a "Distinct Category" institution under UGC guidelines presents a transformative opportunity. This status enables us to redefine the educational landscape and make significant contributions across domains.

1. Academic Excellence:

We will pioneer innovative curriculum frameworks that integrate theoretical knowledge with practical application, ensuring our graduates are equipped with future-ready skills. The focus on niche domains such as AI, sustainability, and health informatics will set benchmarks for academic programs globally.

2. Research and Innovation:

The institution will become a hub for cutting-edge research by fostering collaborations with global universities and industries. Initiatives like establishing centers of excellence in AI, AgriTech, and sustainability will propel advancements and commercialization of research outputs.

3. Cultural and Societal Impact:

Leveraging programs in Indian knowledge systems, we will preserve and promote cultural heritage while integrating traditional wisdom into modern contexts. This approach will enhance societal appreciation for indigenous practices globally.

4. Global Partnerships:

Strategic collaborations with international academic and industrial partners will facilitate student exchange, joint research projects, and co-certification programs. This global engagement will enhance our reputation as an international education hub.

5. Technological Transformation:

With an emphasis on digital ecosystems, the institution will deploy advanced learning management systems, AI-powered student services, and data-driven decision-making in administration.

This vision enables the institution to drive national development goals, address global challenges, and contribute meaningfully to academia and society. The "Distinct Category" status will be a catalyst for achieving these aspirations, ensuring transformational growth and long-lasting impact.

The 'Distinct Category' status offers a unique platform to make transformative contributions across domains. It allows us to scale efforts, institutionalize innovations, and establish global leadership.

1. Building a Transformative Academic Ecosystem

- ▶ **Customized Programs:** Introducing micro-credentials and stackable degree options to cater to professionals, lifelong learners, and international students.
- ▶ **Faculty Excellence:** Recruitment of globally recognized faculty and continuous professional development programs will ensure top-tier instruction.

2. Advancing National and Global Impact Through Research

- ▶ **Patents and IP Development:** Incentivizing faculty and students for innovation and patent filings will establish the institution as a leader in IP generation.
- ▶ **Solutions for Global Challenges:** Focused research on global issues like climate change, food security, and AI ethics will foster worldwide recognition.

3. Strengthening Community Engagement and Social Responsibility

- ▶ **Local Impact, Global Lessons:** Programs such as B.Sc in Organic Farming and PG Diploma in Solar PV and Battery Management Systems will address local challenges while providing scalable models for global application.
- ▶ **Social Entrepreneurship:** Supporting community-centric projects and startups will enhance the institution's societal impact.

4. Creating a Digital and Technological Edge

- ▶ **Smart Campus Initiative:** Utilizing AI-powered administrative systems, green energy solutions, and sustainable infrastructure will create a model campus.
- ▶ **AI-Driven Learning:** Personalized learning platforms using AI will redefine student engagement and outcomes.

5. Recognition and Awards

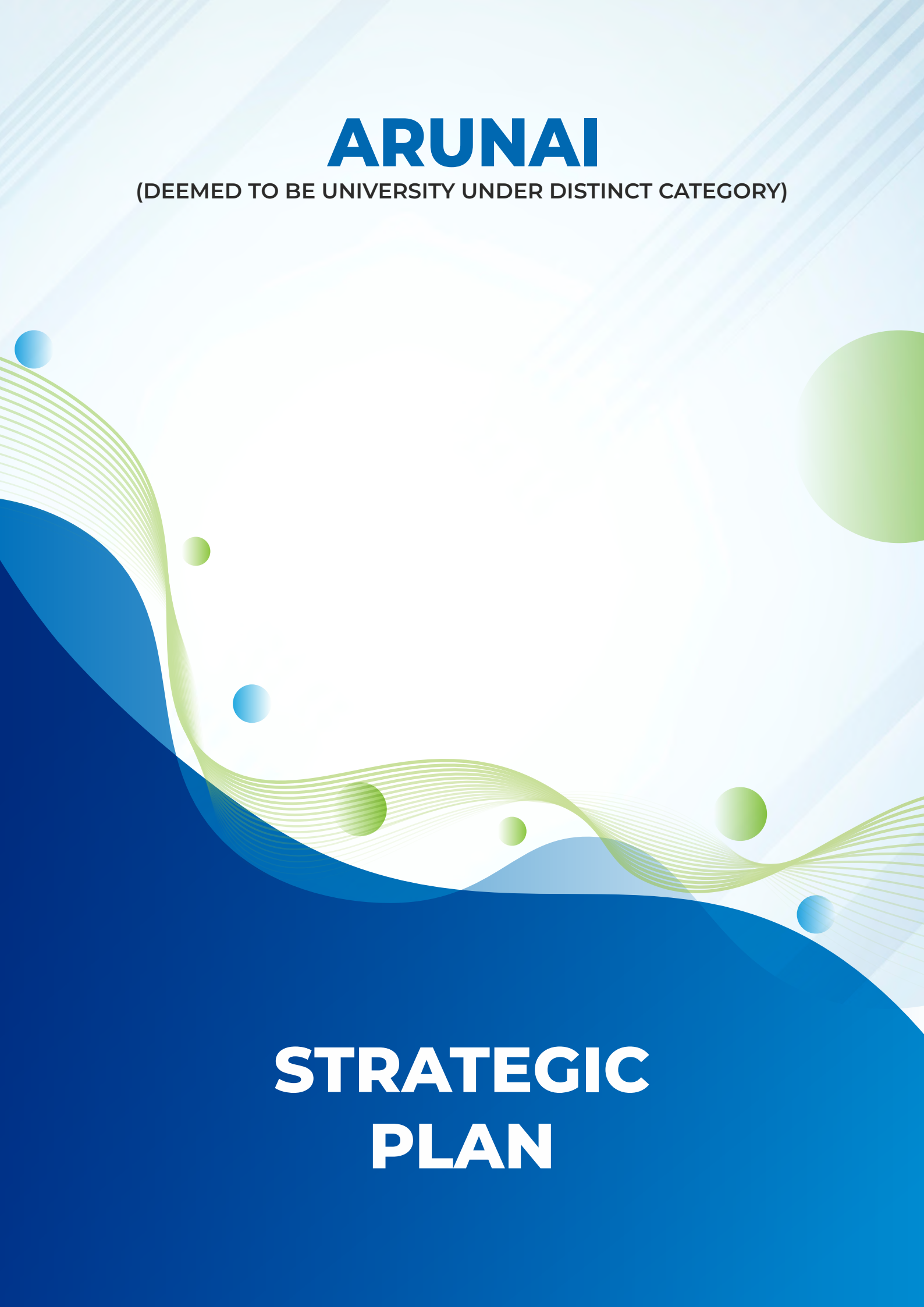
- ▶ **Awards and Certifications:** Active participation in global platforms like UN initiatives, Green Campus certifications, and AI-based innovation awards will build credibility.
- ▶ **Publications in High-Impact Journals:** Encouraging faculty and students to publish in top-tier journals will enhance academic prestige.

The applicant management shall put multidimensional efforts and the institution will not only emerge as a leader in academic and research excellence but also as a transformative force contributing to global knowledge, sustainability, and innovation. The “Distinct Category” status will amplify these aspirations, enabling impactful and sustainable contributions to national priorities and international collaborations. This journey will place the institution at the forefront of education and research, recognized globally for its visionary approach and meaningful outcomes.

3.31. RISK MANAGEMENT AND CONTINGENCY PLANNING

- ▶ Institutional mechanisms for risk identification, mitigation, and management are described in the Strategic Plan document of the proposed University.



The background features a dynamic design with flowing blue and green waves. Scattered throughout are several circles in blue and green, some of which are composed of multiple concentric lines, creating a sense of motion and depth.

ARUNAI

(DEEMED TO BE UNIVERSITY UNDER DISTINCT CATEGORY)

STRATEGIC PLAN

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ARUNAI TO BE UNIVERSITY

Tiruvannamalai Tamil Nadu (india)

(DEEMED TO BE UNIVERSITY UNDER DISTINCT CATEGORY)



1. EXECUTIVE SUMMARY

Arunai Engineering College (AEC), established in 1993, in association with another two higher education institute of the same management, Kamban College of Arts and Science for Women, and Karan Arts and Science College, aims to transition into Arunai University, a Distinct Deemed University under the UGC's 2023 regulations. This strategic plan outlines the university's mission to cultivate technically proficient and ethically grounded professionals through a holistic, impactful curriculum. Positioned on the Chittoor-Cuddalore highway in Tamil Nadu, AEC provides a conducive learning environment, focusing on student-centered development and moral education.

VISION

"To be a pioneering university of distinction, renowned nationally and internationally for its extraordinary, unique programs that integrate technical excellence with ethical and holistic development. We are committed to cultivating globally competent professionals who are not only skilled and innovative but also deeply rooted in values and dedicated to social impact. By fostering transformative learning in a serene, inspiring environment, we empower individuals to excel in their fields, uplift communities, and serve as ambassadors of integrity and wisdom on the world stage."

This vision highlights trust's commitment to unique, high-impact programs that elevate its reputation and reach, enhancing its appeal and influence both in India and globally.

MISSION

"To advance knowledge and societal impact through multi-disciplinary distinctive programs that blend professional ability with moral education, enabling our graduates to become valued leaders who drive positive change both locally and internationally."

STRATEGIC PHASES

- 1. Short-Term (2025–2030):** Foundation-building through the development of distinct, interdisciplinary academic programs, advanced infrastructure, faculty recruitment, and international partnerships.
- 2. Medium-Term (2030–2035):** National and global expansion with diversified programs, a robust research ecosystem, deeper international collaborations, and increased student wellness and community impact.
- 3. Long-Term (2035–2040):** Recognition as a global educational leader, prioritizing sustainable development, enduring partnerships, and a thriving alumni network that drives societal impact.

CORE PLANS

- a. Academic Excellence:** Establish programs that address emerging fields, incorporating ethics and social responsibility.
- b. Research and Innovation:** Foster interdisciplinary research hubs and incubation centers, supporting innovation aligned with societal needs.
- c. Global Outreach:** Expand international collaborations for joint research and student/faculty exchanges.
- d. Community Engagement:** Launch impactful local outreach programs aligned with regional development goals.
- e. Sustainability:** Integrate eco-friendly infrastructure and sustainable practices across the campus.

KEY PERFORMANCE INDICATORS (KPIs)

KPIs will guide the university's progress, focusing on academic program quality, research output, international collaborations, student satisfaction, and sustainability goals.

The budget for Arunai University's transformation totals ₹75 crores over three phases, covering foundational infrastructure, global partnerships, research centers, sustainable development, and faculty/student initiatives. Phase 1 (2025-2030) focuses on academic excellence, faculty recruitment, and digital and physical infrastructure. Phase 2 (2030-2035) expands programs, research, and international partnerships. Phase 3 (2035-2040), aims to achieve global recognition, sustainability, and long-term program resilience.

To manage risks, the university plans proactive mitigation strategies across key areas. Regular industry feedback and regulatory consultations will keep curricula relevant. Research hubs and partnerships are supported by grants, skill training, and collaborations to ensure utilization and innovation. For infrastructure, eco-friendly construction and flexible tech upgrades are prioritized to mitigate delays and obsolescence. Faculty recruitment challenges are addressed with competitive packages and development opportunities, while community engagement programs are designed in collaboration with local leaders to ensure relevance and impact. Sustainable development initiatives are incentivized through awareness campaigns and pilot projects to foster campus-wide adoption.

Conclusion

This strategic roadmap aims to elevate Arunai University as a beacon of ethical, impactful education with a commitment to national and global societal progress.

Introduction

Arunai Engineering College (AEC), established in 1993, in association with another two higher education institute of the same management, Kamban College of Arts and Science for Women and Karan Arts and Science College, aims to transition into Arunai University, a Distinct Deemed University under the UGC's 2023 regulations. This strategic plan outlines the university's mission to cultivate technically proficient and ethically grounded professionals through a holistic, impactful curriculum. Positioned on the Chittoor-Cuddalore highway, in Tamil Nadu, The institutes provides a conducive learning environment, focusing on student-centered development and moral education with a mission to develop technically proficient, morally upright, and socially responsible professionals, The institute has consistently delivered a high standard of education, supported by a holistic approach to student development. Located on the Chittoor-Cuddalore national highway, the campus provides a serene, inspiring learning environment that fosters intellectual curiosity and personal growth.

In alignment with its vision "to shape and transform youth with utmost care, innovative and dedicated service for morals and education," the institutes are now poised to ascend to a new level of excellence by transforming into a "Distinct Deemed University." This transition is inspired by the goals set forth in the University Grants Commission (UGC) Regulations for Deemed Universities, 2023. By achieving this status, the institution aims to become a pioneering force in higher education, offering unique programs that seamlessly integrate technical excellence, ethical values, and holistic development, ultimately contributing to the nation's advancement and societal well-being.

This Detailed Project Report (DPR) encapsulates the strategic vision and operational framework that will guide transformation over the next fifteen years. The report is structured to provide a comprehensive overview of the strategic and rolling implementation plans that will underpin this journey.

These include:

- a) Academic Plan:** Envisioning a curriculum that blends technical depth with interdisciplinary breadth, aiming to equip students with skills for emerging fields and societal challenges.
- b) Faculty Recruitment Plan:** Building a team of accomplished, research-driven educators with international exposure, to create a knowledge ecosystem that promotes innovation and critical thinking.
- c) Student Admission Plan:** Establishing an admission strategy that attracts diverse, high-caliber students from across the globe, enhancing the university's intellectual and cultural diversity.

Introduction

- d) Research Plan:** Creating an impactful research ecosystem, fostering collaboration with industry and academic institutions worldwide, and focusing on areas that address societal challenges and drive technological advancement.
- e) Campus Information and Communication Technology Plan:** Adopting cutting-edge ICT infrastructure to create a digitally empowered campus, facilitating seamless teaching, learning, and administrative processes.
- f) Infrastructure Development Plan:** Expanding and enhancing campus facilities to support a dynamic learning environment, inclusive of state-of-the-art laboratories, libraries, and recreational spaces.
- g) Finance Plan:** Ensuring the financial sustainability and growth of the institution through diversified funding sources, including industry collaborations, alumni contributions, and government grants.
- h) Administrative Plan:** Streamlining administrative processes to support efficient and effective operations, ensuring transparency, accountability, and alignment with institutional goals.
- i) Governance Plan:** Establishing a governance framework that emphasizes strategic oversight, ethical leadership, and stakeholder engagement, ensuring the institution's integrity and commitment to its mission.

With clear annual milestones and action plans, this DPR details a transformative journey that aims to elevate Arunai Engineering College into a leading institution recognized for its academic rigor, ethical standards, and contribution to societal progress. The report also includes an assessment of institutes current academic and physical infrastructure, laying the groundwork for the envisioned development trajectory. By adhering to this roadmap, AEC aspires to achieve national and international distinction, nurturing future leaders who embody technical prowess, integrity, and a commitment to societal impact.

VISION

"To be a pioneering university of distinction, renowned nationally and internationally for its extraordinary, unique programs that integrate technical excellence with ethical and holistic development. We are committed to cultivating globally competent professionals who are not only skilled and innovative but also deeply rooted in values and dedicated to social impact. By fostering transformative learning in a serene, inspiring environment, we empower individuals to excel in their fields, uplift communities, and serve as ambassadors of integrity and wisdom on the world stage."

This vision highlights trust's commitment to unique, high-impact programs that elevate its reputation and reach, enhancing its appeal and influence both in India and globally.

MISSION

"To advance knowledge and societal impact through multi-disciplinary distinctive programs that blend professional ability with moral education, enabling our graduates to become valued leaders who drive positive change both locally and internationally."

3. FIFTEEN-YEAR STRATEGIC PLAN (2025-26 to 2039-40)

PHASE 1:

SHORT-TERM GOALS (2025–2030)

Objective: Establish the foundation for a distinctive, high-impact academic environment with global outreach.

1. Academic Excellence and Distinct Programs

- ▶ Develop and launch unique, interdisciplinary programs focused on emerging fields (e.g., AI, biotechnology, sustainable engineering) that integrate ethics and social impact.
- ▶ Set up an international curriculum advisory board to ensure global standards and alignment with industry needs.

2. Infrastructure and Learning Environment

- ▶ Build state-of-the-art research labs and centers of excellence for distinct programs.
- ▶ Create a serene, student-centered campus environment with amenities that support holistic development.
- ▶ Create the Digital infrastructure with latest technologies and facilities.

3. Faculty Development and Recruitment

- ▶ Attract and retain world-class faculty with expertise in both technical and ethical education.
- ▶ Implement regular training in pedagogy, research, and interdisciplinary collaboration.

4. Community Engagement and Social Impact

- ▶ Launch community outreach programs, including local workshops and social impact initiatives, that involve students and faculty in addressing real-world challenges.

PHASE 2:

MEDIUM-TERM GOALS (2030–2035)

Objective: Expand the university's national and international reputation and increase program offerings.

1. Program Diversification and Expansion

- ▶ Introduce additional specialized programs aligned with future skills, such as smart technology, green infrastructure, and ethical AI.
- ▶ Begin offering certifications, online programs, and continuing education options to attract international students.

2. Research and Innovation Ecosystem

- ▶ Establish an interdisciplinary research center that encourages collaborative research across domains, including technical innovation, ethics, and social sciences.
- ▶ Create an incubation center to support student and faculty startups focused on societal benefits.

3. Strengthening International Partnerships

- ▶ Deepen existing international collaborations and initiate student exchange programs to build a diverse, multicultural campus environment.
- ▶ Actively participate in global academic and research forums to enhance visibility and attract talent worldwide.

4. Enhanced Student Experience

- ▶ Implement mentorship programs for students to support both academic and personal growth.
- ▶ Develop support systems for student wellness, including mental health resources and skill-building workshops.

5. Community Impact and Outreach Expansion

- ▶ Scale up community-based projects that align with regional development goals, focusing on rural upliftment and technology integration.
- ▶ Partner with NGOs and government agencies for larger, impact-driven projects.

PHASE 3:

LONG-TERM GOALS (2035–2040)

Objective: Achieve recognition as a global leader in technical education with a strong emphasis on ethics and social responsibility.

1. Global Academic Recognition and Rankings

- ▶ Attain top-tier positions in national and international university rankings for distinctiveness in education and impactful research.
- ▶ Position the university as a preferred destination for international students and researchers, reaching a significant percentage of foreign student enrollment.

2. Innovation and Leadership in Research

- ▶ Establish the university as a hub for groundbreaking research with contributions to both industry and society.
- ▶ Lead global research initiatives in specialized areas, fostering innovation and sustainable development solutions.

3. Sustainable Development and Social Responsibility

- ▶ Become a role model in sustainable campus practices, including carbon neutrality, eco-friendly infrastructure, and community-centric sustainability initiatives.
- ▶ Ensure that all university programs, from technical to social disciplines, contribute to sustainable development goals.

4. Enduring Global Partnerships and Influence

- ▶ Build lasting relationships with global educational institutions, industry leaders, and government bodies to promote academic and research exchange.
- ▶ Facilitate international conferences, symposiums, and thought-leadership forums on pressing global issues, led by the university's experts and students.

5. Alumni Network and Societal Impact

- ▶ Establish a robust alumni network that supports university initiatives, fosters mentorship for students, and amplifies the university's global reach.
- ▶ Highlight and leverage alumni success stories to inspire future generations and uphold the university's legacy of social impact and ethical leadership.

4. THE PROJECT PLANNING

PHASE	YEAR	GOAL	PROJECT	Cost (INR Cr)	COMMITTEE	RESOURCES REQUIRED
Short-Term	2025-2026	Academic Excellence and Distinct Programs	Develop and launch interdisciplinary programs	5.00	Academic & Research Committee	Faculty, Digital Infrastructure
	2025-2028	Faculty Development and Recruitment	Hire expert faculty & conduct training	3.00	HR & Training Committee	Trainers, Digital Resources
	2026-2028	Infrastructure and Learning Environment	Build research labs and centers of excellence	6.00	Infrastructure Committee	Lab Equipment, Faculty
	2028-2030	Community Engagement & Social Impact	Launch social impact & outreach programs	2.00	Community Engagement Committee	Local Networks, Faculty Involvement
Medium-Term	2028-2030	Program Diversification & Expansion	Introduce specialized courses	4.00	Curriculum Development Committee	Faculty, Online Learning Tools
	2030-2032	Research and Innovation Ecosystem	Establish interdisciplinary research center	10.00	Research & Innovation Committee	Research Grants, Labs
	2032-2034	Strengthening International Partnerships	Initiate student & faculty exchange programs	5.00	International Relations Committee	Travel Grants, MoUs
	2034-2035	Enhanced Student Experience	Implement mentorship & wellness programs	4.00	Student Welfare Committee	Counseling Centers, Workshops
Long-Term	2035-2036	Global Academic Recognition & Rankings	Improve ranking through academic initiatives	7.00	Accreditation Committee	Research Publications, PR Strategies
	2036-2038	Innovation & Leadership in Research	Drive groundbreaking projects	8.00	Innovation & Research Committee	R&D Funds, Patents
	2038-2039	Sustainable Development & Social Responsibility	Implement eco-friendly campus policies	5.00	Sustainability Committee	Renewable Energy Infrastructure
	2039-2040	Alumni Network & Societal Impact	Build a strong alumni network	3.00	Alumni Engagement Committee	Events, Networking Portals
Total				75.00		

5. THE HEAD-WISE EXPENDITURE FOR THE PLANED PROJECTS

Year	Project	Cost of HR (INR Cr)	Cost of Machinery & Equipment (INR Cr)	Cost of Infrastructure (INR Cr)	Cost of Contingencies (INR Cr)	Total Cost (INR Cr)
2025-2026	Develop and launch interdisciplinary programs	2.00	1.50	0.80	0.70	5.00
2025-2028	Hire expert faculty & conduct training	1.50	0.80	0.40	0.30	3.00
2026-2028	Build research labs and centers of excellence	2.50	2.50	0.60	0.40	6.00
2028-2030	Launch social impact & outreach programs	0.80	0.60	0.30	0.30	2.00
2028-2030	Introduce specialized courses	1.50	1.50	0.60	0.40	4.00
2030-2032	Establish interdisciplinary research center	4.00	4.50	0.90	0.60	10.00
2032-2034	Initiate student & faculty exchange programs	2.00	1.50	0.80	0.70	5.00
2034-2035	Implement mentorship & wellness programs	1.50	1.20	0.80	0.50	4.00
2035-2036	Improve ranking through academic initiatives	3.00	2.50	0.90	0.60	7.00
2036-2038	Drive groundbreaking projects	3.50	3.50	0.60	0.40	8.00
2038-2039	Implement eco-friendly campus policies	2.00	1.50	0.90	0.60	5.00
2039-2040	Build a strong alumni network	1.00	1.00	0.50	0.50	3.00
Total	—	25.30	22.60	8.10	5.60	75.00

SUMMARY:

- Cost of HR: 25.30 Cr
- Cost of Machinery & Equipment: 22.60 Cr
- Cost of Infrastructure: 8.10 Cr
- Cost of Contingencies: 5.60 Cr
- Total Cost: 75.00 Cr

6. KEY PERFORMANCE INDICATORS

YEAR	PROJECT	KEY PERFORMANCE INDICATORS (KPIs)
2025-2026	Develop and launch interdisciplinary programs	- Number of new programs introduced - Student enrollment in new programs - Industry collaborations formed
2025-2028	Hire expert faculty & conduct training	- Number of faculty recruited - Number of training sessions conducted - Improvement in faculty research output
2026-2028	Build research labs and centers of excellence	- Number of labs established - Research publications from new labs - Faculty and student research participation
2028-2030	Launch social impact & outreach programs	- Number of community projects initiated - Student participation in outreach activities - Impact assessment reports
2028-2030	Introduce specialized courses	- Number of new courses introduced - Enrollment in new courses - Industry-relevant certifications included
2030-2032	Establish interdisciplinary research center	- Number of research projects initiated - Funding received for research - Research collaborations with industry & academia
2032-2034	Initiate student & faculty exchange programs	- Number of exchange agreements signed - Number of students & faculty participating in exchanges - Impact of international exposure
2034-2035	Implement mentorship & wellness programs	- Number of mentorship programs conducted - Student participation rates - Improvement in student well-being indicators
2035-2036	Improve ranking through academic initiatives	- Improvement in university ranking (national/international) - Increase in research publications - Alumni and employer feedback
2036-2038	Drive groundbreaking projects	- Number of patents filed - Number of industry-funded research projects - Awards and recognitions received
2038-2039	Implement eco-friendly campus policies	- Percentage reduction in carbon footprint - Renewable energy usage on campus - Water and waste management efficiency
2039-2040	Build a strong alumni network	- Number of active alumni members - Number of alumni engagement events - Alumni contributions (funding, mentorship)

7. MAJOR STEPS IN THE DEVELOPMENT OF THE PROJECT

Year	Project	Major Steps in Development
2025-2026	Develop and launch interdisciplinary programs	1. Identify key interdisciplinary domains. 2. Design curriculum with industry and academic experts. 3. Obtain approvals from academic bodies. 4. Recruit faculty with expertise in new domains. 5. Launch and promote programs to students.
2025-2028	Hire expert faculty & conduct training	1. Identify required faculty positions in emerging fields. 2. Conduct national & international faculty recruitment drives. 3. Organize faculty orientation and pedagogy training. 4. Provide incentives for research and development. 5. Monitor faculty performance and student feedback.
2026-2028	Build research labs and centers of excellence	1. Identify research focus areas based on future trends. 2. Procure state-of-the-art equipment. 3. Construct and set up research labs. 4. Recruit research staff and faculty. 5. Establish collaborations for research funding.
2028-2030	Launch social impact & outreach programs	1. Identify community issues and areas of impact. 2. Develop collaboration with NGOs and government bodies. 3. Train students and faculty for community projects. 4. Implement pilot outreach projects in local areas. 5. Evaluate impact and scale up programs.
2028-2030	Introduce specialized courses	1. Conduct market research on industry demands. 2. Develop course content with academic and industry experts. 3. Obtain necessary accreditations and approvals. 4. Implement courses in online and offline formats. 5. Monitor course effectiveness and update content.
2030-2032	Establish interdisciplinary research center	1. Define research themes aligned with global challenges. 2. Secure funding and partnerships for research initiatives. 3. Establish state-of-the-art research infrastructure. 4. Recruit interdisciplinary research teams. 5. Launch and promote research projects.
2032-2034	Initiate student & faculty exchange programs	1. Identify global partner institutions. 2. Develop agreements and MoUs for exchange. 3. Establish scholarships and funding for students. 4. Promote exchange programs through awareness campaigns. 5. Evaluate impact on student learning and faculty development.
2034-2035	Implement mentorship & wellness programs	1. Develop mentorship models tailored for students. 2. Recruit mentors from faculty and industry. 3. Set up student wellness centers with professional counselors. 4. Organize mental health and skill-building workshops. 5. Monitor student progress and feedback.
2035-2036	Improve ranking through academic initiatives	1. Strengthen faculty research and publication output. 2. Improve student employability and industry partnerships. 3. Focus on academic quality and global accreditations. 4. Increase student and faculty international collaborations. 5. Promote university achievements through PR strategies.
2036-2038	Drive groundbreaking projects	1. Identify breakthrough research areas. 2. Secure grants and funding from government & industries. 3. Establish a research incubation hub. 4. Encourage faculty and students to file patents. 5. Promote commercialization of research outcomes.

2038-2039	Implement eco-friendly campus policies	1. Conduct sustainability audit of campus. 2. Develop policies for carbon footprint reduction. 3. Implement renewable energy solutions. 4. Promote water conservation and waste management. 5. Create student-led sustainability initiatives.
2039-2040	Build a strong alumni network	1. Develop an alumni database and engagement strategy. 2. Organize networking events and reunions. 3. Encourage alumni mentorship and scholarships. 4. Highlight alumni success stories for inspiration. 5. Establish alumni contributions for institutional development.

8. DETAILS FOR PROGRAM/PROJECT IMPLEMENTATION

Project Name	Year	Objectives	Key Steps	Key Performance Indicators (KPIs)	Core Committee
Develop and launch interdisciplinary programs	2025-2026	Introduce cutting-edge programs in emerging fields. - Align curriculum with industry needs and global standards.	1. Identify interdisciplinary domains. 2. Design curriculum with experts. 3. Obtain regulatory approvals. 4. Recruit faculty. 5. Launch and promote programs.	- Number of new programs. - Student enrollment. - Industry collaborations.	Academic & Research Committee
Hire expert faculty & conduct training	2025-2028	- Recruit highly qualified faculty. - Train faculty in advanced pedagogy and research skills.	1. Identify required faculty positions. 2. Conduct recruitment drives. 3. Organize training workshops. 4. Provide incentives for research. 5. Monitor faculty performance.	- Faculty recruited. - Training sessions conducted. - Research output improvement	HR & Training Committee
Build research labs and centers of excellence	2026-2028	Establish state-of-the-art research infrastructure. - Enhance research capabilities of faculty and students.	1. Identify research areas. 2. Procure lab equipment. 3. Construct and set up labs. 4. Recruit research staff. 5. Secure research grants.	- Number of labs established. - Research publications. - Funding secured.	Infrastructure Committee

Launch social impact & outreach programs	2028-2030	- Engage students and faculty in community development. - Address local social challenges.	1. Identify key social issues. 2. Collaborate with NGOs. 3. Train students for community work. 4. Implement pilot projects. 5. Scale up based on impact.	- Community projects initiated. - Student participation. - Social impact assessment.	Community Engagement Committee
Introduce specialized courses	2028-2030	- Expand curriculum to include future-ready skills. - Offer certifications aligned with industry needs.	1. Conduct market research. 2. Develop course content. 3. Obtain accreditations. 4. Implement online & offline courses. 5. Assess effectiveness.	- New courses introduced. - Enrollment growth. - Industry partnerships.	Curriculum Development Committee
Establish interdisciplinary research center	2030-2032	- Foster collaborative, multi-domain research. - Increase research funding and output.	1. Define research themes. 2. Secure funding. 3. Set up infrastructure. 4. Recruit research teams. 5. Launch projects and monitor outcomes.	- Research projects initiated. - Funding received. - Industry collaborations.	Research & Innovation Committee
Initiate student & faculty exchange programs	2032-2034	- Provide international exposure to students & faculty. - Enhance global learning opportunities.	1. Identify international partners. 2. Develop MoUs. 3. Set up funding for exchanges. 4. Promote opportunities. 5. Evaluate impact.	- Exchange agreements signed. - Number of participants. - Global exposure impact.	International Relations Committee
Implement mentorship & wellness programs	2034-2035	- Support student academic and personal development. - Improve student well-being.	1. Develop mentorship models. 2. Recruit mentors. 3. Establish wellness centers. 4. Organize workshops. 5. Monitor student progress.	- Mentorship programs implemented - Student well-being improvement - Feedback scores.	Student Welfare Committee

Improve ranking through academic initiatives	2035-2036	- Enhance academic reputation globally. - Improve research and student employability	1. Strengthen research output. 2. Improve faculty quality. 3. Increase industry collaborations. 4. Participate in global accreditations. 5. Develop PR strategies.	- National & international rankings. - Research citations. - Employer satisfaction.	Accreditation Committee
Drive groundbreaking projects	2036-2038	- Promote cutting-edge research in innovation. - File patents and develop commercial applications.	1. Identify breakthrough research areas. 2. Secure industry & government grants. 3. Establish innovation incubation hubs. 4. Encourage patent filing. 5. Launch commercialization efforts.	- Patents filed. - Industry research projects. - Awards & recognitions.	Innovation & Research Committee
Implement eco-friendly campus policies	2038-2039	- Develop a green campus. - Reduce environmental impact of university operations.	1. Conduct sustainability audits. 2. Implement carbon reduction policies. 3. Install renewable energy systems. 4. Improve water & waste management. 5. Engage students in sustainability initiatives.	- Reduction in carbon footprint. - Renewable energy use. - Campus sustainability score.	Sustainability Committee
Build a strong alumni network	2039-2040	- Strengthen alumni engagement and contributions. - Develop mentorship and funding networks.	1. Build an alumni database. 2. Organize alumni networking events. 3. Encourage mentorship & student support. 4. Develop alumni success campaigns. 5. Establish alumni endowments.	- Active alumni participation. - Alumni funding & mentorship programs. - University reputation growth.	Alumni Engagement Committee

9. BRIEF DESCRIPTION OF THE PROJECT IMPLEMENTATION

1. Develop and Launch Interdisciplinary Programs (2025-26)

The objective of this project is to introduce cutting-edge programs in emerging fields, ensuring alignment with industry needs and global standards. The key steps involve identifying interdisciplinary domains, designing a curriculum with industry and academic experts, obtaining necessary regulatory approvals, recruiting specialized faculty, and launching the programs with promotional activities. The success of this project will be measured by the number of new programs introduced, student enrollment rates, and the number of industry collaborations formed. The Academic & Research Committee will oversee the implementation, ensuring high-quality academic standards.

2. Build Research Labs and Centers of Excellence (2026-27)

This project aims to establish state-of-the-art research facilities that will enhance the institution's research capabilities. The implementation involves identifying key research areas, procuring necessary equipment, constructing and setting up research labs, recruiting faculty and research staff, and securing external funding. The effectiveness of the project will be assessed based on the number of labs established, research publications produced, and the amount of research funding secured. The Infrastructure Committee will be responsible for the development and operational setup.

3. Hire Expert Faculty & Conduct Training (2027-28)

To enhance teaching and research quality, this project focuses on recruiting world-class faculty and providing regular training in advanced pedagogy and interdisciplinary collaboration. The major steps include identifying faculty requirements, conducting national and international recruitment drives, organizing faculty training workshops, providing incentives for research productivity, and monitoring faculty performance. The key performance indicators will include the number of faculty recruited, the number of training programs conducted, and improvements in faculty research output. The HR & Training Committee will manage recruitment and training initiatives.

4. Launch Social Impact & Outreach Programs (2027-28)

The goal of this project is to involve students and faculty in real-world problem-solving by launching community outreach programs. The implementation steps include identifying pressing social issues, establishing partnerships with NGOs and government bodies, training students for community engagement, executing pilot projects, and expanding successful initiatives. The effectiveness will be measured by the number of community projects initiated, student participation rates, and the impact assessment of outreach activities. The Community Engagement Committee will oversee program execution.

5. Introduce Specialized Courses (2028-29)

To meet evolving industry demands, this project focuses on expanding the curriculum with specialized courses. Key steps include conducting market research on skill requirements, developing course content with academic and industry experts, obtaining necessary accreditations, implementing courses in both online and offline modes, and continuously updating content based on feedback. The success of this project will be assessed through the number of new courses introduced, student enrollment, and industry certifications linked to the courses. The Curriculum Development Committee will manage the implementation.

6. Establish Interdisciplinary Research Center (2029-30)

This project aims to encourage collaborative research across multiple disciplines, contributing to both technical innovation and societal challenges. The implementation includes defining research themes, securing funding and partnerships, setting up infrastructure and labs, recruiting interdisciplinary research teams, and launching collaborative projects. The project's success will be measured by the number of research projects initiated, external funding received, and industry-academic partnerships established. The Research & Innovation Committee will lead the project.

7. Initiate Student & Faculty Exchange Programs (2030-31)

To provide global learning opportunities, this project will establish exchange programs with international universities. The key steps involve identifying potential international partner institutions, developing formal exchange agreements, setting up funding and scholarships for participants, promoting the program to students and faculty, and evaluating the impact of the exchanges. The effectiveness will be measured by the number of signed agreements, the number of students and faculty participating, and the academic and cultural impact of the exchanges. The International Relations Committee will be responsible for execution.

8. Implement Mentorship & Wellness Programs (2031-32)

This project aims to enhance student support systems by introducing mentorship and wellness programs. The implementation steps include developing structured mentorship models, recruiting mentors from faculty and industry, establishing wellness centers with professional counselors, organizing skill-building and mental health workshops, and monitoring student progress through feedback and impact assessments. The success will be determined by the number of mentorship programs conducted, student participation rates, and improvements in student well-being indicators. The Student Welfare Committee will oversee this initiative.

9. Improve Ranking Through Academic Initiatives (2032-33)

This project focuses on elevating the institution's academic standing in national and international rankings. The steps include strengthening faculty research output, improving student employability, increasing industry collaborations, obtaining international accreditations, and implementing strategic PR campaigns. The project's success will be measured by improvements in rankings, the number of high-impact research publications, and employer feedback on graduates. The Accreditation Committee will lead these efforts.

10. Drive Groundbreaking Projects (2033-34)

To promote innovation and cutting-edge research, this project will support high-impact research initiatives. The implementation steps include identifying breakthrough research areas, securing government and industry funding, establishing innovation incubation centers, encouraging faculty and students to file patents, and commercializing research outcomes. The key performance indicators will include the number of patents filed, industry-funded research projects, and awards or recognitions received. The Innovation & Research Committee will manage this initiative.

11. Implement Eco-Friendly Campus Policies (2034-35)

This project aims to transform the campus into an environmentally sustainable institution. Key steps include conducting sustainability audits, implementing carbon footprint reduction policies, installing renewable energy solutions, enhancing water conservation and waste management, and promoting student-led sustainability initiatives. The effectiveness will be assessed based on reductions in carbon emissions, the percentage of renewable energy usage, and sustainability ratings. The Sustainability Committee will be responsible for implementation.

12. Build a Strong Alumni Network (2035-36)

This project focuses on strengthening alumni engagement to support institutional growth. The major steps involve building an alumni database, organizing networking events and reunions, encouraging alumni mentorship for students, highlighting alumni success stories, and developing alumni contribution programs, such as scholarships and endowments. The success of this initiative will be determined by the number of active alumni members, alumni participation in mentorship programs, and financial contributions received from alumni. The Alumni Engagement Committee will lead the efforts.

10. RISKS AND RISKS MITIGATING STRATEGIES

1. Develop and Launch Interdisciplinary Programs (2025-26)

Risks:

- Low student enrollment due to lack of awareness.
- Difficulty in securing faculty with interdisciplinary expertise.
- Regulatory delays in course approvals.

Risk Mitigation Strategies:

- Conduct extensive outreach campaigns targeting students and parents.
- Collaborate with industry experts and visiting professors for faculty support.
- Work closely with regulatory bodies to ensure compliance and expedite approvals.

2. Build Research Labs and Centers of Excellence (2026-27)

Risks:

- High cost of infrastructure and equipment.
- Slow adoption of research culture among faculty and students.
- Limited industry funding and collaborations.

Risk Mitigation Strategies:

- Secure external research grants and industry sponsorships.
- Provide incentives and research funding to faculty.
- Establish partnerships with industries and research organizations for joint projects.

3. Hire Expert Faculty & Conduct Training (2027-28)

Risks:

- Shortage of qualified faculty in emerging fields.
- High faculty attrition rates due to better opportunities elsewhere.
- Resistance to adopting new pedagogical methods.

Risk Mitigation Strategies:

- ▶ Offer competitive salaries and research incentives to attract faculty.
- ▶ Implement faculty development programs and long-term benefits for retention.
- ▶ Conduct continuous training and workshops on modern teaching methodologies.

4. Launch Social Impact & Outreach Programs (2027-28)**Risks:**

- ▶ Lack of student and faculty participation.
- ▶ Difficulty in measuring social impact.
- ▶ Limited funding for outreach initiatives.

Risk Mitigation Strategies:

- ▶ Integrate community service into the curriculum to ensure participation.
- ▶ Use clear impact assessment frameworks for evaluating outreach success.
- ▶ Seek sponsorships and government support for social initiatives.

5. Introduce Specialized Courses (2028-29)**Risks:**

- ▶ Rapid technological changes making course content obsolete.
- ▶ Low industry acceptance of new courses.
- ▶ Poor student response due to unfamiliar course content.

Risk Mitigation Strategies:

- ▶ Regularly update course content in collaboration with industry experts.
- ▶ Offer certification programs with industry validation.
- ▶ Conduct awareness programs and career counseling for students.

6. Establish Interdisciplinary Research Center (2029-30)**Risks:**

- ▶ Difficulty in securing research funding.
- ▶ Faculty reluctance to collaborate across disciplines.
- ▶ Insufficient research infrastructure.

Risk Mitigation Strategies:

- ▶ Apply for government and private-sector research grants.
- ▶ Provide incentives for interdisciplinary research collaborations.
- ▶ Gradually build research infrastructure through phased investments.

7. Initiate Student & Faculty Exchange Programs (2030-31)**Risks:**

- ▶ High cost of international exchange programs.
- ▶ Low student and faculty participation.
- ▶ Visa and regulatory restrictions for cross-border education.

Risk Mitigation Strategies:

- ▶ Establish scholarships and funding opportunities.
- ▶ Promote exchange benefits through awareness campaigns.
- ▶ Partner with universities in countries with easier visa policies.

8. Implement Mentorship & Wellness Programs (2031-32)

Risks:

- Low student engagement due to lack of interest.
- Shortage of qualified mentors and counselors.
- Difficulty in assessing program effectiveness.

Risk Mitigation Strategies:

- Introduce mentorship as a credit-based program to increase participation.
- Collaborate with alumni and industry experts for mentorship roles.
- Use periodic surveys and feedback mechanisms to assess program success.

9. Improve Ranking Through Academic Initiatives (2032-33)

Risks:

- Delay in measurable improvements in ranking.
- Difficulty in attracting top faculty and researchers.
- Limited global recognition of institutional research.

Risk Mitigation Strategies:

- Strengthen research output through faculty incentives.
- Focus on faculty recruitment from reputed institutions.
- Increase international collaborations and joint research publications.

10. Drive Groundbreaking Projects (2033-34)

Risks:

- High failure rate of innovative research projects.
- Difficulty in securing patents and commercialization opportunities.
- Limited industry interest in funding early-stage research.

Risk Mitigation Strategies:

- Encourage a culture of innovation by providing funding for pilot projects.
- Establish an intellectual property (IP) cell to support patent filings.
- Strengthen industry partnerships to ensure applied research relevance.

11. Implement Eco-Friendly Campus Policies (2034-35)

Risks:

- High initial cost of implementing sustainable infrastructure.
- Resistance from students and staff in adapting to new policies.
- Difficulty in maintaining sustainability standards.

Risk Mitigation Strategies:

- Phase-wise implementation of green initiatives to distribute costs.
- Conduct awareness campaigns to encourage community participation.
- Establish a dedicated sustainability committee for monitoring compliance.

12. Build a Strong Alumni Network (2035-36)

Risks:

- Low alumni engagement and participation.
- Difficulty in maintaining up-to-date alumni records.
- Limited financial contributions from alumni.

Risk Mitigation Strategies:

- Organize frequent alumni networking events and mentorship programs.
- Develop a digital alumni portal for continuous engagement.
- Offer recognition programs for alumni who contribute financially or through mentorship.

11. THE PROGRAMS/PROJECT IMPLEMENTATION STRATEGIES

The consolidated implementation strategy for Arunai University's projects and programs from 2025 to 2040 is described below. Each strategy aims to establish a cohesive framework for achieving the university's long-term goals effectively and sustainably.

1. Curriculum Development and Academic Excellence Programs

Strategy: Establish a core academic committee focused on curriculum development and academic standards. Include regular industry feedback loops, international advisory input, and academic reviews to align with evolving industry needs.

Implementation Steps:

- Form an International Curriculum Advisory Board.
- Conduct bi-annual curriculum reviews to incorporate feedback.
- Design interdisciplinary modules that integrate emerging fields and skill-based learning.

2. Research Labs and Centers of Excellence

Strategy: Prioritize research areas with high potential societal impact and align these with Arunai University's mission of producing ethical and globally competent leaders.

Implementation Steps:

- Develop specialized research labs by 2027, focusing on technological advancement and social impact.
- Establish centers of excellence with sustainable funding and international collaborations.
- Create incentive programs for faculty to engage in active research, publication, and innovation.

3. Strategic International Partnerships and Exchange Programs

Strategy: Build a network of international partners to provide students and faculty with opportunities for global exposure, collaboration, and cultural exchange.

Implementation Steps:

- Sign formal MOUs with universities and research organizations.
- Design student and faculty exchange programs with cultural training modules.
- Allocate scholarships and support services to encourage participation.

4. Community Outreach and Social Impact Initiatives

Strategy: Establish dedicated community engagement programs to reinforce Arunai University's commitment to societal development.

Implementation Steps:

- Launch social impact projects in collaboration with local NGOs and government agencies.
- Involve students in community-based research, fieldwork, and internships.
- Develop measurable outcomes for each project to assess community benefits and refine engagement strategies.

5. Faculty Development and Recruitment Programs

Strategy: Recruit highly qualified faculty, and develop ongoing training programs to build academic leadership and research capacity.

Implementation Steps:

- a. Implement training sessions and workshops on innovative pedagogy and research methods.
- b. Collaborate with reputed academic institutions for faculty exchange and professional development.
- c. Monitor faculty performance and engagement through Key Performance Indicators (KPIs) related to teaching quality, research, and service.

6. Infrastructure Expansion and Technological Upgrades

Strategy: Develop a phased infrastructure expansion plan to accommodate growth in student intake, research labs, and residential facilities.

Implementation Steps:

- a. Prioritize high-impact infrastructure projects, such as research labs, campus facilities, and technology centers.
- b. Integrate eco-friendly and sustainable practices in all new construction and upgrades.
- c. Regularly review infrastructure needs and make adjustments based on feedback from stakeholders.

7. Student-Centered Learning and Mentorship Programs

Strategy: Shift towards a student-centric model that prioritizes personalized learning experiences, academic support, and mentorship.

Implementation Steps:

- a. Implement mentorship programs that connect students with alumni, faculty, and industry experts.
- b. Develop an online platform for students to access resources, feedback, and academic guidance.
- c. Create flexible, personalized learning pathways that allow students to focus on their strengths and interests.

8. Internationalization of Programs and Global Rankings Initiative

Strategy: Strengthen Arunai University's international reputation through targeted programs that focus on global relevance, research excellence, and partnerships.

Implementation Steps:

- a. Pursue international accreditation and ranking improvement strategies by enhancing faculty research output, graduate employability, and student diversity.
- b. Develop unique programs that cater to global markets, including dual-degree and certification programs.
- c. Promote the university at global education fairs and conferences to attract international students and faculty.

9. Innovation, Incubation, and Emerging Fields Programs

Strategy: Foster a culture of innovation by establishing incubation centers, research hubs in emerging fields, and partnerships with tech industry leaders.

Implementation Steps:

- a. Set up an incubation center for startups and early-stage innovation projects.
- b. Introduce programs in emerging fields like AI, data science, and sustainable technology.
- c. Establish industry partnerships to provide students and faculty with real-world experience and funding for high-potential projects.

10. Sustainable Development and Campus Sustainability Initiatives

Strategy: Integrate sustainability into all aspects of campus life, including infrastructure, academics, and operations.

Implementation Steps:

- Form a sustainability committee to oversee initiatives like energy-efficient buildings, waste reduction, and water conservation.
- Educate the campus community about sustainable practices and their importance.
- Implement a sustainability reporting system to track progress and refine strategies.

11. Alumni Engagement and Mentorship Network Development

Strategy: Build a strong alumni network that contributes to mentorship, funding, and professional development for current students and faculty.

Implementation Steps:

- Launch a global alumni engagement platform, offering networking, career resources, and mentorship opportunities.
- Establish a fund for alumni-supported scholarships, research grants and community projects.
- Organize regular alumni events, both online and on-campus, to maintain connections and encourage active participation.

12. University-Wide Risk Management and Assessment Program

Strategy: Develop a centralized risk management and assessment framework to identify, analyze, and mitigate risks associated with strategic projects.

Implementation Steps:

- Form a risk management committee to conduct regular assessments across departments.
- Introduce training for faculty and staff in risk identification and response strategies.
- Implement a risk reporting system that enables real-time monitoring and mitigation planning.

13. Global Leadership and Thought Leadership Conferences

Strategy: Host international conferences to position Arunai University as a thought leader in technical, ethical, and holistic education.

Implementation Steps:

- Organize annual or biennial global conferences on topics of emerging importance, inviting global thought leaders.
- Engage students in conference organization to develop skills in event management and professional networking.
- Publish conference proceedings and white papers to establish Arunai University's contribution to global knowledge.

14. Long-Term Strategic Vision and Evaluation Framework (2040)

Strategy: Create a strategic vision and evaluation framework that continuously aligns the university's development with its mission, market needs, and societal impact.

Implementation Steps:

- Formulate a 5-year strategic plan aligned with the long-term 2040 vision.
- Conduct annual reviews of strategic progress using KPIs and stakeholder feedback.
- Incorporate emerging educational trends and innovations into the strategic vision to remain adaptable and forward-thinking.

ARUNAI

(DEEMED TO BE UNIVERSITY UNDER DISTINCT CATEGORY)

INSTITUTE DEVELOPMENT PLANS

The background of the page features an abstract geometric design. It includes several overlapping shapes: a large blue triangle on the left, a yellow triangle on the right, and a white triangle at the bottom right. There are also smaller blue and yellow shapes scattered throughout. The overall design is modern and clean.

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INSTITUTE DEVELOPMENT PLANS ARUNAI

(DEEMED TO BE UNIVERSITY UNDER DISTINCT CATEGORY)
(2025-2040)

1. PREAMBE

The Institute Development Plan (IDP) is a strategic framework designed to guide the growth and development of higher educational institutions. In line with the directives of the University Grants Commission (UGC), India, the IDP aims to foster holistic and sustainable development within institutes by addressing key areas such as academic programs, infrastructure, research, governance, and community engagement. The plan serves as a roadmap to align institutional goals with national policies in higher education and global trends.

The rationale behind the IDP is to ensure that institutions develop a clear vision and mission that reflect the evolving needs of society. It emphasizes the importance of structured planning to enhance the quality of education, promote innovation, and build competencies among students and faculty members. By implementing an IDP, institutions can identify their strengths and weaknesses, set achievable targets, and develop action plans to meet both short-term and long-term objectives.

Moreover, the IDP encourages institutes to adopt best practices in administration, promote industry-academia linkages, and enhance the employability of students. The plan also focuses on improving access to education, promoting equity, and ensuring inclusivity in the higher education system. This approach supports institutions in adapting to the changing landscape of education while maintaining their distinct identity and contributing meaningfully to the larger educational ecosystem.

Institute Development Plans (IDPs) play a crucial role in the implementation of the National Education Policy (NEP) 2020, which envisions transforming India's higher education system to meet the demands of the 21st century. In the context of NEP-2020, IDPs serve as strategic blueprints for institutions to align their educational practices, infrastructure, and governance with the policy's objectives. The IDP framework emphasizes key aspects such as multidisciplinary education, flexibility in learning, research and innovation, and the promotion of digital and technological advancements. It helps institutions systematically plan for achieving holistic student development, enhancing faculty capacities, and fostering industry-academia linkages. The IDP also enables institutions to work towards accreditation, quality improvement, and internationalization while ensuring equity, inclusion, and sustainability. By integrating NEP-2020 guidelines, IDPs ensure that institutions become centers of excellence that cater to the diverse needs of learners, contributing to India's goal of becoming a global knowledge super power.

Introduction

Arunai Engineering College (AEC), along with other two institutes of the management, Kamban College of Arts and Science for Women and Karan Arts and Science College, has been a beacon of quality education and a transformative force in the region. With a mission to develop proficient, morally upright, and socially responsible professionals, AEC has consistently delivered a high standard of education, supported by a holistic approach to student development. Located on the Chittoor-Cuddalore national highway, the campus provides a serene, inspiring learning environment that fosters intellectual curiosity and personal growth.

In alignment with its vision "to shape and transform youth with utmost care, innovative and dedicated service for morals and education," the management Sarswathi Ammal Educational Trust is now poised to ascend to a new level of excellence by transforming into a "Distinct Deemed University." This transition is inspired by the goals set forth in the University Grants Commission (UGC) Regulations for Deemed Universities, 2023. By achieving this status, the institution aims to become a pioneering force in higher education, offering unique programs that seamlessly integrate technical excellence, ethical values, and holistic development, ultimately contributing to the nation's advancement and societal well-being.

This Detailed Project Report (DPR) encapsulates the strategic vision and operational framework that will guide the transformation over the next 15 years. The report is structured to provide a comprehensive overview of the strategic and rolling implementation plans that will underpin this journey. These include:

- a) Academic Plan:** Envisioning a curriculum that blends technical depth with interdisciplinary breadth, aiming to equip students with skills for emerging fields and societal challenges.
- b) Faculty Recruitment Plan:** Building a team of accomplished, research-driven educators with international exposure, to create a knowledge ecosystem that promotes innovation and critical thinking.
- c) Student Admission Plan:** Establishing an admission strategy that attracts diverse, high-caliber students from across the globe, enhancing the university's intellectual and cultural diversity.

Introduction

- d) Research Plan:** Creating an impactful research ecosystem, fostering collaboration with industry and academic institutions worldwide, and focusing on areas that address societal challenges and drive technological advancement.
- e) Campus Information and Communication Technology Plan:** Adopting cutting-edge ICT infrastructure to create a digitally empowered campus, facilitating seamless teaching, learning, and administrative processes.
- f) Infrastructure Development Plan:** Expanding and enhancing campus facilities to support a dynamic learning environment, inclusive of state-of-the-art laboratories, libraries, and recreational spaces.
- g) Finance Plan:** Ensuring the financial sustainability and growth of the institution through diversified funding sources, including industry collaborations, alumni contributions, and government grants.
- h) Administrative Plan:** Streamlining administrative processes to support efficient and effective operations, ensuring transparency, accountability, and alignment with institutional goals.
- i) Governance Plan:** Establishing a governance framework that emphasizes strategic oversight, ethical leadership, and stakeholder engagement, ensuring the institution's integrity and commitment to its mission.

With clear annual milestones and action plans, this DPR details a transformative journey that aims to elevate Arunai Engineering College and other two sister institutions, into a leading institution recognized for its academic rigor, ethical standards, and contribution to societal progress. The report also includes an assessment of project institutes current academic and physical infrastructure, laying the groundwork for the envisioned development trajectory. By adhering to this roadmap, the management aspires to achieve national and international distinction, nurturing future leaders who embody technical prowess, integrity, and a commitment to societal impact.



Vision & Mission

VISION

"To be a pioneering university of distinction, renowned nationally and internationally for its extraordinary, unique programs that integrate technical excellence with ethical and holistic development. We are committed to cultivating globally competent professionals who are not only skilled and innovative but also deeply rooted in values and dedicated to social impact. By fostering transformative learning in a serene, inspiring environment, we empower individuals to excel in their fields, uplift communities, and serve as ambassadors of integrity and wisdom on the world stage."

This vision highlights AEC's commitment to unique, high-impact programs that elevate its reputation and reach, enhancing its appeal and influence both in India and globally.

MISSION

"To advance knowledge and societal impact through distinctive programs that blend professional ability with moral education, enabling our graduates to become valued leaders who drive positive change both locally and internationally."

STRATEGIC GOALS

1. Academic Excellence and Distinct Programs
2. Infrastructure and Learning Environment
3. Faculty Development and Recruitment
4. Community Engagement and Social Impact
5. Program Diversification and Expansion
6. Research and Innovation Ecosystem
7. Strengthening International Partnerships
8. Enhanced Student Experience
9. Community Impact and Outreach Expansion
10. Global Academic Recognition and Rankings
11. Innovation and Leadership in Research
12. Sustainable Development and Social Responsibility
13. Enduring Global Partnerships and Influence
14. Alumni Network and Societal Impact

6. MAPPING OF UNIVERSITY GOALS WITH THE UGC RECOMMENDED STRATEGIC GOALS

SR. NO.	UNIVERSITY GOALS AS PER VISION AND MISSION	STRATEGIC GOALS AS PER UGC IDP GUIDELINES
1	Academic Excellence and Distinct Programs	Promote academic quality and program differentiation through outcome-based education, distinctive curricula, and interdisciplinary approaches.
2	Infrastructure and Learning Environment	Develop and maintain state-of-the-art infrastructure and resources that foster an effective and holistic learning environment.
3	Faculty Development and Recruitment	Implement a comprehensive faculty development plan to recruit, retain, and support faculty in research and academic advancement.
4	Community Engagement and Social Impact	Encourage community engagement by aligning institutional activities with societal needs, promoting social responsibility and impact.
5	Program Diversification and Expansion	Expand academic offerings by introducing new and emerging fields of study to meet industry demands and student interests.
6	Research and Innovation Ecosystem	Cultivate a robust research ecosystem that supports innovation, interdisciplinary research, and solution-driven projects addressing societal challenges.
7	Strengthening International Partnerships	Establish strategic international partnerships to enhance academic and research collaboration, exchange programs, and joint projects.
8	Enhanced Student Experience	Provide a comprehensive and enriching student experience by enhancing academic support, co-curricular activities, and personal development programs.
9	Community Impact and Outreach Expansion	Broaden the institution's societal outreach and impact through extension activities, awareness programs, and community partnerships.
10	Global Academic Recognition and Rankings	Pursue initiatives to achieve national and international recognition, accreditations, and rankings for academic and research excellence.
11	Innovation and Leadership in Research	Drive leadership in research by focusing on innovative projects, establishing centers of excellence, and supporting entrepreneurial initiatives.
12	Sustainable Development and Social Responsibility	Promote sustainable development principles and integrate social responsibility into institutional activities, research, and community programs.
13	Enduring Global Partnerships & Influence	Build long-term global alliances with academic institutions and industry leaders to enhance the university's international presence and influence.
14	Alumni Network and Societal Impact	Strengthen the alumni network to foster meaningful connections and enable alumni to contribute positively to the institution and society

7. THE ENABLES TO ACHIEVE THE STRATEGIC GOALS

SR. NO.	UNIVERSITY GOALS	ENABLES	REMARKS
1	Academic Excellence and Distinct Programs	Outcome-based curriculum, experienced faculty, academic resources, and quality assurance	Focus on delivering a differentiated curriculum with measurable learning outcomes to promote excellence.
2	Infrastructure and Learning Environment	Modern facilities, smart classrooms, ICT infrastructure, green campus initiatives	Develop a supportive and sustainable learning environment through advanced infrastructure and technology.
3	Faculty Development and Recruitment	Faculty development programs, research incentives, international collaborations	Recruit and retain qualified faculty through training, research opportunities, and global exposure.
4	Community Engagement and Social Impact	Community outreach programs, social responsibility initiatives, local partnerships	Engage actively with the community to address local issues and foster social impact through service learning.
5	Program Diversification and Expansion	Industry collaboration, interdisciplinary curriculum, regulatory approvals	Introduce new programs that address current industry trends and student interests in emerging fields.
6	Research and Innovation Ecosystem	Research funding, centers of excellence, interdisciplinary research initiatives	Develop an ecosystem that supports innovation and research in areas addressing societal and global needs.
7	Strengthening International Partnerships	MoUs with foreign institutions, joint programs, faculty/student exchange programs	Enhance global collaboration to enrich academic and research capabilities with international perspectives.
8	Enhanced Student Experience	Student support services, co-curricular activities, career guidance, wellness programs	Ensure an enriching and holistic student experience, focusing on both academic success and personal growth.
9	Community Impact and Outreach Expansion	Extension programs, community training, awareness campaigns	Expand outreach activities to positively impact local communities, leveraging institutional knowledge.
10	Global Academic Recognition and Rankings	Accreditation, benchmarking, international collaborations	Pursue recognitions that validate quality and enhance the university's global standing and attractiveness.
11	Innovation and Leadership in Research	Innovation labs, incubation centers, entrepreneurship development	Promote leadership in research by supporting innovation, startup culture, and entrepreneurial projects.
12	Sustainable Development and Social Responsibility	Sustainability programs, eco-friendly campus initiatives, awareness drives	Integrate sustainable practices across the campus, focusing on environmental responsibility and social values.
13	Enduring Global Partnerships and Influence	Strategic global alliances, faculty exchange, collaborative research projects	Build influential and enduring partnerships that support long-term global academic and research engagement.
14	Alumni Network and Societal Impact	Alumni association, mentoring programs, alumni-led initiatives	Leverage alumni expertise and networks to contribute to the university and foster societal impact.

8. DEVELOPMENTAL OBJECTIVES OF ARUNAI DEEMED TO BE UNIVERSITY - TIRUVANNAMALAI (TAMIL NADU)

GOALS OF ARUNAI UNIVERSITY/ STRATEGIC GOALS	DEVELOPMENTAL OBJECTIVES FOR ARUNAI UNIVERSITY
Institutional Visibility and Reputation	Achieve inclusion in the top 500 bracket of international university rankings by enhancing global visibility and reputation through high academic standards, partnerships, and strategic initiatives.
Research and Innovation	Achieve 200% growth in research outcomes, including quality publications in Quartile Journals, patents, sponsored projects, technology products, and books. - Establish a robust research culture, incentivizing faculty and students to engage in cutting-edge research and innovation.
Multidisciplinary and Interdisciplinary Research	Develop six multidisciplinary research centers and introduce five interdisciplinary programs to foster innovation, addressing complex global and societal challenges.
Industry-Academia Collaboration and Skill Development	Establish 10 collaborative centers of excellence that provide essential skills for enhancing academic pursuits and industry alignment. Strengthen linkages with industry to ensure the curriculum remains relevant to evolving industrial demands.
Entrepreneurship and Innovation Ecosystem	Set up an industry-partnered incubation center to nurture entrepreneurship and innovation, fostering an entrepreneurial ecosystem that provides mentorship and resources for startups.
Encouraging Entrepreneurship and Higher Studies	Encourage 25% of the learner population to opt for entrepreneurship or pursue higher studies, creating pathways for advanced education and entrepreneurial success.
Global Faculty and Talent Mobility	Attract 5% of faculty from reputed international universities, promoting global academic and research excellence. - Strengthen global talent exchange and foster diverse perspectives in the teaching-learning process.
Digital Transformation and Infrastructure	Develop a comprehensive digital ecosystem that enhances communication and collaboration between stakeholders, using advanced learning management systems, digital tools, and technologies.
Collaborations and Partnerships	Form 50 collaborations and MoUs with reputed institutions in India and abroad to enhance academic programs, research collaborations, and student exchange opportunities. Expand international and national networks, facilitating knowledge sharing and joint research initiatives.

EXPLANATION OF THE DEVELOPMENTAL OBJECTIVES

1. Global Acclaim and Ranking:

- ▶ Elevate ARUNAI University's profile by participating in globally recognized ranking systems (e.g., QS, ABET), focusing on academic quality, international collaborations, and research output.

2. Exponential Growth in Research Outcomes:

- ▶ Foster an environment that supports increased research publications in high-impact journals, securing patents, and developing technologies that contribute to the knowledge economy.
- ▶ Provide incentives and funding for faculty and student research, promoting collaborations with global institutions.

3. Multidisciplinary and Interdisciplinary Programs:

- ▶ Create academic and research programs that encourage collaboration across disciplines, blending science, engineering, management, social sciences, and humanities for solving real-world problems.
- ▶ Establish centers of excellence in emerging areas like AI, data science, renewable energy, biotechnology, and entrepreneurship.

4. Industry-Collaborative Centers of Excellence:

- ▶ Partner with industry leaders to create 10 collaborative centers that provide hands-on training, internships, and industry-driven projects, ensuring students graduate with practical skills and industry-readiness.

5. Entrepreneurship and Incubation Ecosystem:

- ▶ Launch an industry-supported incubation center to provide students and faculty with resources, mentorship, and seed funding for startups.
- ▶ Promote entrepreneurial thinking and skills through hackathons, workshops, and partnerships with venture capitalists.

6. Student Entrepreneurship and Higher Studies Focus:

- ▶ Encourage 25% of students to either launch their own startups or pursue advanced degrees (master's, doctoral programs) in leading institutions, globally and nationally.
- ▶ Facilitate pathways and support systems like career counseling, research grants, and incubator programs.

7. Attracting International Faculty:

- ▶ Create a dynamic, diverse academic community by recruiting faculty from top global universities, enhancing the university's teaching and research capabilities.
- ▶ Promote faculty exchange programs to encourage knowledge transfer and global academic practices.

8. Building a Digital Ecosystem:

- ▶ Integrate cutting-edge digital technologies across academic and administrative processes to enhance the learning environment, including the deployment of AI, cloud-based solutions, and advanced LMS platforms.
- ▶ Ensure that all stakeholders—students, faculty, and staff—can communicate, collaborate, and access learning resources seamlessly in a secure digital ecosystem.

9. Collaborations and Global Partnerships:

- ▶ Forge at least 50 collaborations with internationally recognized institutions, research bodies, and industries, allowing students and faculty to benefit from exchange programs, dual degrees, and joint research initiatives.
- ▶ Actively pursue internationalization strategies that elevate ARUNAI's standing in the global academic community.

9. THE PROJECT PLANNING (2025-2040)

PHASE	YEAR	GOAL	PROJECT	COST (INR CR)	COMMITTEE	RESOURCES REQUIRED
Short-Term	2025-30	Academic Excellence and Distinct Programs	Develop and launch interdisciplinary programs	5.00	Academic & Research Committee	Faculty, Digital Infrastructure
		Infrastructure and Learning Environment	Build research labs and centers of excellence	6.00	Infrastructure Committee	Lab Equipment, Faculty
		Faculty Development and Recruitment	Hire expert faculty & conduct training	3.00	HR & Training Committee	Trainers, Digital Resources
		Community Engagement & Social Impact	Launch social impact & outreach programs	2.00	Community Engagement Committee	Local Networks, Faculty Involvement
Medium-Term	2030-35	Program Diversification & Expansion	Introduce specialized courses	4.00	Curriculum Development Committee	Faculty, Online Learning Tools
		Research and Innovation Ecosystem	Establish interdisciplinary research center	10.00	Research & Innovation Committee	Research Grants, Labs
		Strengthening International Partnerships	Initiate student & faculty exchange programs	5.00	International Relations Committee	Travel Grants, MoUs
		Enhanced Student Experience	Implement mentorship & wellness programs	4.00	Student Welfare Committee	Counseling Centers, Workshops
Long-Term	2035-40	Global Academic Recognition & Rankings	Improve ranking through academic initiatives	7.00	Accreditation Committee	Research Publications, PR Strategies
		Innovation & Leadership in Research	Drive groundbreaking projects	8.00	Innovation & Research Committee	R&D Funds, Patents
		Sustainable Development & Social Responsibility	Implement eco-friendly campus policies	5.00	Sustainability Committee	Renewable Energy Infrastructure
		Alumni Network & Societal Impact	Build a strong alumni network	3.00	Alumni Engagement Committee	Events, Networking Portals
Total				75.00		

10. THE HEAD-WISE EXPENDITURE FOR THE PLANED PROJECTS

YEAR	PROJECT	Cost of HR (INR Cr)	Cost of Machinery & Equipment (INR Cr)	Cost of Infrastructure (INR Cr)	Cost of Contingencies (INR Cr)	Total Cost (INR Cr)
2025-26	Develop and launch interdisciplinary programs	2.00	1.50	0.80	0.70	5.00
2025-28	Hire expert faculty & conduct training	1.50	0.80	0.40	0.30	3.00
2026-28	Build research labs and centers of excellence	2.50	2.50	0.60	0.40	6.00
2028-30	Launch social impact & outreach programs	0.80	0.60	0.30	0.30	2.00
2028-30	Introduce specialized courses	1.50	1.50	0.60	0.40	4.00
2030-32	Establish interdisciplinary research center	4.00	4.50	0.90	0.60	10.00
2032-34	Initiate student & faculty exchange programs	2.00	1.50	0.80	0.70	5.00
2034-35	Implement mentorship & wellness programs	1.50	1.20	0.80	0.50	4.00
2035-36	Improve ranking through academic initiatives	3.00	2.50	0.90	0.60	7.00
2036-38	Drive groundbreaking projects	3.50	3.50	0.60	0.40	8.00
2038-39	Implement eco-friendly campus policies	2.00	1.50	0.90	0.60	5.00
2039-40	Build a strong alumni network	1.00	1.00	0.50	0.50	3.00
Total	—	25.30	22.60	8.10	5.60	75.00

SUMMARY:

- ▶ Cost of HR: 25.30 Cr
- ▶ Cost of Machinery & Equipment: 22.60 Cr
- ▶ Cost of Infrastructure: 8.10 Cr
- ▶ Cost of Contingencies: 5.60 Cr

TOTAL COST: 75.00 CR

11. SUMMARY OF THE INSTITUTIONAL DEVELOPMENT PLAN (IDP) FOR ARUNAI UNIVERSITY

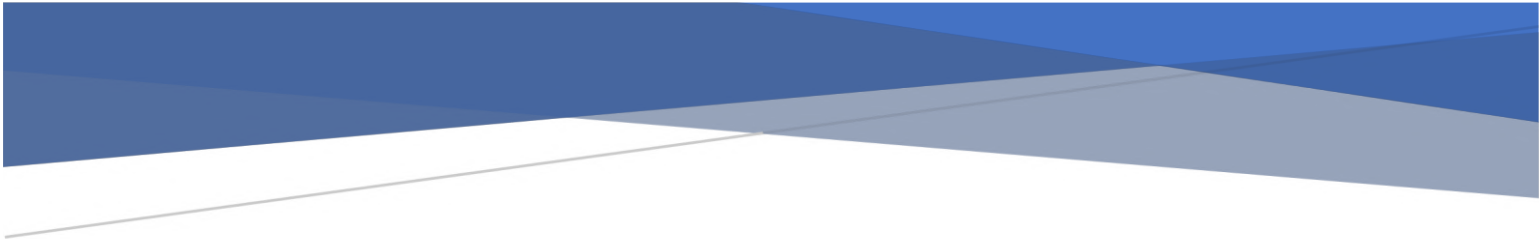
The Institutional Development Plan (IDP) for ARUNAI University, aims to strategically position the university among the top global institutions by 2040. Saraswathi Ammal Educational Trust, founded with a vision of providing world-class education, has played a vital role in shaping the educational landscape of Tamilnadu, India. The trust operates multiple institutions across various disciplines, emphasizing excellence, innovation, and social responsibility. ARUNAI University, established as a forward-thinking institution, envisions becoming a globally recognized hub for education, research, and entrepreneurship. Its mission is to create a learner-centric, inclusive community that fosters intellectual, experiential, and lifelong learning, while contributing meaningfully to society.

Aligned with its vision and mission, the university's key goals are to achieve global recognition in the top 500 rankings, attain significant growth in research output, expand its multidisciplinary and interdisciplinary offerings, nurture entrepreneurship, and establish impactful collaborations with national and international institutions. Strategic goals are set to support these aspirations, including the establishment of research centers, industry-partnered incubation hubs, a robust digital ecosystem, and the recruitment of global faculty. The plan focuses on fostering intellectual rigor, creativity, and innovation among students and faculty while maintaining a strong social mission of community development and inclusion.

The IDP has been developed in three phases (short-term, medium-term, and long-term) over a 15-year period from 2025 to 2040, with a detailed roadmap that includes the gradual implementation of projects and initiatives. These range from enhancing global rankings and research outputs in the short term to creating centers of excellence and incubation hubs in the medium term, and achieving global leadership in research, education, and entrepreneurship in the long term. The plan also outlines the developmental objectives, such as establishing 10 collaborative centers, achieving 150% research growth, and expanding student entrepreneurship to 25% by 2040. Each phase is structured to address key university goals, with enablers such as funding, human resources, and infrastructure, designed to drive the plan's success.

The total project cost has been optimized to ₹75 Crores, ensuring a balanced allocation of resources across different strategic initiatives. The projects have been designed to create a high impact while ensuring financial sustainability. By the end of 2040, ARUNAI University aspires to stand as a global leader in higher education, research, and entrepreneurship, fostering innovation and creating societal impact. This Institutional Development Plan not only sets a clear roadmap for achieving academic excellence but also reflects the university's commitment to holistic development, inclusive growth, and societal contributions.

The management and staff of the ARUNAI University are fully committed to the successful implementation of the Institutional Development Plan (IDP) in alignment with the recommendations of the UGC, India. The university leadership, under the guidance of the Saraswathi Ammal Educational Trust, recognizes the critical importance of fostering a culture of excellence, innovation, and inclusivity to meet both academic and societal needs. This unwavering commitment is reflected in the dedication to building robust systems and processes, aligning resources, and maintaining a strong focus on the institution's long-term goals. The faculty and administrative teams will work collaboratively to realize the university's vision through consistent efforts in enhancing teaching, research, and entrepreneurship. With a clear understanding of the strategic goals outlined in the IDP, the management has pledged to provide the necessary infrastructure, professional development opportunities, and financial resources to ensure the smooth execution of the plan. By fostering a climate of accountability and active participation, the university's stakeholders are determined to meet the global standards of education, research, and social responsibility as envisaged in the IDP, and to position ARUNAI University as a leading institution both nationally and internationally.



ACADEMIC PLAN

ARUNAI DEEMED TO BE UNIVERSITY

THE PROPOSED PROGRAMS TO BE OFFERED AND THE ACADEMIC PLANNING

The Proposed Under-Graduate, Post-Graduate and Research programs.

TABLE NO.1: UNDERGRADUATE AND POST-GRADUATE PROGRAMS

Stream	S. No	Course Name
Environment and Sustainability	1.	B.Tech. Climate Science & Engineering
	2.	M.Tech. Sustainability in Building Maintenance
Artificial Intelligence and Computers	3.	M.Tech. Natural Language Processing
	4.	M.Tech. AI in VLSI technology
Agriculture and Biotechnology	5.	M.Tech. Agro-Electronics
	6.	B.Sc. Organic Farming
	7.	M.Sc. Clinical Research and Health Informatics (Integrated Course)
Indian Knowledge System	8.	B.A. Tamil Culture and Civilization
	9.	M.A. Thozil Sar Kalaigal
Commerce and Modern Trade	10.	B.Com. Financial Risk Management and Taxation
Skill Development Courses	11.	PG Diploma Solar PV and Battery Management Systems
	12.	PG Diploma AIML in Health Care Management

TABLE NO. 2: RESEARCH PROGRAMS/DOMAINS

Sr.No	Name of the Program	Available Vertical Specialization
1.	Agricultural Technologies	M.Tech (Agro Electronics) – Proposed Distinct Course B.Tech (Agricultural Engineering) – Existing Course
2.	VLSI	M.Tech. AI in VLSI technology – Proposed Distinct Course B. Tech (AIDS) – Existing Course B.E (AIML) – Existing Course B.E (ECE) – Existing Course
3.	Computer Science and Engineering (Specialization in AI)	M.Tech. Natural Language Processing – Proposed Distinct Course M.Tech. AI in VLSI technology – Proposed Distinct Course M.E Computer Science and Engineering B. Tech (AIDS) – Existing Course B.E (AIML) – Existing Course B.E Computer Science and Engineering – Existing Course
4.	Sustainable Technologies	M.Tech. Sustainability in Building Maintenance – Proposed Distinct Course B.E. Civil Engineering – Existing Course
5.	Health Informatics	M.Sc. Clinical Research and Health Informatics (Integrated Course) – Proposed Distinct Course M.Tech Biotechnology – Existing Course B.Sc. Microbiology – Existing Course

		B.Tech Biotechnology – Existing Course
6.	Tamil Literature and Tradition	M.A. Thozil Sar Kalaigal – Proposed Distinct Course M.A Tamil – Existing Course B.A. Tamil Culture and Civilization – Proposed Distinct Course B.A Tamil – Existing Course

The Academic Plan to introduce the Under-graduate, Post-graduate and Research Programs and Enrolment Plan (2025-2030)

Table 3: Year-wise Introduction Academic Plan and Enrolment Plan (2025-2030)

Year	Undergraduate Programs	Enrolment Per Year	Postgraduate Programs	Enrolment Per Year	Skill Development Programs	Enrolment Per Year
2025	B.Sc. Organic Farming	60	M.Tech. Sustainability in Building Maintenance	15	PG Diploma Solar PV and Battery Management Systems	30
2026	B.A. Tamil Culture and Civilization	60	M.Tech. Agro-Electronics	15	PG Diploma AIML in Health Care Management	30
2027	B.Com. Financial Risk Management and Taxation	60	M.Sc. Clinical Research and Health Informatics (Integrated Course)	15	Nil	
2028	B.Tech. Climate Science & Engineering	60	M.Tech. Natural Language Processing	15	Nil	
2029	Nil		M.Tech. AI in VLSI Technology	15	Nil	
2030	Nil		M.A. Thozil Sar Kalaigal	15	Nil	

Table 4: Academic Plan for Research Work (2025-2035)

Year	Research Program	Research Focus & Specialization*
2025	Agricultural Technologies	Development of smart agro-electronics solutions for precision farming
2026	VLSI	AI-driven VLSI design and automation for semiconductor advancements
2027	Computer Science and Engineering (AI Specialization)	Natural Language Processing (NLP) applications in intelligent systems
2028	Sustainable Technologies	Smart infrastructure and sustainable building maintenance systems
2029	Health Informatics	AI-driven clinical research and digital health record management
2030	Tamil Literature and Tradition	Digital preservation and AI-based analysis of Tamil literature
2031	Agricultural Technologies	Integration of IoT and AI in agricultural automation
2032	VLSI	Advanced chip design for AI and embedded systems
2033	Computer Science and Engineering (AI Specialization)	AI ethics and explainable AI models in computing
2034	Sustainable Technologies	Circular economy and renewable energy solutions for smart cities
2035	Health Informatics	Genomics-based AI research for personalized medicine

***The detailed description of the avenues of research under each domain is provided in the DPR.**

RECRUITMENT PLAN-TABLE-1

Year	Undergraduate Programs	Total Enrolment Per Year	Year-1	Year-2	Year-3	Year-4	Year-5
1	Under-graduate Program (4)	240	60	60	60	60	
2	Post-graduate Program (6)	90	15	15	15	15	15
3	Post-graduate Diploma (2)	60	30	30			
Total			105	105	75	75	15

RECRUITMENT PLAN: TABLE-2

Year	Undergraduate Programs	Total Enrolment Per Year	Total Enrolment Year-1	Total Enrolment Year-2	Total Enrolment Year-3	Total Enrolment Year-4	Total Enrolment Year-5
1	Under-graduate Program (4)	240	105	210	285	360	375
2	Post-graduate Program (6)	90					
3	Post-graduate Diploma (2)	60					
Total Enrolment			105	210	285	360	375
Total faculty Requirement with 1:20			6	11	14	18	19

NOTE: THE FACULTY MEMEBERS FOR THE RESEARCH AND DEVELOPMENT COMPONENT WILL BE RECRUITED ON THE CONTRACTUAL BASIS AS PER THE FINANCIAL PLANS SHOWN IN THE STRAEGIC PLAN.